

Bio-Plex™ Manager 3.0 Software User Guide



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Table of Contents

1. Bio-Plex System Overview.....	1
Advantages of the Bio-Plex System	2
For More Information.....	3
 2. Bio-Plex Manager Software Overview.....	 5
Note on this User Guide.....	5
Software Versions.....	5
Workflow	6
Types of Files	7
Protocol File	7
Results File	8
Key Software Features.....	8
 3. Installation and Specifications	 9
System Components.....	9
Bio-Plex Manager	10
System Requirements.....	10
Installation	11
Microsoft.net.....	11
BioPlex.mdb File	11
Hardware Protection Key	11
Uninstalling	12
Technical Support	12

4. Getting Started	13
Starting Bio-Plex Manager	13
Required Screen Resolution.....	13
Communication to the Array Reader and Microplate Platform	14
COM Port Settings	14
Disconnecting and Reconnecting.....	15
Menu and Toolbars	16
Status Bar	16
Quick Guide	17
Sample Needle Adjustment.....	17
 5. Bio-Plex Suspension Array System Controls	 21
MCV Plate II	22
Start Up 	23
Optics Warm Up and Shut Down 	24
Calibration 	25
Opening the Calibration Dialog Box	25
Calibration Microsphere Control Numbers	26
Calibration Set Up	28
Performing the Calibration.....	29
Calibration Log	30
Wash Between Plates 	31
Remove Air Bubbles 	32
Unclog 	33

Validation	34
Validation Kit	34
Setting Up a Validation Run	35
Performing a Validation Run	36
Validation Results	37
Validation Log	38
Platform Heater 	40
Instrument Information 	42
Eject/Retract Plate 	42
Additional Instrument Functions	43
Wash	43
Drain	43
Prime	43
Sanitize	43
Alcohol Wash	44
Alcohol Flush	44
Back flush	44
Cancel Operation	44
Shut Down 	44
6. Preparing a Protocol	47
Creating/Opening Protocol Files 	47
Saving Protocols	47
Reducing the File Size of Protocols	48
Protocol Window	49
Step 1. Describe Protocol 	50
Step 2. Select Analytes 	50

- Selecting Analytes 51
- Customizing Analytes and Panels 52
 - Creating a New Panel of Analytes 52
 - Editing a Panel of Analytes 54
 - Deleting and Renaming Custom Panels 55
- Combining Panels of Analytes 56
- Step 3. Format Plate  56
 - Plate Formatting 58
 - Defining Unknown Sample Wells 58
 - Well Numbering and Replicate Groups 59
 - Defining Standard Wells 62
 - Defining Control Wells 63
 - Defining Blank Wells 64
 - Deleting or Changing Well Formatting 65
 - Plate Grouping 65
 - Defining a Group 67
 - Defining the Reference 67
 - Printing the Plate Format 68
- Step 4. Enter Standards Info  68
 - Selecting External Standards 70
 - Standards Information 71
 - Entering Concentrations (for Current Standards Only) 73
 - Concentration Units 75
 - Regression Method 76
 - Logistic Weighting 77
 - Recovery Percentage Range 78
- Step 5. Enter Controls Info  79
- Step 6. Enter Sample Info (optional)  82
- 7. Running the Protocol  85**
 - Run Protocol Window 85

Bead Count.....	87
Advanced Settings	87
Sample Size	88
Automatically Saving Results	88
Sampling Errors	88
PMT Voltage	90
Doublet Discriminator Gate Range	91
Plate Loading Guidelines	91
Running the Protocol.....	92
Status Bar	93
Manually Stopping a Reading	94
Generating Results from a Protocol	94
Bead Map and Histogram 	94
Histogram.....	96
DD Gate Range.....	97
Selecting the Channel Type	98
Scaling the Histogram Data.....	99
Magnifying and Resizing the Histogram.....	99
Bead Map.....	100
Changing the X- and Y-Axes of the Map.....	101
Scaling the Bead Map Data.....	102
Magnifying and Resizing the Bead Map.....	102
Filtering the Bead Map Display	103
Raw Data Table 	104
Bead Statistics	105
Sampling Errors	106
Other Table Formatting.....	106
Copying the Raw Data	106
Rerun/Recovery Mode	107

8. Analyzing the Results..... 109

Opening a Results File  109

Saving a Results File..... 109

 Reducing the Size of a Results File.....109

Results Window 110

Viewing/Changing the Protocol Settings..... 112

Raw Data  113

 Exporting Raw Data 114

 Printing Raw Data 114

Report Table  115

 Report Table Columns 115

 Report Table Options 117

 Replicate Groups..... 118

 Selecting Outliers 119

 Obs/Exp*100 Column..... 120

 Conc in Range Column 120

 Table Formatting 121

 Exporting the Report Table 122

 Copying the Report Table 124

 Printing the Report Table 124

Standard Curve  125

 Regression Methods 127

 Logistic-5PL 128

 Logistic-4PL 129

 Linear 129

 Cubic Spline..... 130

 Point-to-Point 130

Linear versus Logistic Regression Methods	130
Copying the Standard Curve	131
Printing the Standard Curve	131
Run Info 	132
Appendix 1. Troubleshooting Guide.....	133
Appendix 2. Glossary.....	139
Appendix 3. Table Error Codes	147
Appendix 4. Basic Concepts	149
Microspheres	149
Reporter fluorochromes.....	149
Fluidics	150
Excitation	150
Appendix 5. Microsphere Handling.....	151
Microsphere Dispersion.....	151
Probe Sonicator.....	151
Bath Sonicator	151
Enumeration of Microsphere Suspensions	152
Microsphere separation methods	152
Centrifugation	152
Vacuum	152
Microsphere Agitation During Assay	153
Microsphere Stability and Storage	153
Appendix 6. Product Numbers	155

Appendix 7. Reporting Problems 157

Appendix 8. Software Warranty..... 159

1. Bio-Plex System Overview

The Bio-Plex suspension array system is a flow-based dual laser system for simultaneously identifying and quantitating up to 100 different analytes in a single biomolecular assay. The system detects and measures molecules bound to the surfaces of fluorescent microspheres, providing highly accurate, real-time digital analysis of serum or culture media samples as small as 50 μ l. The Bio-Plex system allows for quantitative analysis of a wide variety of cell biology assays, including immunoassays, complex genetic assays, and enzymatic assays.

The Bio-Plex suspension array system is fully integrated, and consists of the Bio-Plex array reader and microplate platform, validation and calibration reagents, a selection of cytokine and phosphoprotein assays, an amine coupling kit, sample preparation reagents, and a computer running Bio-Plex Manager software.



Figure 1. Bio-Plex array reader and microplate platform.

The assays contain sets of microscopic color-coded beads, each of which is conjugated with a different reactant.

Reactants can include:

- DNA
- Enzyme substrates
- Receptors
- Antigens
- Antibodies

These can be used to create, for example, a capture sandwich immunoassay.

To perform a multiplex reading, samples are mixed with conjugated microsphere and reactant mixtures and fluorescent reporter molecules are added. The assays are loaded into the wells of a 96-well microtiter plate and the plate is inserted into the Bio-Plex microplate platform. The platform and Bio-Plex array reader are controlled by a computer running Bio-Plex Manager software.

The detection system in the array reader uses two lasers to analyze the microspheres in a flow stream. The first laser identifies each microsphere and its associated analyte based on the fluorescent signature of the microsphere, and the second measures the amount of analyte using the reporter molecules attached to the analytes. When the reading is complete, Bio-Plex Manager displays the raw data and generates detailed summary reports.

Advantages of the Bio-Plex System

With the Bio-Plex system, you can:

- Simultaneously quantitate up to 100 different analytes from culture media and serum samples as small as 50 μ l.
- Automatically analyze all the samples in a 96-well microtiter plate—yielding up to 9,600 data points—in about 30 minutes.
- Instantly customize your experiments by mixing Bio-Plex assays or creating your own assays.
- Analyze results, prepare reports, and print and/or export data immediately after each reading.

For More Information...

For more information on the principles and concepts of the Bio-Plex suspension array system, see Appendix 4 on page 149, or visit our Web site at <http://www.bio-rad.com/bioplexsystem>.

2. Bio-Plex Manager Software Overview

Note on this User Guide

All references to “Bio-Plex Manager” in this user guide refer to the Workstation version of the software. The Desktop and Network Desktop versions of the software include the analysis functions described in this manual, but not the system control functions. (See the next section for details.)

This user guide assumes that you are familiar with your computer and standard Microsoft® Windows® commands and functions, such as opening, closing, and saving files, and moving and clicking your mouse.

Software Versions

Bio-Plex Manager software comes in three versions:

- **Bio-Plex Manager Workstation** is designed to collect, analyze and output data from the Bio-Plex system. It runs as a Windows® 2000 or XP application on a computer that is directly connected to the array reader and microplate platform. A hardware protection is required to run this program (see page 11).
- **Bio-Plex Manager Desktop** is designed to analyze data from the Bio-Plex system. It runs as a Windows® 2000 or XP application on a desktop computer with a hardware protection key attached (see page 11). You can analyze data using all of the features available in Bio-Plex Manager Workstation; however, Bio-Plex Manager Desktop does not control the Bio-Plex system hardware.
- **Bio-Plex Manager Network Desktop** is a version of Bio-Plex Manager Desktop that uses a single hardware protection key installed on a network file server to provide multiple licenses to network users. The network key

can be configured to support 5, 10 or 20 users, depending on the type of license purchased. It runs as a Windows® 2000 or XP application on a computer linked to a server. You can use this version to analyze data using all of the features available in Bio-Plex Manager Workstation; however, Bio-Plex Manager Network Desktop does not control the Bio-Plex system.

-
- **Desktop** and **Network Desktop** versions do not include access to the instrument controls described in Chapter 5 or the controls for running the Protocol described in Chapter 7.
-

Each version of the software runs as Windows® 2000 or XP application, and features a standard Windows® interface, with pulldown menus, toolbars, and keyboard shortcuts. The computer included with the Bio-Plex suspension array system comes preinstalled with Windows® 2000 and Bio-Plex Manager Workstation.

-
- All references to “Bio-Plex Manager” in this user guide refer to the Workstation version of the software. The data analysis functions described herein also apply to the Desktop and Network Desktop versions of the software.
-

Workflow

Using Bio-Plex Manager, you perform a reading by following these steps:

1. Start up and calibrate the array reader.
2. Select the analytes that you want to detect, specify the layout of the 96-well plate, and enter information about your standards.
3. Take the reading.
4. Review the results.
5. Print and/or export the data.

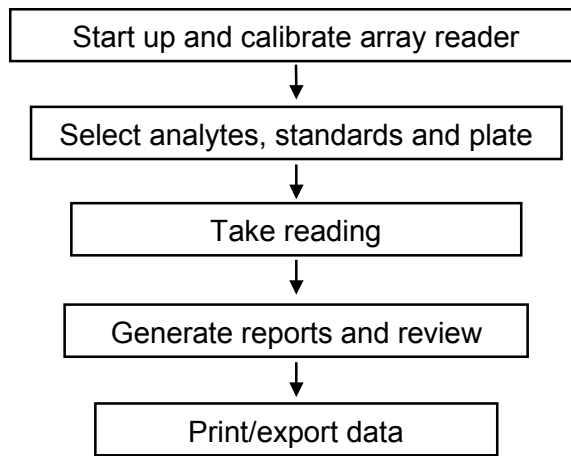


Figure 2. Workflow of Bio-Plex Manager

The *Quick Guide*, which displays automatically at startup, is designed to guide you through all the major functions of the software. Commands such as **Start Up** and **Calibrate** open dialog boxes that contain step-by-step instructions to guide you through the procedures.

Types of Files

There are two types of files created and used by Bio-Plex Manager. A **Protocol file** contains the settings for reading a microplate. A **Results file** contains the data from the reading and any formatting associated with that data.

Protocol File

The Protocol file, which has the file extension **.pbx**, contains the parameters of a Bio-Plex reading. It includes such information as the analytes selected, the plate wells read, sample info, standards info, etc. At the completion of a reading, the settings specified in the Protocol file are copied into the Results file, along with the reading data.

You can use the same Protocol settings for multiple readings. Data from the most recent reading is stored in the Protocol file, and you can generate a new Results file from this data.

Results File

The Results file, which has the file extension **.rbx**, contains the data from a Bio-Plex reading, the Protocol parameters used to collect that data, and analysis tools for interpreting the data. After a reading is complete, the Results file will open. You can also generate new Results files from saved Protocols.

The Results file contains all the information necessary to analyze the raw Bio-Plex data, including a description of the assays, well types, concentrations, dilutions, and the regression method used for calculating the concentrations of unknowns from standards. Note that you can edit many of the parameters copied from the original Protocol file—including the plate format, analyte selection, standards info, etc.—and recalculate your results. However, you can only change the parameters of wells that have already been read, and you can only calculate results based on the analytes that were present in the sample.

Key Software Features

Key features of Bio-Plex Manager software include:

- Graphical user interface
- Complete control of the Bio-Plex array reader and microplate platform
- Automated maintenance, calibration and validation functions
- Automatic error detection and logging system
- Bead map and histogram
- Raw data reports
- Data reduction
- Logistic curve-fitting algorithms
- Calculation of unknown concentrations
- Export to Microsoft® Excel and text files
- Microplate platform heater control

3. Installation and Specifications

System Components

A certified Bio-Rad service engineer will install the complete Bio-Plex system at your site, including the array reader, platform, and computer. The setup procedure is described in the Bio-Plex array reader hardware manual.

The Bio-Plex suspension array system includes the following components:

- Bio-Plex array reader
- Microplate platform
- Maintenance, calibration and validation plate II (MCV Plate II, which includes fluidics validation, is required for use with Bio-Plex Manager 3.0)
- A Pentium®-class PC, preinstalled with Windows® 2000, Microsoft® Excel, Microsoft® Internet Explorer, Microsoft.net, and Bio-Plex Manager Workstation software
- Software package, containing Bio-Plex Manager CD-ROM, user guide and a HASP protection key
- Instruction manual for Bio-Plex array reader and microplate platform
- Sample tube holders (two sizes, large and small)
- Sheath fluid bottle
- Sample needles (one short and two long)
- Protective shield
- Needle adjustment tool
- Waste bottle
- Communications cable to connect the array reader to the computer
- Communications cable to connect the microplate platform to the computer
- Computer monitor
- Power cords

- Computer keyboard
- Computer mouse

The Bio-Plex suspension array system comes with these reagents:

- Bio-Plex calibration kit (CAL1 and CAL2)
- Bio-Plex validation kit 3.0 (optical, fluidics, reporter, and classify components)
- Bio-Plex sheath fluid

See Appendix 6 on page 155 for a complete list of product numbers for the system.

Bio-Plex Manager

Bio-Plex Manager Workstation software, Windows® 2000, and Microsoft.net are preinstalled on the computer included with your Bio-Plex system. All versions of the software—Workstation, Desktop, and Network Desktop (see page 5)—can be installed from the Bio-Plex Manager CD as described below.

System Requirements

Component	Minimum	Recommended
Operating system	Windows® 2000 or XP	Windows® 2000 or XP
Processor	Pentium III or equivalent, 733 MHz	Pentium III or higher, 1 GHz or higher
Hard disk space	4 GB	16 GB
System memory	128 MB	256 MB
Screen resolution	1024 x 768	1024 x 768
Screen colors	256 colors	24-bit True Color
Serial Ports (Workstation only)	2 RS232 serial ports or 1 RS232 serial port and 1 USB port with USB-to-serial converter	2 RS232 serial ports
Parallel Ports (Workstation only)	1 25-pin parallel port	1 25-pin parallel port
Other installed components	Internet Explorer 5.01 Excel 97 Microsoft.net (included on Bio-Plex Manager CD)	Internet Explorer 6.0 or later Excel 2000 or XP Microsoft.net (included on Bio-Plex Manager CD)

Installation

To install, reinstall or upgrade the software, insert the Bio-Plex Manager CD into the CD-ROM drive on your computer.

-
- Before reinstalling Bio-Plex Manager, we recommend that you first *uninstall* any existing version of Bio-Plex Manager on your computer.
-

The **Bio-Plex Manager Installation Program** will open, displaying a navigation screen for performing the installation.

Microsoft.net

Microsoft.net must be installed on your computer before installing Bio-Plex Manager. If your computer does not already have Microsoft.net, select the Microsoft.net installation from the list of options in the Bio-Plex Manager CD installer screen. After installing Microsoft.net, return to the screen to perform the Bio-Plex Manager installation. Microsoft.net is required to support the logistic curve fitting features of Bio-Plex Manager.

BioPlex.mdb File

During installation, you will be prompted to save the application database (**BioPlex.mdb**) to a location on your hard drive or a file server. This database contains both calibration and validation information for your Bio-Plex instrument. You can save this file in any location; the default location is the Bio-Plex Manager application folder. If your computer is connected to multiple Bio-Plex instruments, each instrument must have a separate BioPlex.mdb file saved in a different location.

Hardware Protection Key

A hardware protection key (HPK), or HASP key, is required to run all versions of Bio-Plex Manager. You must attach the HPK to the 25-pin parallel port on your computer before you can run the software.

Before attaching the HPK, turn off your computer. If you have a printer attached to the parallel port, turn that off as well, and then disconnect the printer cable from the parallel port.

Attach the HPK to the parallel port, and then reattach the printer cable (if necessary) to the key itself and restart your computer and printer.

The HPK has a driver that is automatically installed when you install Bio-Plex Manager.

Uninstalling

To uninstall Bio-Plex Manager from your computer, use the Windows **Add/Remove Programs** function. Click on the Windows **Start** button, select **Settings**, select **Control Panel**, double-click on **Add/Remove Programs**, and follow the instructions for removing the program.

Technical Support

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4. Getting Started

Before starting Bio-Plex Manager, make sure the Hardware Protection Key (see page 11) is attached to your computer, and switch on the Bio-Plex array reader and the microplate platform. This turns on the optics inside the array reader and enables communication between the array reader and software.

-
- Avoid using other applications while Bio-Plex Manager Workstation is communicating with the array reader. Running other applications may interrupt the communication link between Bio-Plex Manager Workstation and the array reader.
-

Starting Bio-Plex Manager

To start Bio-Plex Manager, click on the application icon on your desktop or select **Bio-Plex Manager** from the **Programs** directory on your Windows *Start* menu.



Figure 3. Bio-Plex Manager desktop icon.

The software will open and connect to the Bio-Plex array reader and platform.

Required Screen Resolution

Your computer screen resolution must be set to at least 1024 x 768 pixels for correct display of the Bio-Plex Manager interface. Some dialog boxes will not display properly at lower resolution. If your display is currently set to a lower resolution:

1. Right-click on an empty region of the Windows Desktop.

2. Select **Properties** from the context menu.
3. In **Display Properties**, select the **Settings** tab.
4. Drag the **Screen Area** slider to the right (**More**) until you have selected **1024 by 768 pixels**. Click on **OK** to accept the settings.

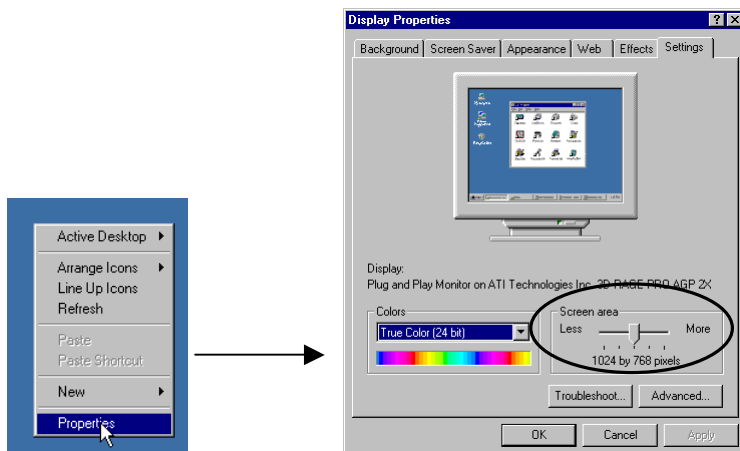


Figure 4. Changing the screen settings.

Communication to the Array Reader and Microplate Platform

The computer running Bio-Plex Manager software is connected to the Bio-Plex array reader and the microplate platform by two separate serial cables. See the **Hardware Manual** for a description of the cable connections.

COM Port Settings

The computer COM ports assigned to the serial cables are set when Bio-Plex Manager software is initially installed. The default settings are **COM1** for the array reader and **COM2** for the microplate platform.

Communication to the array reader and platform is established when you first open Bio-Plex Manager. If you have a problem connecting to the array reader or platform, you can change the port settings from within Bio-Plex Manager itself.

-
- To check the connection status of the array reader and/or platform, go the status bar at the bottom of the main application window. The array reader status should be "Ready." If the status reads "Check Link," you should check your cables and COM port settings.
-

To check or change the COM port settings, go to the *Instrument* menu, select *Set Up*, and select **Communications**. The *Communication Method* dialog box will open.

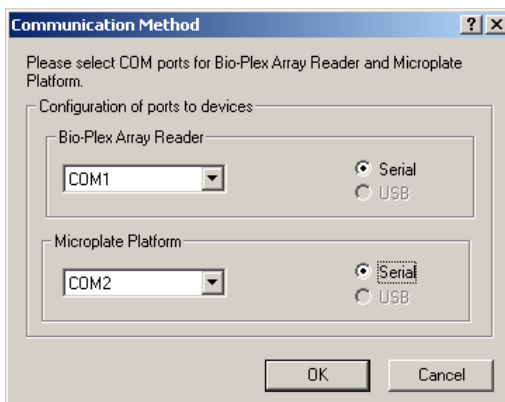


Figure 5. *Communication Method* dialog box.

In the dialog, select the appropriate COM ports for the array reader and platform from the pulldown lists, and click on **OK**. Note that both connection types should be set to **Serial**.

Disconnecting and Reconnecting

The serial connections to the array reader and platform are closed when you exit Bio-Plex Manager. If you need to disconnect from the array reader and platform without shutting down Bio-Plex Manager software (e.g., to troubleshoot the instrument), go the *Instrument* menu and select **Disconnect**.

To attempt to reconnect to the array reader and platform while the software is open, select **Reconnect** from the *Instrument* menu.

See the Bio-Plex system hardware manual for specific instructions.

Menu and Toolbars

Bio-Plex Manager includes a menu bar and main toolbar at the top of the application window.

The pulldown menus contain all the major functions of the software. Note that these menus change depending on whether you are displaying a Protocol window, a Results window, or neither.

The main toolbar includes the major instrument controls, including **Start Up**, **Shut Down**, **Wash**, **Unclog**, etc. The various Protocol and Results windows also contain their own toolbars, with commands specific to those windows. These are described in greater detail in the following chapters.

Status Bar

The instrument **status bar** shows the current state of the instrument (Calibration, Warm Up, Ready, Pressurizing, etc.). It includes a **Cancel** button for canceling the current operation (see page 44).



Figure 6. Instrument status bar.

When the array reader is performing an operation, the time remaining for the operation will be shown in the instrument status bar.

The Bio-Plex Manager software status bar is below the instrument status bar, and provides information about the command under the cursor. This information is continuously updated as you move your cursor over the software window.

Quick Guide

When Bio-Plex Manager first opens, a *Quick Guide* toolbar is displayed in the upper right corner of the screen. The *Quick Guide* is designed to guide you through the typical workflow when using the Bio-Plex system, from start up and calibration through shut down.

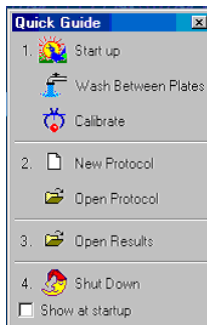


Figure 7. Bio-Plex Quick Guide.

By default, the *Quick Guide* opens automatically when you start Bio-Plex Manager. To disable this feature, deselect the **Show at Startup** checkbox at the bottom of the guide.

To open the *Quick Guide* after it has been closed, select **Quick Guide** from the *View* menu.

Sample Needle Adjustment

See the Bio-Plex Hardware Manual for instructions on installing the sample needle.

The height of the Bio-Plex array reader sample needle must be adjusted when (1) the style of microtiter plate has changed or (2) when the sample needle is replaced. The sample needle height is the same for flat bottom or filter plates. Note that, for round bottom plates, the needle must be raised so that it is slightly above the bottom of the round bottom well.

- If you adjust the sample needle for use with round bottom plates, you will not be able to use the MCV Plate II without readjusting the needle to filter-plate height.
-

With the Bio-Plex array reader and platform switched on, go to the *Instrument* menu, go to the *Set Up* submenu, and select **Adjust Needle**. The *Adjust Needle* dialog box will open.

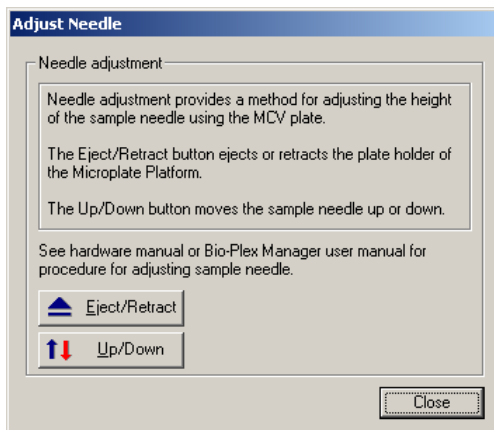


Figure 8. *Adjust Needle* dialog box.

Then follow the steps below:

1. Click on the **Eject/Retract** button to eject the plate holder, and place the MCV Plate II on the microplate platform with the white arrow pointing toward the array reader.
2. While holding the platform door open, click on the **Eject/Retract** button again to retract the plate.
3. Tape the access door of the microplate platform open. You must be able to see inside the access door.



Warning: DO NOT stick your hands and fingers inside the platform at any time.

4. Click on the **Up/Down** button in the *Adjust Needle* dialog to lower the needle into the needle adjustment well in the MCV Plate II.

5. Using the hex key needle adjustment tool (supplied with the array platform), loosen the screw at the top of the needle so that the needle housing can move up and down freely.
6. Move the knurled dial on the needle arm and manually move it downward so that it just touches the bottom of the needle adjustment well. Move the needle up and down gently a couple of times to verify that the needle is just touching the bottom of the well.

➤ The needle should not push the MCV Plate II down when it is in the "down" position. If the MCV Plate II moves when the needle drops down, readjust the needle height by repeating the previous steps.

7. Tighten the needle set screw so that it is no longer possible to manually move the needle up and down.
8. Finally, remove the tape so that the platform door can open and close normally, and click on the **Close** button to close the *Adjust Needle* dialog box.

See the Bio-Plex hardware manual for specific instructions.

5. Bio-Plex Suspension Array System Controls

This chapter covers the Bio-Plex Manager commands that control basic system functions, such as instrument start up, warm up, calibration, validation, and shut down.

-
- The Start Up, Warm Up, and Calibration functions on the Bio-Plex array system must be performed prior to running an assay, and all together take about 45 minutes.
-



When performing the following operations, the sheath fluid container and the waste fluid container should be closely monitored.



The sheath fluid bottle must be placed at the same level as the Bio-Plex array reader (unless you are using the HTF; see caution below), and the fluid level should be below the air inlet connection and above the sheath outlet connection. Always check the sheath fluid level before starting a run or procedure.



If you are using the HTF (high-throughput fluidics), it should sit on the counter next to the Bio-Plex array reader, while the sheath fluid cube should be placed ~3–4 feet below the reader (e.g., on the floor).



The waste fluid container receives waste from the system. **Do not allow the waste container to overflow!** Empty the waste bottle each time the sheath fluid bottle is filled. The waste container should not be placed on top of the instrument. All waste containers should have vented caps.

MCV Plate II

Bio-Plex Manager 3.0 requires the use of MCV Plate II (MCV stands for maintenance, calibration and validation). MCV Plate II has been modified from the previous MCV plate to include more comprehensive validation functions. This plate contains wells marked for the different types of fluids used in validation, washing, calibration, and other functions.

- The MCV Plate II is required for complete validation of the Bio-Plex system using Bio-Plex Manager 3.0 and Bio-Plex Validation Kit 3.0.

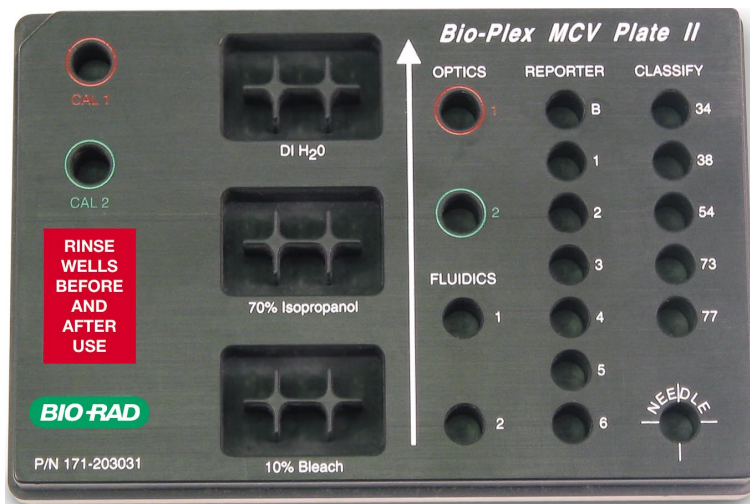


Figure 9. MCV Plate II.

When a particular procedure (e.g., Calibration) requires you to add solutions to the MCV Plate II, the dialog box describing the procedure will include a diagram of the MCV Plate II with the wells to be loaded highlighted in blinking yellow.

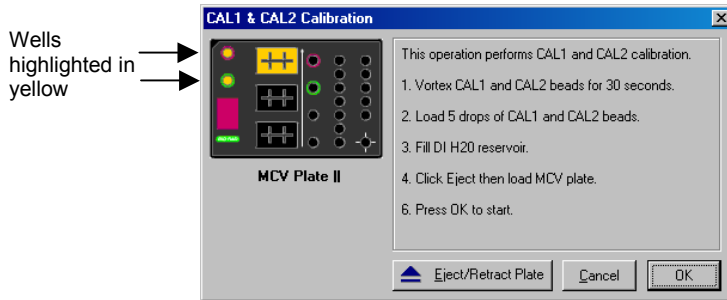


Figure 10. Highlighted wells in diagram indicate wells to be filled.

Start Up

Start up is a series of fluidics functions that prepares the Bio-Plex array reader to read Bio-Plex assays. This process requires the MCV Plate II included with the system, distilled water, and 70 percent isopropanol. Start up takes approximately 10 minutes.

-
- You do not need to wait for the Bio-Plex array reader optics to warm up before performing the start up procedure.
-

Click the **Start Up** button on the *Quick Guide* or main toolbar  or select **Start Up** from the *Instrument* menu and follow the step-by-step directions in the dialog box for preparing the MCV Plate II.

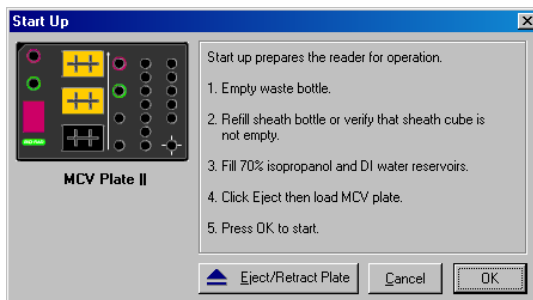


Figure 11. *Start Up* dialog box.

Insert the prepared plate into the microplate platform and click on **OK** to begin the start up process.

Optics Warm Up and Shut Down

To ensure accurate and reproducible results, the optics (i.e., the lasers) in the Bio-Plex array reader must warm up for at least 30 minutes prior to calibration and reading assays. Optics warm up begins when you first turn on the array reader.

You can proceed with the array reader start up procedure described in the previous section while the optics are still warming up. However, if you try to perform calibration or an assay reading, you will receive a warning message and the reading will be delayed until warm up is complete.

You can cancel the warm up procedure using the **Cancel Operation** command on the *Instrument* menu or the **Cancel** button in the status bar; however, this is not recommended. Results for identical readings may vary if the optics have not reached optimal operating temperature.

If the array reader is idle for more than four hours, the optics will automatically power down (though the array reader itself will remain on). Depending on the length of the shut down, a full 30-minute warm up period may be required before more readings can be taken.

To begin warm up after automatic shut down, click on the **Warm Up** button  on the main toolbar or select **Warm Up** from the *Instrument* menu.

If you attempt to perform a reading after automatic shut down, you will receive a warning message and the optics will begin warming up.

Calibration

Calibration of the Bio-Plex array reader is essential for optimal performance and day-to-day reproducibility of results. The Bio-Plex array reader should be calibrated each day after the start up procedure is complete and the optics have warmed up. Also, the array reader must be re-calibrated if the instrument temperature changes by more than 2° C during the course of the day. The array reader includes a temperature gauge; if the temperature changes by more than 2° C, a message box will prompt you to recalibrate.

➤ Before calibrating, make sure that optics warm up is complete.

The calibrators (CAL1 and CAL2 beads) are microspheres with stable fluorescent intensities in the RP1, CL1, and CL2 wavelength ranges. The calibration process uses these microspheres to adjust voltage settings for optimal and consistent microsphere classification and reporter readings over time and across different instruments. Current calibrated settings are automatically applied to any new session.

The calibration procedure requires the Bio-Plex MCV Plate II, the bottle of CAL1 beads, the bottle of CAL2 beads, and DI water. The CAL1 beads calibrate the array reader's doublet discriminator and classification channels, while the CAL2 beads calibrate the array reader's reporter channel for reporter fluorescence detection.

Opening the Calibration Dialog Box

To begin calibration, click on the **Calibrate** button  on the *Quick Guide* or main toolbar, or select **Calibrate** from the *Instrument* pull-down menu. The *Calibrate* dialog box will open.

The screenshot shows the 'Calibrate' dialog box. At the top, there's a section for 'Enter user name' with the text 'Srinivas'. To the right, 'Last calibration' shows 'Date: 16-Feb-01, 10:02 AM' and 'Temp (Celsius): 21.57'. Below this is 'Select Calibration type' with three radio buttons: 'CAL1 & CAL2' (selected), 'CAL1 Only', and 'CAL2 Only'. The 'Select control numbers' section has two columns. The left column is for 'CAL1 Control Number' with a dropdown showing 'CAL1/89876', 'Add...' and 'Delete' buttons, and three target fields: 'DD Target' (5684.00), 'CL1 Target' (3759.00), and 'CL2 Target' (3911.00). The right column is for 'CAL2 Control Number' with a dropdown showing 'CAL2/89877', 'Add...' and 'Delete' buttons, and one target field: 'RP1 Target' (16894.00). At the bottom are buttons for 'Eject/Retract', 'OK', and 'Close'.

Figure 12. *Calibrate* dialog box.

At the top of the dialog box, enter your name in the *Enter user name* field. The time and date of the last calibration are listed, as is the temperature at the time of that calibration. If the temperature has not changed by more than 2° C in a single day, it is not necessary to recalibrate the instrument.

Next, select the calibration type by clicking on the **CAL1 & CAL2**, **CAL1 only**, or **CAL2 only** button. You should perform both CAL1 and CAL2 calibration daily.

Calibration Microsphere Control Numbers

Under *Select Control Numbers*, select the control numbers for your CAL1 and CAL2 microspheres from the pulldown lists. The target values for the control numbers will appear in the appropriate fields.

When you receive a new supply of CAL1 and/or CAL2 calibration microspheres, you must add their control numbers and target values to the *Calibrate* dialog. These numbers are printed on the bottles containing the beads.

-
- It is critical that you enter the correct target values for your CAL1 and CAL2 calibrators. Entering incorrect values will result in an incorrectly calibrated array reader, which will adversely impact assay results.
-

To add a new CAL1 control number, click on the **Add** button under *CAL1 control number* in the *Calibrate* dialog. The *Add New CAL1 Control Number* dialog box will open.

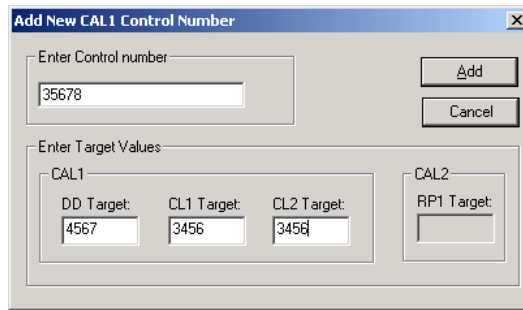


Figure 13. Adding a new CAL1 control number.

In the dialog, enter the control number from the CAL1 bottle in the field. Next, enter the DD, CL1, and CL2 target values for the control number, as printed on the bottle, in the appropriate fields.

-
- If you are using a Luminex-type assay from another manufacturer, do not use the DD target value specified in the manufacturer's manual. Use only the DD target value on the Bio-Plex CAL1 bottle.
-

When you are done, click on **Add** to close the dialog box and save your changes.

To add a new CAL2 control number, click on the **Add** button under *CAL2 control number* in the *Calibrate* dialog box. The *Add New CAL2 Control Number* dialog box will open.

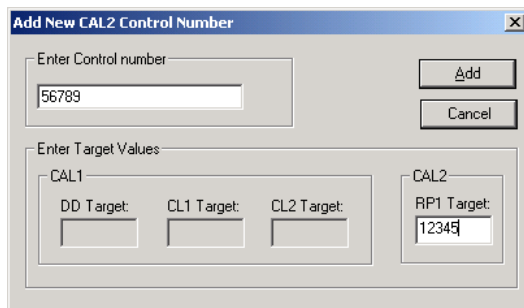


Figure 14. Adding a new CAL2 control number.

In the dialog, enter the control number from the CAL2 bottle in the field. Next, enter the High RP1 or Low RP1 target value for the control number, as printed on the bottle, in the *RP1* field.

-
- The CAL2 calibration bottle label lists two different RP1 target values: Low RP1 and High RP1. Bio-Plex phosphoprotein assays require calibration using the High RP1 target value, while Bio-Plex cytokine assays may be calibrated with either the Low or High RP1 target value. Other Luminex-type assays typically require calibration with the Low RP1 target value.
-

When you are done, click on **Add** to close the dialog box and save your changes. Your new control numbers will be added to the selection lists in the main *Calibrate* dialog.

To delete a particular control number, first select it in the *Calibrate* dialog box, then click on the **Delete** button.

Calibration Set Up

When you have specified your control numbers of calibration microspheres and specified the type of calibration you want to perform (CAL1 & CAL2, CAL1 only, or CAL2 only), click on **OK** in the *Calibrate* dialog.

Another dialog box will list step-by-step instructions for preparing the MCV Plate II.

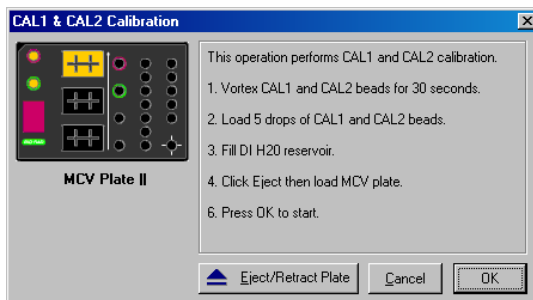


Figure 15. Preparing the MCV Plate II for CAL1 & CAL2 calibration.

When preparing the MCV Plate II, note the following:

- **Important:** Before vortexing, remove the calibration beads from 2–8°C (36–46°F) storage and allow to warm to room temperature. Vortex each bottle for 30 seconds. Proper resuspension of the microspheres is essential for efficient calibration.
- Never dilute the calibration beads, and be careful to limit their exposure to light. Store at 2–8° C immediately following calibration.
- Load 5 drops of beads (approximately 200 µl) per reservoir (CAL1 or CAL2).
- The DI H₂O well of the MCV Plate II holds about 3 ml of distilled water.

Performing the Calibration

Click on the **Eject/Retract Plate** button to eject the microplate platform plate carrier, place the MCV Plate II in the carrier with the white arrow facing toward the platform, then retract the plate. Click on **OK** to start the calibration process.

As calibration proceeds, the status bar at the bottom of the Bio-Plex Manager window will monitor the progress of calibration. The number of beads per second should be 100 or higher; fewer beads per second may indicate a problem with the fluidics system. See the troubleshooting guide at the end of this manual.

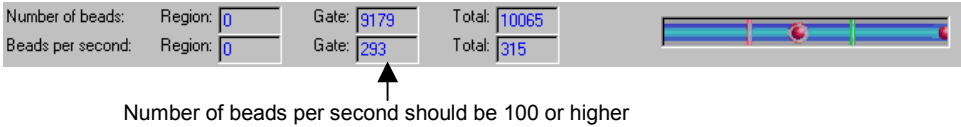


Figure 16. Status bar during calibration.

Because calibration beads are highly concentrated, calibration is followed by three wash cycles using distilled water.

An alert box will notify you whether calibration has succeeded or failed. If the calibration process failed, check the MCV Plate II to verify that the correct beads were added to the wells. Then perform a fluidics **Wash** (page 43) and repeat the above steps. For more information, consult the **Troubleshooting** guide on page 133.

Calibration Log

The calibration log provides a list of past calibration dates and times, results, instrument settings, and other data. This data is stored in a log file called **bioplex.mdb**, which by default is saved in the main Bio-Plex Manager application folder on your computer.

To open the calibration log, go to the *View* menu and select **Calibration Log**. The *Calibration Log* viewer will open.

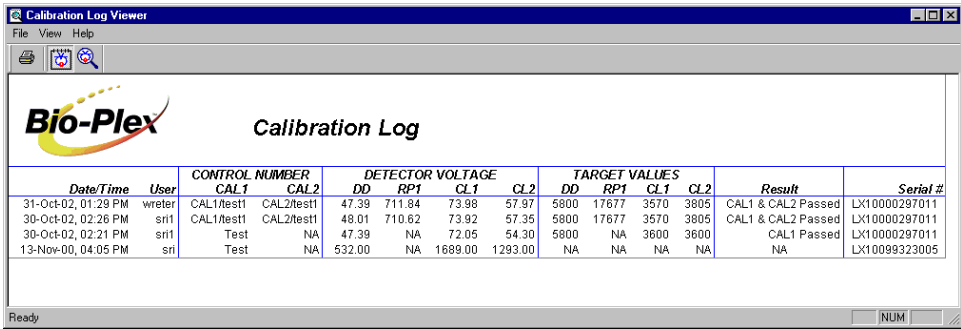


Figure 17. Calibration Log viewer.

The log report displayed in the viewer includes the date, operator, lot numbers, detector voltage, target values, results, and instrument serial number for each calibration in table format.

The possible entries in the **Result** column are Pass, Fail, Fail—low bead number, or NA (if the calibration was performed on a previous version of Bio-Plex Manager).

-
- If you see a drastic change in a detector's voltage from one calibration to the next, it could indicate a problem with the instrument. A steadily increasing detector voltage may indicate that the laser is decreasing in intensity.
-

On the toolbar, click on **Calibration Detail**  to display additional information for each calibration, including detector gain. Click on **Calibration Log**  to return to the default view.

For CAL1-only calibrations, the RP1 column will display a entry of NA (not applicable). For CAL2-only calibrations, the DD, CL1, and CL2 columns will display entries of NA.

To print the calibration report, select **Print** from the *File* menu. **Print Preview** displays the report as it will appear in a printout. **Print Setup** allows you to select some standard print settings.

Wash Between Plates

You should wash the fluidics lines between each plate reading to prevent traces of sample or other debris from building up inside the system. Select **Wash Between**

Plates  from the *Quick Guide*, main toolbar, or *Instrument* menu. A dialog box will guide you through the steps for preparing the MCV Plate II for a wash procedure.

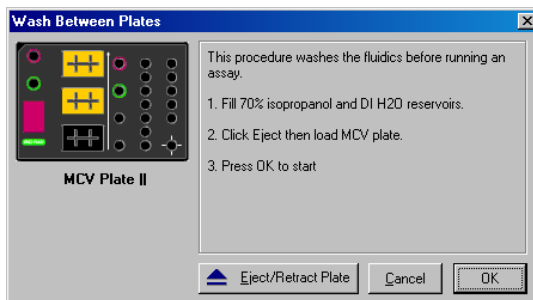


Figure 18. Steps for washing between plate readings.

Add 70% isopropanol and distilled water to the appropriate reservoirs, insert the plate in the microplate platform, and click on **OK**. This procedure performs a series of different fluidics operations and will take several minutes. After initial system pressurization (5–20 seconds), the time remaining in the operation will be displayed in the Bio-Plex Manager status bar.

Remove Air Bubbles

Microscopic air bubbles in the cuvette may cause a sudden shift in the bead regions during an assay reading. If the array reader detects such a shift, the reading will stop and you will be prompted to perform an alcohol wash to force air bubbles out of the system.

Select **Remove Bubbles**  from the main toolbar or *Instrument* menu. A dialog box will guide you through the steps for preparing the MCV Plate II for the procedure.

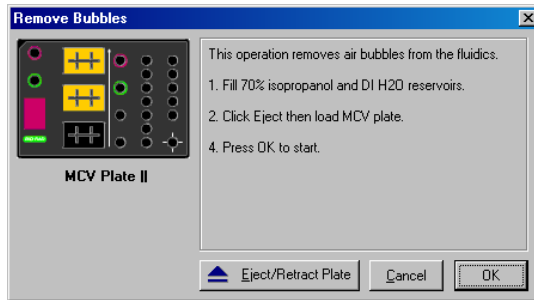


Figure 19. Steps for removing air bubbles from the fluidics system.

Add 70% isopropanol and distilled water to the appropriate reservoirs, insert the plate in the microplate platform, and click on **OK**. This procedure performs a series of different fluidics operations and will take several minutes. After initial system pressurization (5–20 seconds), the time remaining in the operation will be displayed in the Bio-Plex Manager status bar.

Unclog

If the array reader detects an unusually low bead count during a reading, you will be prompted to perform an **Unclog** operation to remove possible obstructions from the

fluidics lines. Select **Unclog**  from the main toolbar or *Instrument* menu. An instruction box will guide you through preparation of the MCV Plate II.

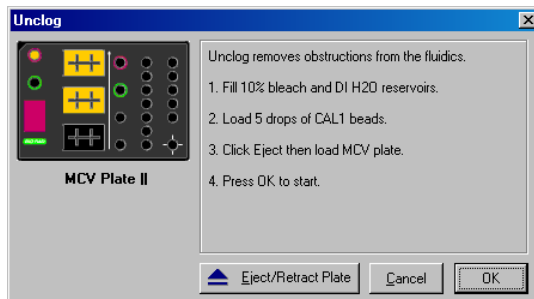


Figure 20. Steps for unclogging the fluidics system.

Add 10% bleach solution and distilled water to the appropriate reservoirs, add 5 drops of CAL1 beads to the CAL1 reservoir, insert the plate in the microplate platform, and click on **OK**. This procedure performs a series of fluidics operations and reads a sample of CAL1 beads to verify that the fluidics are operating properly. It will take several minutes. After initial system pressurization (5–20 seconds), the time remaining in the operation will be displayed in the Bio-Plex Manager status bar.

Validation

Validation of the Bio-Plex array reader is a formal process for documenting that the instrument is fit for its intended use and that it is kept in a state of maintenance and calibration.

-
- Validation is performed after Start Up and Calibration have been performed.
-

You should perform a validation reading:

- Once a month
- Each time you move the Bio-Plex array reader, or
- If there are problems with the array reader that cannot be solved by other procedures (calibrating, washing, unclogging, etc.).

Validation Kit

The Bio-Plex Validation Kit 3.0 is used to perform validation of the Bio-Plex system. It must be used in conjunction with MCV Plate II. The Validation Kit consists of reagents and procedures to evaluate:

- Optical alignment
- Reporter channel performance
- Efficiency of multiplexing
- Integrity of fluidics

The Validation Kit validates the operation of all of the primary components of the Bio-Plex array system, and can also be used to discriminate between assay and instrumentation problems. See the kit manual for detailed instructions.

Each Validation Kit has particular specifications for each component of that kit. These specifications are listed in the Validation Kit manual, and are essential for proper validation of your Bio-Plex instrument.

Setting Up a Validation Run

To set up a validation reading, go to the *Instrument* menu and select *Validation*. The *Validation* dialog box will open.

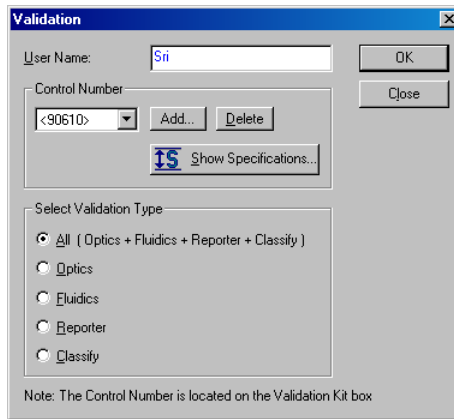


Figure 21. *Validation* dialog box.

Enter your user name in the top field.

Each Validation Kit has a unique Control Number. This Control Number is linked to the specifications for the kit, and must be entered into the software. The control number is printed on the Validation Kit box. Click on the **Add** button, and enter the Control Number in the *Add Control Number* dialog box.

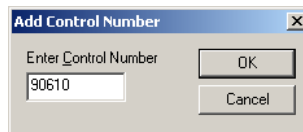


Figure 22. *Add Control Number* dialog box.

When you click on **OK**, the Control Number will be added to the list in the *Validation* dialog.

To delete a Control Number for an old validation kit, select it from the pulldown list in the *Validation* dialog and click on the **Delete** button. Note that this will not delete any past validation results—only the control number from the list.

To view the specifications for a Control Number, select the number from the pulldown list and click on the **Show Specifications** button. The dialog box will list the validation specifications.

The **Validation Specifications** dialog box displays the following data:

Optics			
DD Median:	4774 - 6593	CL1 CV%:	3.00 - 7.00
CL1 Median:	3383 - 4135	CL2 CV%:	4.00 - 8.00
CL2 Median:	3520 - 4302	RP1 CV%:	5.00 - 10.00
RP1 Median:	15205 - 18583		

Fluidics	
Carryover:	< or = 4.0%

Reporter	
Dynamic Range:	4.15 - 4.28
Accuracy:	> 90.00%
Sensitivity:	< 200 MESF
Linearity:	> 0.995
Slope:	0.0593 - 0.0799

Classify	
Bead 34:	> 80.0%
Bead 38:	> 80.0%
Bead 54:	> 80.0%
Bead 73:	> 80.0%
Bead 77:	> 80.0%
DD Efficiency:	> or = 75.0%

Close

Figure 23. *Validation Specifications* dialog box.

Performing a Validation Run

To perform a validation run, make sure the Control Number for the kit you are using is selected, and then select the type of validation you want to perform using the radio buttons in the *Validation* dialog: All, Optics, Fluidics, Reporter, or Classify.

Click on **OK**. The dialog box for the selected type of validation will open.

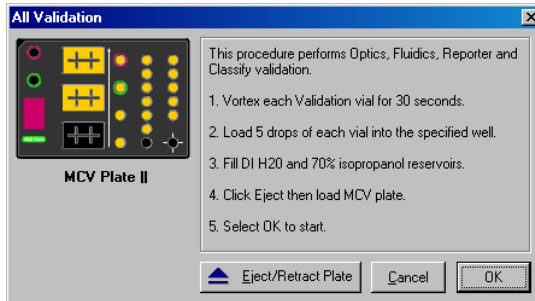


Figure 24. Dialog box for performing all validation types.

See the Validation Kit manual for information about the bottles in the Validation Kit, and instructions on loading the MCV Plate II for validation. When you have prepared the plate according to the instructions, click on **Eject/Retract Plate** in the dialog box, place the plate on the plate carrier, and click on **OK**.

Validation Results

After a run, the *Validation Results* dialog box will automatically open. You can also view this dialog by going to the *View* menu and selecting **Validation Results**.

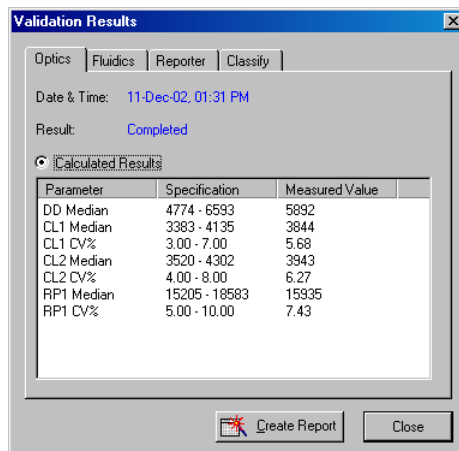


Figure 25. *Validation Results* dialog box.

The *Validation Results* dialog box includes tabs for each type of validation. If you selected and performed All Validations, all the tabs will be available. Otherwise, only the tab for the validation you performed (Optics, Fluidics, Reporter, or Classify) will be available and displayed.

Click on **Create Report** to generate and display a Microsoft® Excel spreadsheet showing the validation results. Note that Microsoft® Excel for Windows® 97, 2000, or XP must be installed on your computer for this function to work (Excel comes preinstalled on all Bio-Plex workstation computers).

See the Validation Kit 3.0 manual for interpreting the results of a validation run.

Validation Log

Each validation run is saved in a *Validation Log* file. This log data is stored in a file called **bioplex.mdb**, which is saved in the main Bio-Plex Manager application folder on your computer. To view this file, go to the *View* menu and select **Validation Log**.

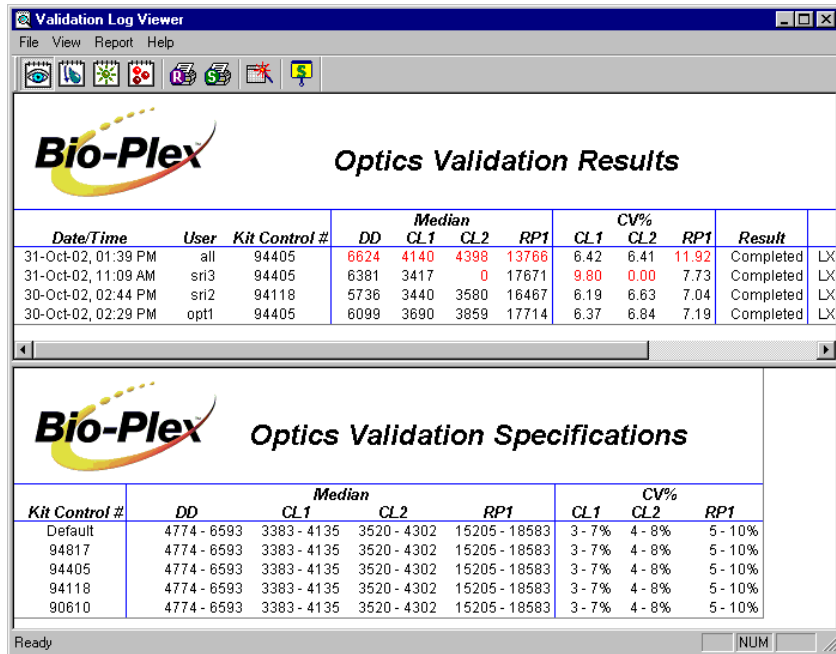


Figure 26. Validation Log viewer.

Each validation reading is listed by date and time. The validation results are listed by date and time in the upper window, and the specifications associated with each Control Number are listed in the lower window.

Use the buttons in the upper left corner of the window to view the different validation types (Optics , Fluidics , Reporter , and Classify .

To print the Results window, click on the **Print Report Results** button . To print the Specifications window, click on the **Print Specifications** button .

To show or hide the specifications data in the *Validation Log* viewer, click on the **Show/Hide Specifications Window** button .

To create a spreadsheet report for a particular validation run, click on the row for the run in the Results window. It will appear selected, as will the specifications associated with the Control Number for that run in the lower window. Then click on

Create Report . A Microsoft® Excel spreadsheet will be automatically generated and displayed showing all the validation data for that run. Note that Microsoft® Excel for Windows® 97, 2000, or XP must be installed on your computer for this function to work.

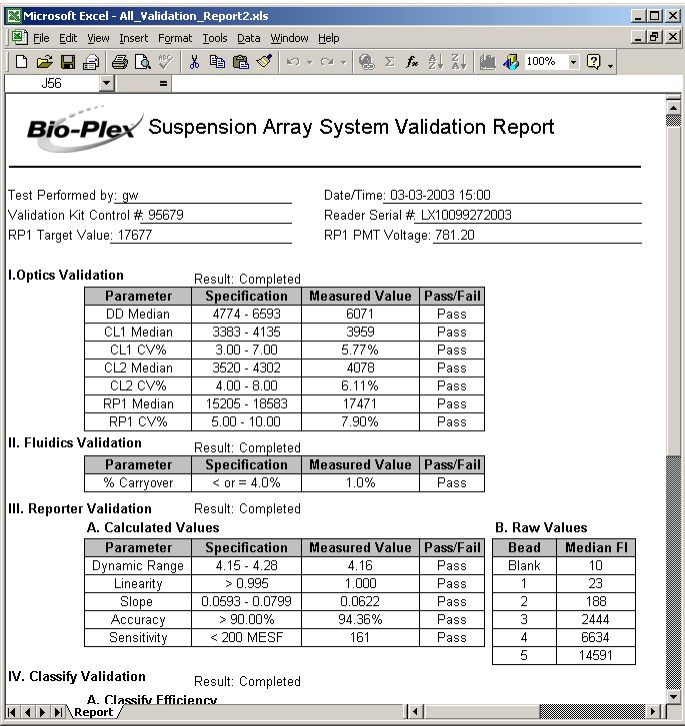


Figure 27. Validation Results spreadsheet (Microsoft® Excel).

Platform Heater

The Bio-Plex suspension array system includes a microplate platform heater for warming samples. The heater has an operating range of 35°C to 60°C (95°F to 140°F), and can be controlled through Bio-Plex Manager.

To access the heater controls, select **Platform Heater** from the *Instrument* menu, or click on the button on the main toolbar .

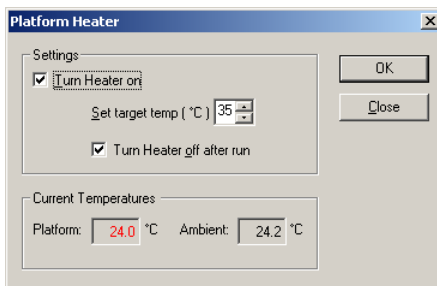


Figure 28. *Platform Heater* dialog.

In the *Platform Heater* dialog, click on the **Turn Heater on** checkbox to turn on the heater, and set the target temperature within the range 35–60°C using the scroll buttons.

To automatically turn the heater off at the end of a run, select the **Turn Heater off after run** checkbox.

The dialog box and the Bio-Plex Manager status bar will indicate the current platform temperature. The status bar will also indicate whether the heater is turned on.



Figure 29. Status bar with heater on.

If the heater is in the process of warming up or cooling down, the temperature will be shown in red. If the heater is on and has reached the specified temperature, the temperature will be shown in green. If the heater is off and is at ambient temperature, the temperature will be displayed in black.

- If you initiate a run before the heater has warmed up or cooled down (i.e., the temperature is shown in red), you will have the option of waiting until the specified temperature is reached or proceeding with the run immediately.

Instrument Information

The Bio-Plex array reader continuously monitors its internal temperature, voltage, pressure, and other systems for diagnostic purposes. You can access this information by clicking on the **Information** button  on the main toolbar or selecting **Information** from the *Instrument* menu.

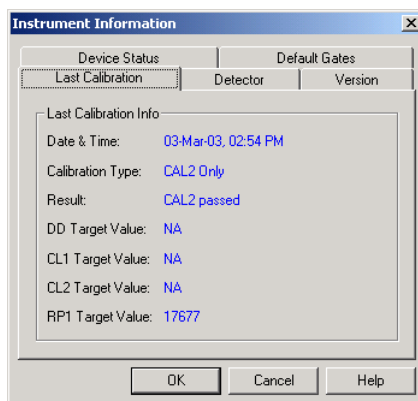


Figure 30. *Instrument Information* dialog box.

Click on the tabs in the dialog to access different information about the instrument, including information about the most recent calibration. See the hardware manual for detailed instrument specifications.

Eject/Retract Plate

The **Eject/Retract Plate** command  on the main toolbar and *Instrument* menu ejects and retracts the plate carrier in the microplate platform. Note that every

function that requires insertion of a plate into the platform (e.g., **Start Up**, **Wash**, **Calibrate**, etc.) includes an **Eject/Retract** button in its dialog box.

Additional Instrument Functions

The *Additional Functions* submenu under the *Instrument* menu contains secondary instrument fluidics functions. Selecting one of these commands will open a dialog box to guide you through the procedure.

-
- Major fluidics functions such as **Wash Between Plates**, **Remove Bubbles**, **Unclog**, etc. consist of combinations of the following functions.
-

Wash

The **Wash** command performs a single fluidics system wash using DI water. It can be used to remove traces of sample in the fluidics pathway. Note that this procedure is less comprehensive than **Wash Between Plates** (page 31)

Select **Wash** from the *Additional Functions* submenu of the *Instrument* menu and follow the steps in the dialog box.

Drain

This command completely drains the fluidics system of the Bio-Plex array reader—for example, if you need to move the instrument. Make sure you have enough room in your waste fluid container to prevent overflow.

Prime

This command primes the fluidics system with sheath fluid.

Sanitize

This command performs a comprehensive cleansing of the fluidics system using 10 percent bleach and DI water.

Alcohol Wash

This command performs a single fluidics system wash using 70 percent isopropanol.

Alcohol Flush

This command performs a comprehensive cleaning of the fluidics system using 70 percent isopropanol.

Back flush

This command flushes sheath fluid through the fluidics system in the opposite direction of sample flow, to remove particle obstructions from the cuvette. Routine back flushes are not recommended; use back flush only as a troubleshooting method.

Cancel Operation

The **Cancel** button, located in the status bar at the bottom of the Bio-Plex Manager window, may be used to cancel many of the instrument functions, including calibration, validation, or any of the fluidics functions. Click on the **Cancel** button to begin the cancellation procedure.

-
- Certain functions may take **2–5 minutes** for the cancellation process to complete.
-



Figure 31. Cancel button in the status bar.

Shut Down

The shut down procedure consists of a series of functions designed to clean the fluidics lines and prevent a build up of debris within the system. To initiate this

process, select **Shut Down**  from the main toolbar or *Instrument* menu, and follow the instructions in the dialog box.

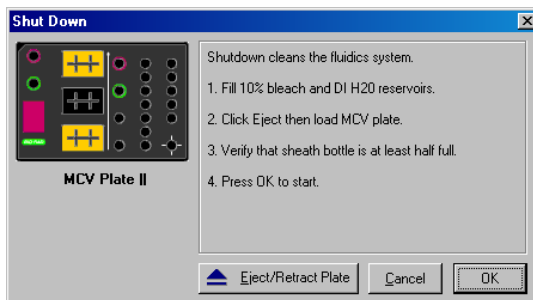


Figure 32. *Shut Down* dialog box.

6. Preparing a Protocol

A **Protocol file** contains all the parameters of a reading. It specifies the type of reading, format of the microtiter plate, analytes to be read, standards and controls (if any), and any other information about the samples that you want to include. You can save Protocol settings and reuse or modify them for other readings.

To perform a reading, first you create the Protocol file. Then, after instrument start up and calibration, you can run the Protocol and a **Results file** will be generated containing the reading data (see the next chapter).

Creating/Opening Protocol Files

To create a new Protocol file, click on the **New** button  on the main toolbar or *Quick Guide* or select **New** from the *File* menu. A new Protocol window will open.

To open an existing Protocol file, click on the **Open** button  on the *Quick Guide* or main toolbar or select **Open** from the *File* menu. A standard Windows *Open* dialog will be displayed, which you can use to locate and select the file. Protocol files have the extension ***.pbx**.

You can generate a new Protocol file from the settings stored in a Results file (see the next chapter for information about Results files). With the Results file open, select **New Protocol from Results** from the *File* menu. A new Protocol window will open.

Saving Protocols

To save the settings in a Protocol, click on the **Save** button  on the main toolbar or select **Save** from the *File* menu. Protocol files are saved with the file extension ***.pbx**.

After you have performed a reading (see **Running the Protocol** on page 85), the data from that reading is also saved with the Protocol file. This data is overwritten in the Protocol file whenever you rerun the Protocol. To preserve the data from multiple readings, you must save separate Results files (see page 109).

Use **Save As** on the *File* menu to save a Protocol under a different name. You can create a new Protocol by opening an existing Protocol, modifying the settings, selecting **Save As**, and specifying a new name for the modified Protocol.

Reducing the File Size of Protocols

Protocol files that include the data from a reading can be very large (>5 megabytes). Much of this file size is due to data necessary to generate the bead map and histogram displays (see page 94). You can save a Protocol *without* this bead map and histogram information, but with all the other numerical data from a reading intact. Note that histogram and bead map information is not necessary for calculating results or generating reports; it is only used for displaying these graphs.

In the *Save As* dialog box, select the **Compressed mode** checkbox to save the file without the bead map and histogram data. All other data will be preserved.

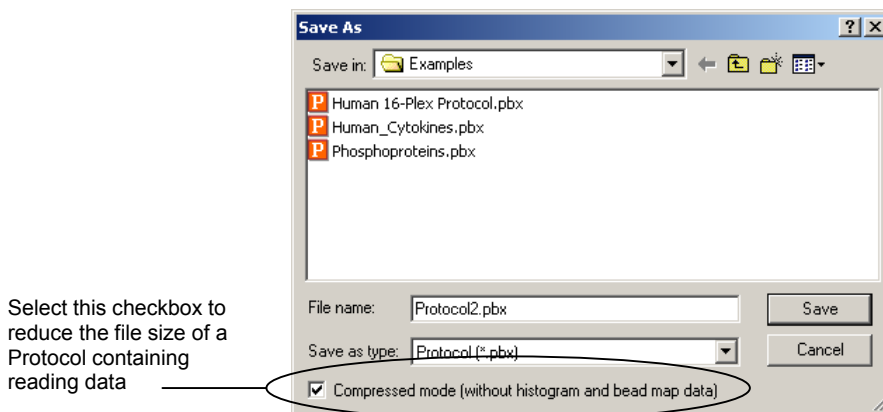


Figure 33. *Save As* dialog for Protocols.

Protocol Window

The Protocol window is divided into a series of subwindows, each containing different controls and settings. Navigate through the different subwindows using the buttons along the left side of the Protocol window.

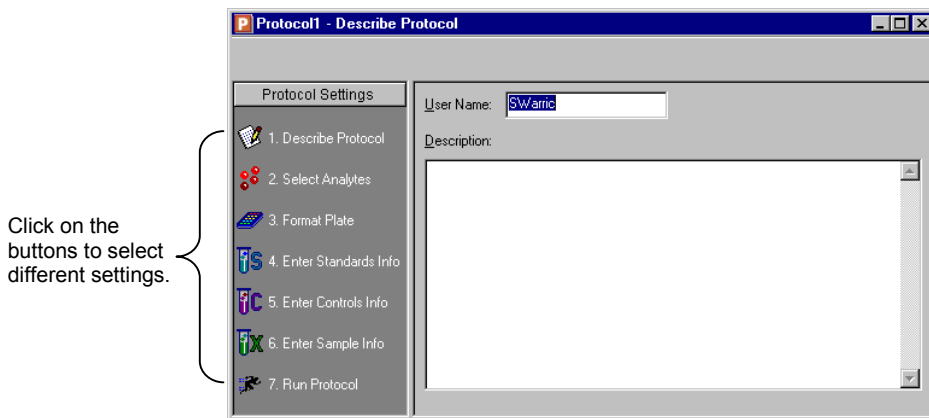


Figure 34. Protocol window.

The buttons are numbered to guide you through the selection process:

1. Describe Protocol (optional)  (see next page).
2. Select Analytes  (see next page).
3. Format Plate  (see page 56).
4. Enter Standards Info  (see page 68).
5. Enter Controls Info  (see page 79).

6. Enter Samples Info  (see page 81).

7. Run Protocol  (see page 85).

Click on each setting button and enter the necessary information, then select the next setting. After all the settings have been specified, you are ready to run the Protocol.

Step 1. Describe Protocol

Entering a description of your Protocol is optional. The Protocol description window is displayed when you first open a Protocol. Otherwise, click on the **Describe Protocol** button to display this window.

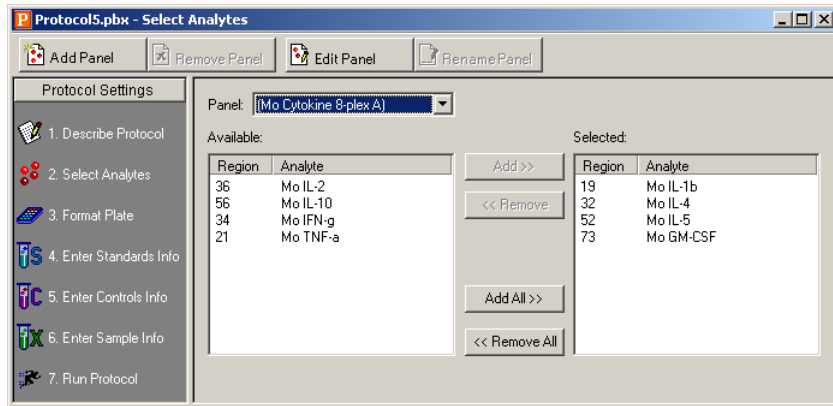
By default, the user name is your computer login name. Enter a new name in the field, if desired, and enter a description of the assay reading into the *Description* field.

Step 2. Select Analytes

In this step, you select the analytes that you want to detect in the reading.

-
- During a reading, the Bio-Plex array reader will always detect *all* the analytes in the sample, including any that you have not selected. However, analytes that were detected but not selected will not appear in the final reports and tables. After a reading, you can go back and correct your selections in the Protocol window and/or the Results file, and the detected analytes will appear in the tables.
-

Click on the **Select Analytes** button to view the analyte selection window.

Figure 35. *Select Analytes* window.

Bio-Plex Manager groups analytes by panels. Preconfigured panels of analytes are built into the software; these panels correspond to typical Bio-Plex assays, and include human, mouse, and rat cytokines and phosphoproteins. You can add to these existing panels of analytes, or create new panels that contain only the analytes that you use in your experiments.

First you select the panel of analytes that you want to detect, and then you select the specific analytes.

Selecting Analytes

Go to the **Panel** pulldown menu and select from the list of preconfigured panels. The analytes in the selected panel will be displayed in the *Available* list.

Each analyte is listed by name and region number. The **region number** refers to the region of a fluorescent color map used to identify the analyte's bead set. Each bead set is embedded with specific quantities of two fluorescent dyes; the combination of these fluorochromes, as detected by the Bio-Plex array reader, places the bead set within a unique region on the color map, thereby identifying the set and its associated analyte.

To move an individual analyte into the *Selected* list, double-click on it. Double-click on an analyte in the *Selected* list to move it back.

Hold down the Shift or Ctrl key and click on multiple analytes in the *Available* list to select them as a group. Then click on the **Add >>** button to move them over to the *Selected* list. Use the **<<Remove** button to return multiple selected analytes back to the *Available* list.

Use **Add All>>** to move all the available analytes into the *Selected* list, and **<<Remove All** to move them back.

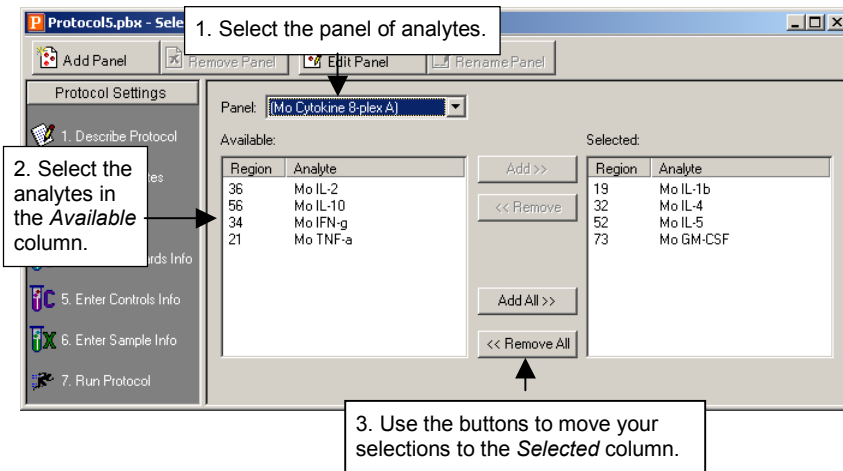


Figure 36. Selecting the analytes to be detected.

Customizing Analytes and Panels

Bio-Plex Manager comes preconfigured with a selection of panels and analytes. You can edit these and/or create new ones.

-
- If you have assay kits for analytes that are not included in Bio-Plex Manager, you will need to add these analytes using the commands described in this section.
-

Creating a New Panel of Analytes

Click on the **Add Panel** button  in the toolbar to create a new panel of analytes.

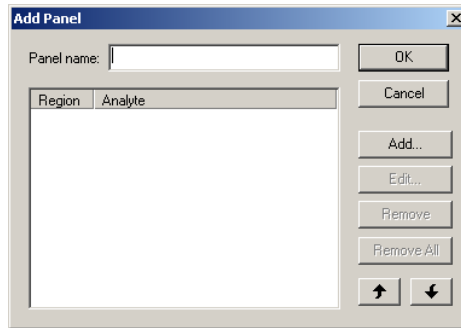


Figure 37. Creating a new panel of analytes.

Enter a name for the panel in the top field, then click on **Add** to add analytes to the panel. The *Add Analyte* dialog box will open.

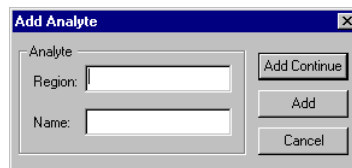


Figure 38. Adding a new analyte.

Enter the bead region number of the first analyte in the **Region** field, and the analyte name in the **Name** field.

-
- The bead region number must be correct for proper detection of analytes. Confirm that this number is correct before proceeding.
-

Click on **Add Continue** to add the analyte to the panel and continue adding more analytes. When you have entered your last analyte, click on the **Add** button to add it to the list and close the *Add Analyte* dialog.

After you have added several analytes to the panel you are creating, you can use the **Sort** buttons at the bottom of the *Add Panel* dialog to resort them. Click on an analyte in the list to select it, then click on the appropriate arrow to move it up or down in the list.

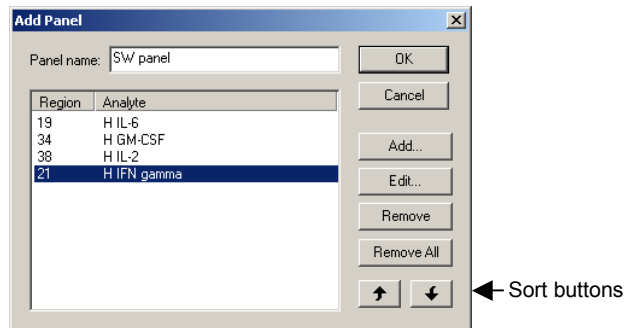


Figure 39. Sorting analytes.

You can also remove an analyte from the list by selecting it then clicking on the **Remove** button, or remove all analytes by clicking on the **Remove All** button.

To edit an analyte in the list, select it, then click on the **Edit** button. A dialog box will open in which you can change the analyte region and name.

When you are finished creating the panel, click on **OK** to save your changes and return to the Protocol window.

Editing a Panel of Analytes

To edit an existing panel of analytes, first select it from the **Panel name** pulldown menu, then click on the **Edit Panel** button . The *Edit Panel* dialog box will open.

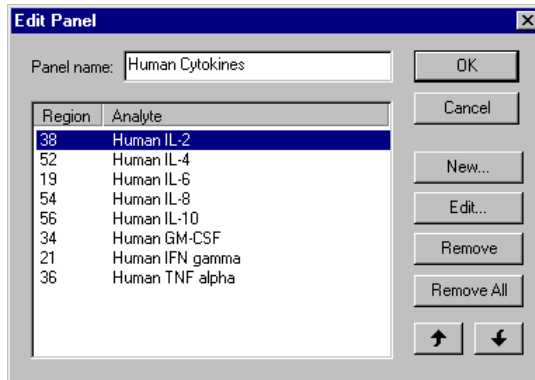


Figure 40. Editing a panel.

Use the **New**, **Edit**, **Remove**, **Remove All**, and **Sort** buttons as described above to change the analytes in the panel.

If you also change the **Panel Name**, the changes will be saved under the new name, and the old panel will not be overwritten.

-
- The preconfigured panels (*Human Cytokines*, *Mouse Cytokines*, etc.) can be edited but not overwritten; therefore, you must change the panel name to preserve your changes.
-

When you are satisfied with your changes, click on **OK**.

Deleting and Renaming Custom Panels

To delete a custom panel of analytes, first select it from the **Panel name** pulldown menu, then click on the **Remove Panel** button .

To rename a custom panel of analytes, first select it from the **Panel name** pulldown menu, then click on the **Rename Panel** button . Enter a new name in the pop-up box, and click on **OK**.

Combining Panels of Analytes

To combine entire panels of analytes (e.g., *Human Cytokines I* and *Human Cytokines II*) without going to the trouble of creating custom panels:

1. Select the panels individually from the pulldown list.
2. Use the **Add All>>** command to move the analytes from each panel into the *Selected* list.
3. Save the entire Protocol as you would normally.

Your analyte selections are saved when you save the Protocol.

Step 3. Format Plate

After you have selected the analytes, you are ready to specify the format of your 96-well microtiter plate using the plate template in the Protocol window. The formatting in the plate template tells the Bio-Plex array reader which wells to read, and tells Bio-Plex Manager how to analyze the different sample types in each well.

-
- Only formatted wells will be read by the Bio-Plex array reader. Be careful to format **all** the wells that you want to read before running a Protocol.
-

Click on the **Format Plate** button in the Protocol window to display the plate template and the formatting controls.

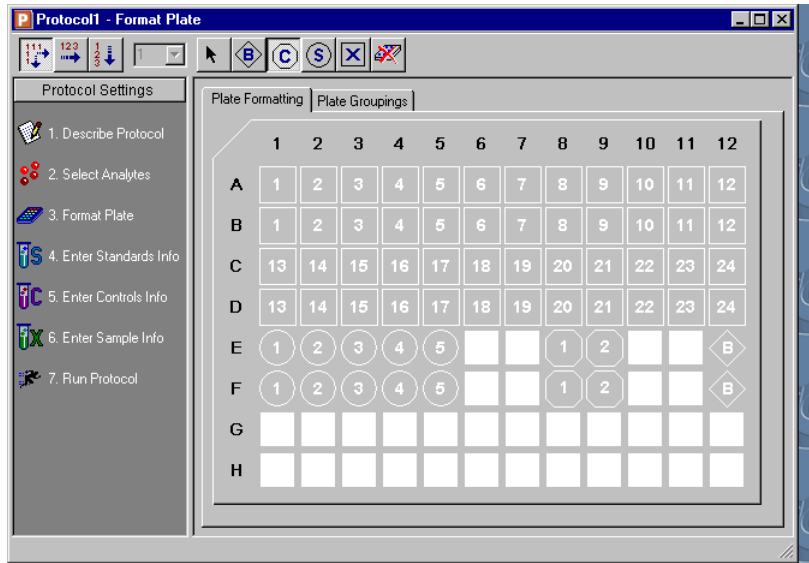


Figure 41. The plate template and formatting toolbar (*Plate Formatting* view).

The *Format Plate* window has two views: the *Plate Formatting* view and the *Plate Groupings* view. You can toggle between these views using the tabs above the plate diagram. As you toggle, the plate template will remain the same, but the available commands will toggle between formatting controls and grouping controls.

- **Plate formatting** tools are used to *define the types of wells* in a plate (sample, control, standard, blank, etc.).
- **Plate grouping** tools are used to *organize the well types into groups*, with one member of each group defined as the Reference (or Primary) member, as in a Western blot kinase assay. The ratio of each member's fluorescent intensity to the fluorescent intensity of the Reference can then be calculated.

➤ Plate formatting is required to perform a reading, because only formatted wells will be read by the Bio-Plex array reader, whereas plate grouping is optional and can be defined later.

Plate Formatting

To format the well types on the plate, make sure that the *Plate Formatting* tab is selected. Then use the buttons on the toolbar above the plate template to define the wells on the microtiter plate. These formatting commands are also located on the *Format Options* menu.

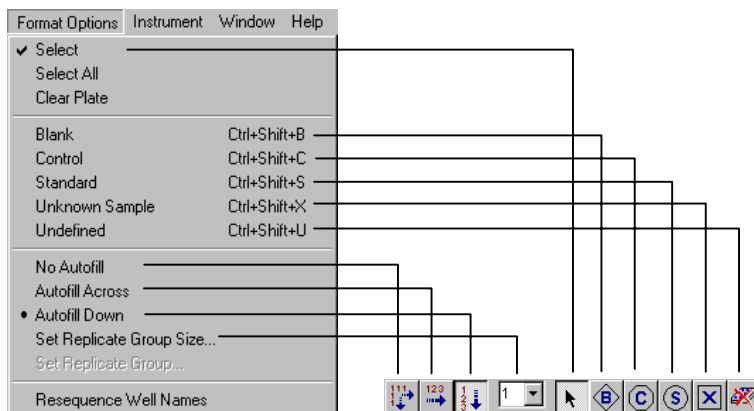


Figure 42. Plate formatting tools (*Plate Formatting* view).

The different well types are:

- Unknown sample
- Standard
- Control
- Blank
- Undefined*

*Remember that all undefined wells will not be read by the array reader.

Defining Unknown Sample Wells

To define wells containing unknown samples, first click on the **Unknown Sample** button , then click or drag on the wells in the template. Use the **Autofill** buttons as described on the following page for defining multiple unknown wells and/or replicate groups of unknowns.

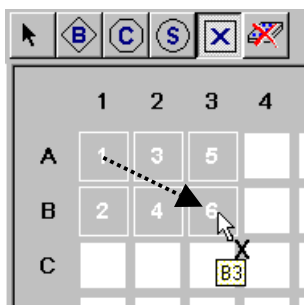


Figure 43. Defining unknown sample wells.

Note that the cursor changes to an *Unknown Sample cursor* as you move it over the template. As you click on each well, the wells are marked with a square and a number at the center. The square identifies the well as containing an unknown sample, and the number identifies the specific sample.

Well Numbering and Replicate Groups

For unknowns, standards and controls, wells with different numbers contain different samples. Wells with the same number contain the same sample, and are defined as a **replicate group**. For each replicate group, a mean value is calculated in the *Report* table, along with the standard deviation and coefficient of variation (see page 117).

Autofilling Well Numbers

To define multiple single wells as different samples, you can click on them

individually, or select **Autofill Across**  or **Autofill Down**  and drag the cursor over the wells to number them sequentially.

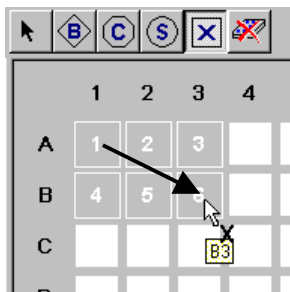


Figure 44. Defining multiple sample wells, left to right.

Note that **Autofill Across** numbers the wells sequentially left to right, then top to bottom, while **Autofill Down** numbers them sequentially top to bottom, then left to right.

Defining a Replicate Group

To define multiple wells as a replicate group, select **Turn Off Autofill**  from the formatting toolbar or *Format Options* menu, then drag your cursor over several wells that contain the same sample. The wells will be labeled with the same number.

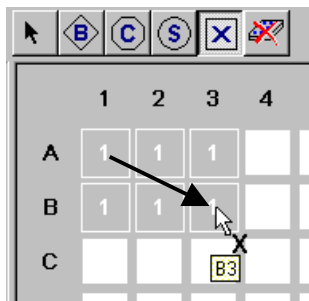


Figure 45. Defining a replicate group of unknowns.

For each replicate group, a mean value is calculated, along with the standard deviation, standard error, and coefficient of variation.

Autofilling Replicate Groups

If you have multiple replicate groups of wells in rows or columns, you can autofill these by selecting **Autofill Across** or **Autofill Down** as before, and then entering the number of wells in the replicate group (up to 96) in the number field next to the autofill buttons. (You can also select a well number up to 10 from the pulldown menu.)



Figure 46. Entering the number of replicate wells in an autofill sequence.

Now when you drag your cursor, the sequential sets of replicate wells will be defined.

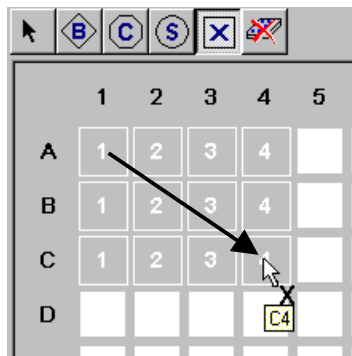


Figure 47. Autofilling a sequence of replicate groups.

For each replicate group, a mean value is calculated, along with the standard deviation, standard error, and coefficient of variation.

Changing a Replicate Group

To change the replicate group of a well or group of wells:

1. Click on the **Select** tool , and click or drag on the wells that you want to change. The wells will appear highlighted.
2. Right-click on the selection, and select **Set Replicate Group** from the context menu.
3. In the pop-up box, select the desired replicate group number from the pulldown list.
4. Click on **OK**. The selection will be changed to the specified group number.

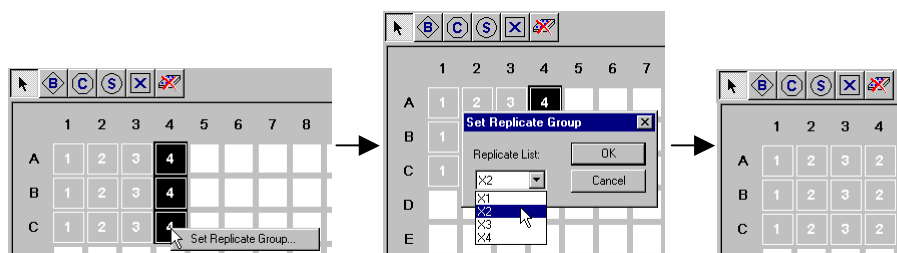


Figure 48. Selecting a different replicate group

Resequencing Well Numbers

If your well numbering is out of sequence and you want to correct it, select **Resequence Well Names** from the *Format Options* menu. All the wells in the plate will be renumbered sequentially, based on whether you have select **Autofill Across** or **Autofill Down**. This is a cosmetic change, and will not affect the analysis in any way.

Defining Standard Wells

Standard wells contain analytes of known concentration, which are used to generate a standard curve of fluorescent intensity versus analyte concentration. The regression equation for the curve is then used to calculate the concentrations of your unknowns.

To define the wells containing standards in your microtiter plate, click on the **Standard** button , then click or drag on the wells in the template. Use the

Autofill buttons as described on page 59 for defining multiple standard wells and/or replicate groups of standards.

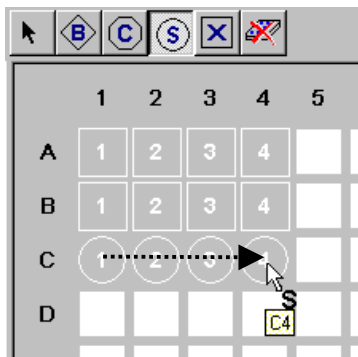


Figure 49. Defining a row of standard wells.

Note that the cursor changes to a *Standard cursor* as you move it over the template. Standard wells are marked in the template by a circle with a number inside it.

-
- After you have defined the standard wells, you are ready to enter the concentrations of the standards as described on page 68.
-

Defining Control Wells

Control wells contain samples of known concentration; the expected versus observed concentrations of the controls can be calculated at the end of the reading.

To define the wells containing controls in your microtiter plate, click on the **Control** button , then click or drag on the wells in the template. Use the **Autofill** buttons as described on page 59 for defining multiple control wells and/or replicate sets of controls.

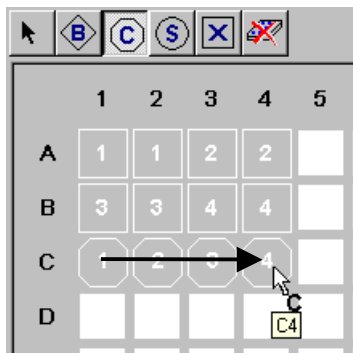


Figure 50. Defining a row of control wells.

Note that the cursor changes to a *Control cursor* as you move it over the template. Control wells are marked in the template by an octagon with a number inside it.

-
- After you have defined the control wells, you are ready to enter the concentrations of the controls as described on page 79.
-

Defining Blank Wells

In certain types of assays such as the Bio-Plex phosphoprotein assay, it may be useful to subtract the assay background from the readings of standards, controls, and unknown samples. To do this, you can prepare "blank" wells containing all the assay components except sample. Wells prepared as blanks are read along with the rest of the assay, and then Bio-Plex Manager subtracts the mean background reading of these wells from the fluorescent intensity values of the wells containing standards, controls, and unknowns.

To format blank wells in the plate template, click on the **Blank** button , then click or drag on the blank well(s) in the template. Note that the cursor changes to a *Blank cursor* as you move it over the template.

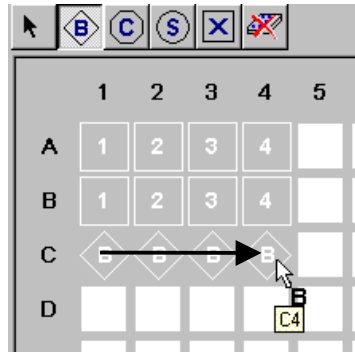


Figure 51. Defining a row of blank wells.

Blank wells are marked in the template with a diamond shape containing a letter B.

Multiple blank wells in a template are treated as a single replicate group. A mean value, standard deviation, standard error, and coefficient of variation are calculated for the group, and the mean is used as the background value.

Deleting or Changing Well Formatting

To delete the formatting from a well or set of wells, select **Undefined**  from the toolbar or *Format Options* menu, then click or drag on the well(s). To clear the formatting from all the wells in the template, select **Clear Plate** from the *Format Options* menu.

To change the formatting of a well or group of wells, simply select the appropriate tool and overwrite the old formatting, as if you were defining the wells for the first time.

Plate Grouping

After you have formatted the wells (see previous section), you can organize them into groups and identify one member of each group as the Reference, or Primary member. You can then calculate the ratio of each member's fluorescent intensity to the fluorescent intensity of its Reference. This type of analysis is similar to that performed in Western blotting.

Click on the *Plate Grouping* tab to display the plate grouping tools.

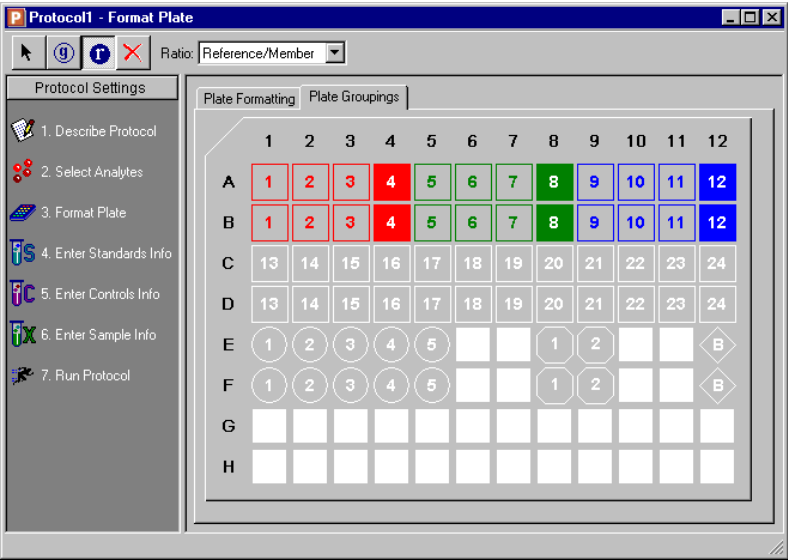


Figure 52. *Plate Grouping* view.

The buttons on the toolbar are used to group the wells on the microtiter plate diagram and identify the Reference well(s). These formatting commands are also located on the *Format Options* menu.

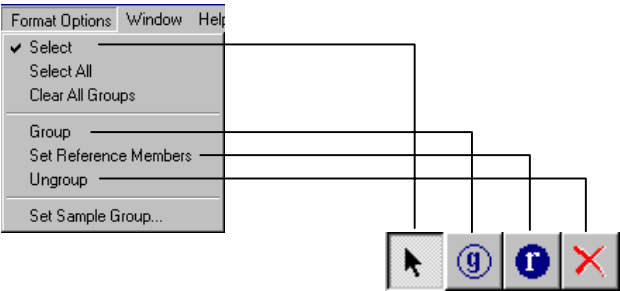


Figure 53. Plate grouping tools (*Plate Grouping* view).

First, select the ratio you want to calculate for each group—**Reference/Member** or **Member/Reference**—using the *Ratio* field pulldown list.

Defining a Group

To define an assay group, click on the **Group** button  and drag the cursor across the defined wells in the assay. Note that:

- Wells must be defined as Samples, Standards, Controls, or Blanks before they can be grouped.
- You can group across well types (e.g., a single group can include Standards, Samples, Controls, and/or Blanks).
- Replicate wells will be automatically placed in the same group (even if you drag over only one well in the replicate set).

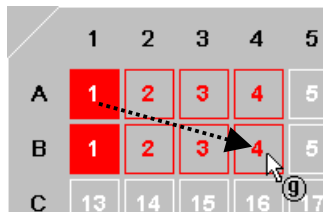


Figure 54. Defining a group of wells.

The grouped wells will change color to indicate their grouping, and the first member of the group will be highlighted to indicate that it is defined as the Reference well.

Each group you create will be designated with a different color.

To remove individual members of a group or the entire group, click on the **Ungroup Samples** button  and click or drag over the members/group.

Note that you can also use the **Select** tool  to first select a group of wells, and then use the **Set Sample Group** command on the *Format Options* menu or context menu to select the group.

Defining the Reference

To change the Reference member of the group, click on the **Reference** button  and click on the desired Reference member in the group. For replicates, all replicate

wells of the same number within the group will be automatically designated as the same Reference.

You can also use the **Select** tool  to first select the Reference well within a group, and then use the **Set Sample as Reference Member** command on the context menu to set the Reference.

Printing the Plate Format

To print the plate format for your records, select **Print** from the *File* menu, or click on the **Print** button  on the main toolbar. For a preview of the printed format, select **Print Preview** from the *File* menu.

Step 4. Enter Standards Info

Standards are analytes of known concentration, which can be used to generate a standard curve of values using one of the several regression methods built into Bio-Plex Manager. This curve can be used to calculate the concentrations of your unknowns.

The standards you select will depend on the performance characteristics of your assay as well as the type of curve-fitting model you want to use. Due to variations in assay preparation, we recommend using a minimum of **eight** standard concentrations to generate the standard curve. See the Bio-Plex cytokine assay manual for recommendations of specific standard concentrations.

The minimum number of non-zero standards required for each type of regression method available in the software is shown in the following table:

Method	Minimum number of standards
Logistic-5PL	6
Logistic-4PL	5
Cubic spine	4
Linear (linear and semi-log)	2
Point to point (linear and semi-log)	2

Before entering the concentration values of your standards, you must select the analytes that you will be analyzing in the reading as described on page 50. If your

current plate contains your standards, you must also identify the standard wells as described on page 61.

Click on the **Enter Standards Info** button in the Protocol window to enter information about your standards.

Protocol1 - Enter Standards Info

Protocol Settings

1. Describe Protocol
2. Select Analytes
3. Format Plate
4. Enter Standards Info
5. Enter Controls Info
6. Enter Sample Info
7. Run Protocol

Standards Info | External Standards Info | Select External Standards

Analyte: **Human IL-2 (38)**

Assign Standards Information

☒ Enter Manually

Std	Description	Conc
S1		32000.00
S2		16000.00
S3		4000.00
S4		500.00
S5		50.00
S6		8.00
S7		4.00

☒ Same conc. values for all analytes

Calculate Concentrations

☐ Enter Automatically

Most Concentrated Standard: ☒ S1 ☐ S7

Concentration of S1:

Dilution Factor: X

Concentration Units:

☒ Same units for all analytes (Units don't impact calculations)

Acceptable Recovery Percentage Range:

☒ Same recovery range for all analytes

Std Regression Curve

Regression Type:

Axis Transformation:

☒ Same regression type for all analytes

Figure 55. Entering information about standards.

The *Enter Standards Info* window allows you to define standards on your current microtiter plate, and/or import external standards data from another plate. Use the tabs above the data fields to select between standards on the current plate and external standards.

- You can define standards on the current plate and import external standards in the same Protocol. This is useful if you want to run standards for some analytes on the current plate, but include external standards for other analytes in your concentration calculations. Note that, if external standards and current standards are defined *for the same analyte*, the external standards will always be used to calculate the concentrations of that analyte.

Selecting External Standards

This section describes how to select a Results file from which to import external standards data. If you are not using external standards, you can skip this section and proceed to **Entering Standards Info** on page 71.

Click on the *Select External Standards* tab. In the window, click on the **Fill Available List** button, navigate to the directory containing the Results file with your standards of interest, and select the file. (Note that the *Open* dialog will only display Results files.)

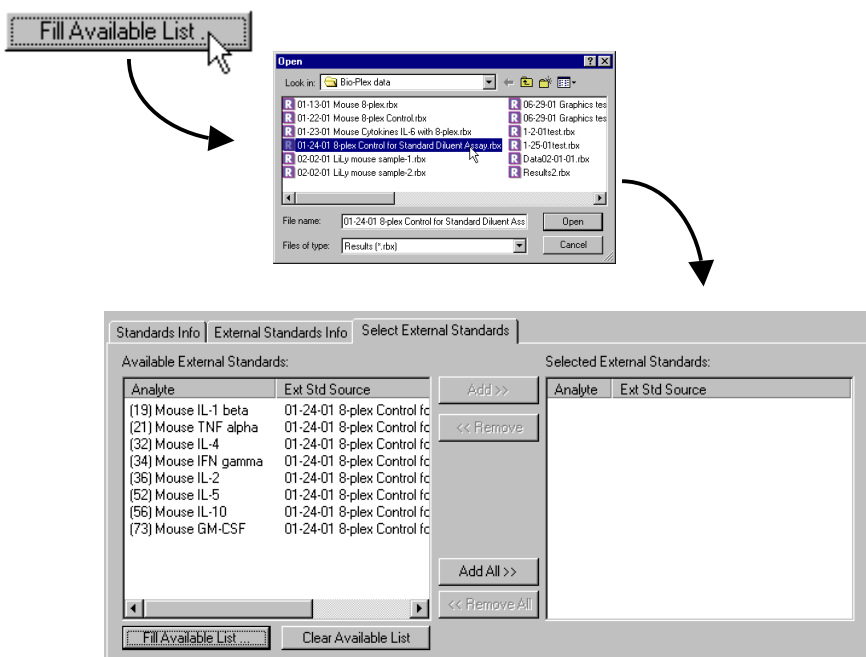


Figure 56. Selecting the Results file containing your external standards.

The *Available External Standards* list will be populated with the standards from the selected file. To add additional standards to the list, repeat the procedure, selecting a different file.

To delete all standards from the *Available* list, click on the **Clear Available List** button.

To make the available external standards accessible in the current Protocol, you must add them to the *Selected External Standards* list (right-hand pane). To select all the standards in the *Available* column, click on the **Add All>>** button. To add available standards individually, double-click on them, or select multiple analytes using Shift+click and Ctrl+click key combinations and click on the **Add>>** button. (Use the **<<Remove** and **<<Remove All** buttons to unselect them.)

-
- Each external standard you select must match the region of a selected analyte in the Protocol. For example, if you selected analytes in regions 19, 32, and 34 in the *Select Analytes* window (page 50), you can only select external standards in those same regions. (Note that the analyte name does not have to match—only the region.) Also, each analyte must come from a different region.
-

When you are finished, the *Selected* list should list all the analytes that you want to use as external standards, and their source file(s).

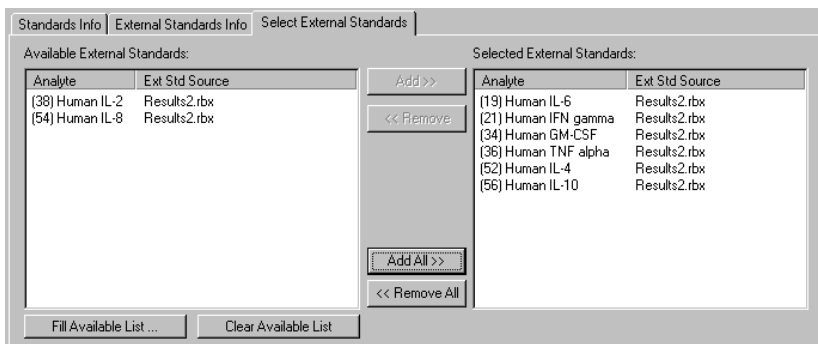


Figure 57. Selected external standards.

When you are done, click on the *External Standards Info* tab to enter additional information about your external standards.

Standards Information

This section describes how to enter information about your standards. To enter information for standard wells defined on the current plate, make sure that the *Standards Info* tab is selected. To enter information for external standards, first select

the external standards as described in the previous section, and then click on the *External Standards Info* tab.

- The fields in under these tabs are the same, except that you cannot alter the concentrations under the *External Standards Info* tab (they will appear grayed out), as shown in Figure 58.

The figure consists of two screenshots of the Bio-Plex Manager software interface, showing the 'Standards Info' and 'External Standards Info' windows.

Top Screenshot: Standards Info window

The 'Standards Info' window has three tabs: 'Standards Info', 'External Standards Info', and 'Select External Standards'. The 'Standards Info' tab is selected. The 'Analyte' dropdown is set to 'Mouse IL-10 (56)'. The 'Assign Standards Information' section has the 'Enter Manually' radio button selected. A table of standards is shown:

Std	Description	Conc
S1		8000.00
S2		6000.00
S3		4000.00
S4		2000.00
S5		1000.00
S6		50.00

The 'Calculate Concentrations' section has the 'Enter Automatically' radio button selected. The 'Most Concentrated Standard' is S1. The 'Concentration of S1' is 1. The 'Dilution Factor' is 2. The 'Calculate' button is visible. The 'Concentration Units' field is empty. The 'Acceptable Recovery Percentage Range' is set to '70 - 130%'. The 'Std Regression Curve' section has the 'Regression Type' set to 'Logistic Weighting...'. The 'Axis Transformation' is set to 'Log(x) - Linear(y)'. The 'Same regression type for all analytes' checkbox is unchecked.

Bottom Screenshot: External Standards Info window

The 'External Standards Info' window has the same three tabs. The 'External Standards Info' tab is selected. The 'Ext. Std Analyte' dropdown is set to 'Human IL-6 (19)'. The 'Assign Standards Information' section has the 'Enter Manually' radio button selected. A table of standards is shown:

Std	Conc
eS1	32000.00
eS2	16000.00
eS3	4000.00
eS4	500.00
eS5	50.00
eS6	8.00
eS7	4.00

The 'Calculate Concentrations' section has the 'Enter Automatically' radio button selected. The 'Most Concentrated Standard' is eS1. The 'Concentration of eS1' is 1. The 'Dilution Factor' is 2. The 'Calculate' button is visible. The 'Concentration Units' field is empty. The 'Acceptable Recovery Percentage Range' is set to '70 - 130%'. The 'Std Regression Curve' section has the 'Regression Type' set to 'Logistic Weighting...'. The 'Axis Transformation' is set to 'Log(x) - Linear(y)'. The 'Same regression type for all analytes' checkbox is unchecked.

Figure 58. *Standards Info* window (top) and *External Standards Info* window (bottom) (external standards concentrations cannot be changed).

The analytes that you selected in the *Select Analytes* window will be available under the **Analyte** pulldown list. Select the first analyte that you will be using as a standard.

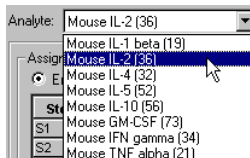


Figure 59. Selecting a standard analyte.

-
- For the settings described on the following pages, you can select different settings for each analyte, or select the same settings for all analytes.
-

Entering Concentrations (for Current Standards Only)

After you have selected an analyte, you are ready to enter the concentrations of the standards on your current plate. If you are using external standards, the concentrations will be automatically entered, and you can skip to the next section on selecting concentration units (page 75). To enter concentrations, make sure the *Standards Info* tab is selected.

To enter the concentrations manually, select **Enter Manually**. To calculate them automatically from a dilution series, select **Enter Automatically**.

Entering Concentrations Manually

If you selected the **Enter Manually** option button, the table will list your standard wells (S1, S2, S3, etc.). These are the wells that you defined in the *Format Plate* window (page 56). For the selected analyte, enter the concentration for each standard in the concentration column.

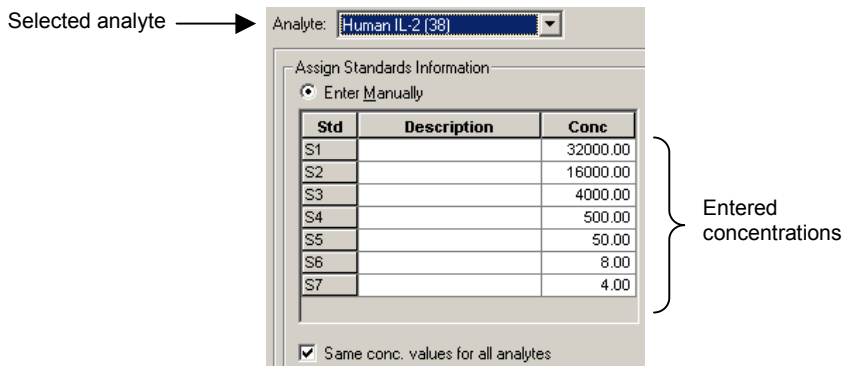


Figure 60. Entering standard concentrations manually.

If the **Same conc. values for all analytes** checkbox is selected, the concentration values you enter will be applied to all the analyte standards. Deselect the checkbox if you want to enter different concentrations for each analyte standard.

You can also enter additional information about your standards in the description column for each well.

-
- You can use the **Cut**, **Copy** and **Paste** commands on the *Edit* menu to copy your descriptions and concentrations between analytes. You can also copy and paste from and to other applications, such as Microsoft® Word or Excel. Click on a cell, row, or column in the table to copy and paste.
-

If you are entering different concentrations for each analyte standard, repeat the procedure by selecting each analyte from the pulldown menu as described above.

Calculating Concentrations Automatically

If a dilution series was created for the standards, the software can automatically calculate the concentrations of each standard. Select the analyte, and then select the **Enter Automatically** option button.

First, select the appropriate **Most Concentrated Standard** well number. (The most concentrated standard must be in either the first standard well or the last.)

Calculate Concentrations

☒ Enter Automatically

Most Concentrated Standard: ☒ S1 ☐ S7

Concentration of S1: 32000

Dilution Factor: 4 X

Calculate

Concentration Units: pg/ml

☒ Same units for all analytes (Units don't impact calculations)

Figure 61. Entering standard concentrations automatically.

Then enter the concentration of the most concentrated standard in the **Concentration of S1** field and the dilution factor for the remaining standards in the **Dilution Factor** field.

-
- Calculations are made by dividing by the dilution factor, so do not enter fractional values for the dilution factor. If each standard is half the concentration of the previous standard, enter 2 for the dilution factor. If each standard concentration is one-fourth the previous, enter 4, and so on.
-

Click on the **Calculate** button when all the information is entered. The concentrations in the table will be automatically updated.

If the **Same conc. values for all analytes** checkbox below the concentrations table is selected, the concentration values will be applied to all the analyte standards. Deselect the checkbox if you want to enter different concentrations for each analyte standard, and repeat the procedure.

Concentration Units

You can enter the units of concentration in the **Concentration Units** field (note that units are not required and will not affect calculations). These units will appear in your reports. If all your analytes have the same concentration units, select the **Same units for all analytes** checkbox.

Regression Method

Using the **Std Regression Curve** pulldown list, choose from among seven different regression methods to generate the standard curve (page 125):

- Logistic-5 PL
- Logistic-4 PL
- Linear
- Cubic spline
- Point-to-point
- Linear (Semi-log)
- Point-to-point (Semi-log)

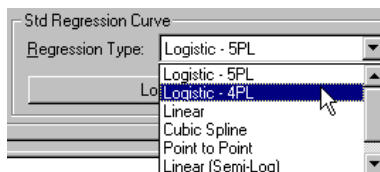


Figure 62. Selecting a regression method.

You can change your selection later in the *Standard Curve* window of the Results file. See page 125 for a more detailed discussion of the different regression methods.

You can also select the regression curve axis scale from four different combinations of linear and logarithmic. Select the axis scale from the **Axis Transformation** pulldown list.

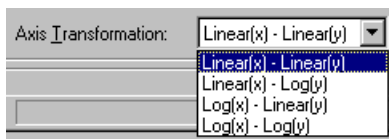


Figure 63. Selecting a regression curve axis transformation.

Note that the axis transformation only changes how the curve is displayed, not the calculated values.

-
- The regression method and axis transformation you select in the Protocol window will be used to calculate your initial results. However, these settings will not affect the reading, and you can change these settings and recalculate your results after a reading is complete.
-

Finally, if you want to use different regression methods for different analytes, deselect the **Same regression type for all analytes** checkbox. Then, when you select a different analyte from the pulldown list, you can select a new regression method as well.

Logistic Weighting

If you select a logistic curve fitting method (Logistic 5PL or Logistic 4PL), the software will perform automatic weighting of the points in the curve. Click on the **Logistic Weighting** button to access the weighting algorithm controls.

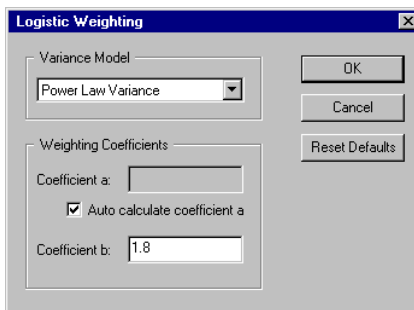


Figure 64. *Logistic Weighting* dialog.

There are eight models for calculating the variance in the logistic weighting algorithm:

- Linear Variance
- Logarithmic Variance
- Exponential Variance
- Power Law Variance (default)
- Linear CV
- Logarithmic CV

- Exponential CV
- Power Law CV

Coefficient a

Weighting coefficient a will be automatically calculated using the following equation:

$$a = .002 * \text{pow}((m_Y\text{Max} + m_Y\text{Min}) / 2 + \text{blank}, .2)$$

Alternatively, you can enter your own value by deselecting the **Auto calculate coefficient a** checkbox and entering a value in the **Coefficient a** field.

Coefficient b

Weighting coefficient b has a default value of 1.8. You can enter a different value in the **Coefficient b** field.

Recovery Percentage Range

Because the standard curve (page 125) is critical for calculating the concentrations of your unknowns, Bio-Plex Manager includes a mechanism for assessing the fit of the curve to the standards. This is the recovery percentage.

For each analyte standard, an observed concentration is back-calculated from the standard curve and the fluorescent intensity. This is divided by the expected concentration as entered in the *Enter Standards Info* window and multiplied by 100 to give the recovery percentage:

$$\text{Recovery percentage} = \text{Observed conc} / \text{Expected conc} * 100$$

Recovery percentages are also used with controls (page 79) to determine the overall accuracy of an assay.

In the *Enter Standards Info* window, use the **Acceptable Recovery Percentage Range** pulldown menu to select an acceptable recovery range.

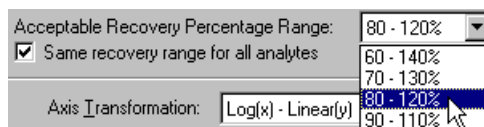


Figure 65. Selecting the recovery percentage range.

For example, if you select the default range of 80–120%, an observed concentration should be within 80–120% of the expected concentration.

The actual recovery rate for each standard and control is shown in the **Obs/Exp*100** column of the *Report* table (see page 120). Concentrations that fall outside this range are flagged as *** in the **Conc in Range** column of the *Report* table.

Step 5. Enter Controls Info

Controls are samples of known concentration whose values are measured during a reading in the same manner as the unknown samples. You can then compare the calculated concentrations of the controls after a reading with the expected concentrations to determine the overall accuracy of the assay. (This is sometimes called the “spiked” recovery method, because you are “spiking” your assay with known quantities of analytes.)

In this step, you enter the concentration values of your controls. First, you must select the analytes that you will be analyzing in the reading as described on page 50, and identify the plate well(s) containing your controls as described on page 61.

-
- Only formatted wells will be read by the Bio-Plex array reader. Be sure to define all your control wells before running a Protocol.
-

Click on the **Enter Controls Info** button in the Protocol window to enter information about your controls.

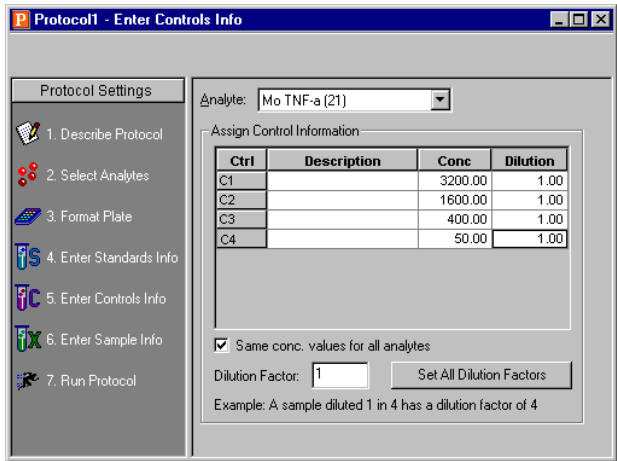


Figure 66. *Enter Controls Info* dialog.

The analytes that you selected in the *Select Analytes* window will be available under the **Analyte** pulldown menu. Select the first analyte that you will be using as a control.

The table will list your control wells (C1, C2, C3, etc.). These are the wells that you defined in the *Format Plate* window (page 56). Enter the concentration of the selected analyte in the **Conc** column for each well.

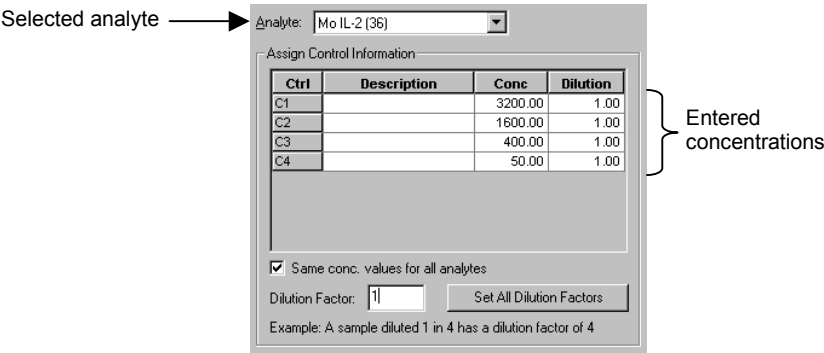
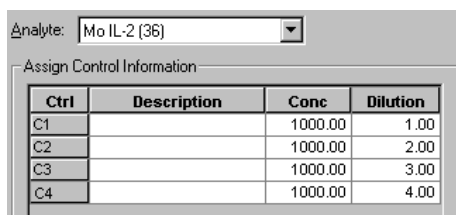


Figure 67. Entering concentrations of the controls.

If the **Same conc. values for all analytes** checkbox is selected, the concentration values you enter will be applied to all the available control analytes. Deselect the checkbox if you want to enter different concentrations for each control analyte.

You can enter additional information about your controls in the description column for each well.

If you are performing a dilution of your controls, you can enter a dilution factor for each control in the **Dilution** column. Note that a control diluted 1 in 2 would have a dilution factor of 2. This feature allows for the performance parallelism studies, whereby a control diluted by different dilution factors is analyzed to verify that the final concentration is the same for all of the samples.



Ctrl	Description	Conc	Dilution
C1		1000.00	1.00
C2		1000.00	2.00
C3		1000.00	3.00
C4		1000.00	4.00

Figure 68. Controls with different dilution factors.

If you are entering different concentrations for each analyte standard, repeat the procedure by selecting each analyte from the pulldown menu as described above.

To enter the same dilution factor for all controls, enter the value in the *Dilution Factor* field, and click on the **Set All Dilution Factors** button. The dilution factor you enter for one analyte is automatically applied to all analytes.

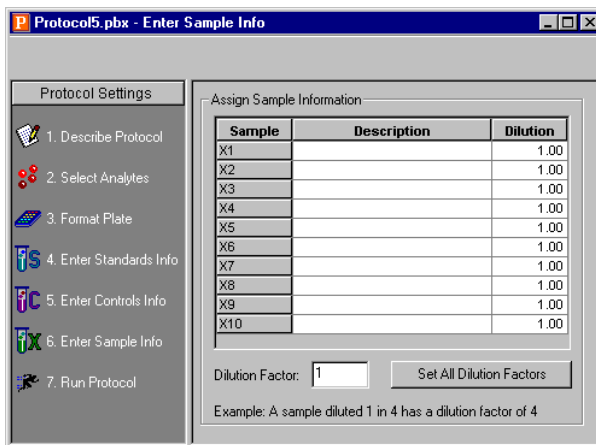
-
- You can use the **Cut**, **Copy** and **Paste** commands on the *Edit* menu to copy your descriptions, concentrations, and/or dilutions between analytes. You can also copy and paste from and to other applications, such as Microsoft® Word or Excel. Click on a cell, row, or column in the table to copy and paste.
-

Step 6. Enter Sample Info (optional)

Before entering information about your unknown samples, select the analytes that you will be analyzing in the reading as described on page 50, and then identify the plate well(s) containing your unknown(s) as described on page 58.

-
- Only formatted wells will be read by the Bio-Plex array reader. Be sure to define all your unknown sample wells before running a Protocol.
-

Click on the **Enter Sample Info** button in the Protocol window to enter information about your unknown samples.



Sample	Description	Dilution
X1		1.00
X2		1.00
X3		1.00
X4		1.00
X5		1.00
X6		1.00
X7		1.00
X8		1.00
X9		1.00
X10		1.00

Dilution Factor:

Example: A sample diluted 1 in 4 has a dilution factor of 4

Figure 69. *Enter Sample Info* dialog.

The information table will list all the unknown sample wells as defined in the plate template for the reading (page 58).

You can enter a description and a dilution factor for each unknown in the appropriate columns. Note that a sample diluted 1 in 2 would have a dilution factor of 2. To enter the same dilution factor for all unknowns, enter the factor in the *Dilution Factor* field and click on the **Set All Dilution Factors** button.

-
- You can use the **Cut**, **Copy** and **Paste** commands on the *Edit* menu to copy your descriptions and/or dilutions from and to other applications, such as Microsoft® Word or Excel. Click on a cell, row, or column in the table to copy and paste.
-

7. Running the Protocol

After you have prepared the Protocol as described in the previous chapter, you are ready to run the Protocol. Click on the **Run Protocol** button  in the Protocol window to select the final settings and run the Protocol.



When performing the following operations, the sheath fluid container and the waste fluid container should be closely monitored.



The sheath fluid bottle must be placed at the same level as the Bio-Plex array reader (unless you are using the HTF; see caution below), and the fluid level should be below the air inlet connection and above the sheath outlet connection. Always check the sheath fluid level before starting a run or procedure.



If you are using the HTF, it should sit on the counter next to the Bio-Plex array reader, while the sheath fluid cube should be placed ~3–4 feet below the reader (e.g., on the floor).



The waste fluid container receives waste from the system. **Do not allow the waste container to overflow!** Empty the waste bottle each time the sheath fluid bottle is filled. Do not place it on the instrument. All waste containers should have vented caps.

Run Protocol Window

The *Run Protocol* window has controls for performing the run, and two views for displaying the status of the run: the *Raw Data* table view and the histogram/bead map view. Click on the **Show/Hide Histogram/Bead Map** button  to display only the *Raw Data* table, or the histogram/bead map display with the *Raw Data* table below it (maximize the window to see both the table and the display).

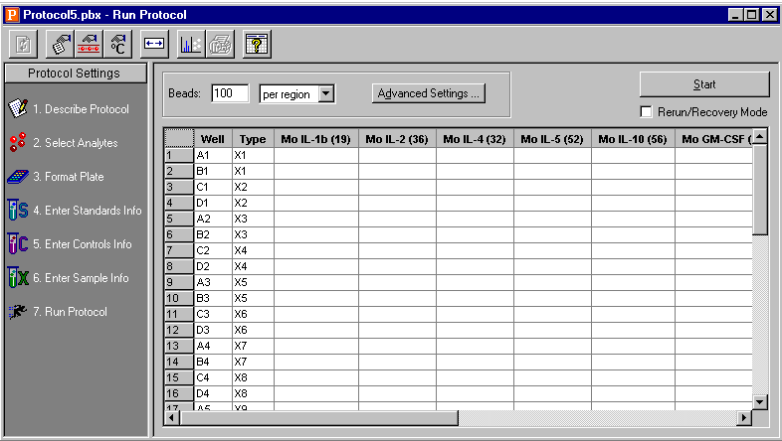


Figure 70. *Run Protocol* window—*Raw Data* table view.

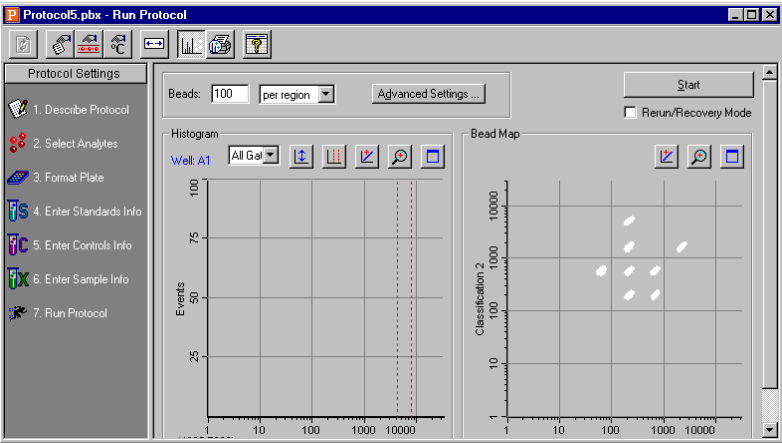


Figure 71. *Run Protocol* window—Histogram/bead map view.

The bead map and histogram (see page 94) provide graphical data about a run, and are continually updated throughout the run. The *Raw Data* table (see page 104) displays the raw data of the reading after it is complete.

The controls for performing the run are described in the following sections.

Bead Count

The **Beads** field lists the number of beads to be detected before a reading is complete. Note that this number can be beads **per region** or **total** detected, depending on the pull-down menu selection.

The default value of **100 beads per region** is recommended. This guarantees that at least 100 beads in each region will be acquired before sample acquisition is complete.

If you want to count beads per region, you can enter a number between 1 and 10,000 in the **Beads** field. For most assays, including the Bio-Plex cytokine assay, use a value of 100 beads per region. For the Bio-Plex phosphoprotein assay, 25 beads per region is recommended to avoid triggering a low bead count warning.

-
- Under *Select Analytes*, be careful to select only those bead types that are present in your samples. If you select a bead type that is not present, the array reader will be unable to count the target number of beads of that type, triggering a warning.
-

If you want to count total beads, you can enter a number between 1 and 100,000 in the **Beads** field. Note that if you choose to measure total beads, the instrument will stop acquisition as soon as it has acquired the total number of beads in any combination of bead sets. If you are detecting multiple analytes, this means that some bead sets may be underrepresented in the final tally.

Advanced Settings

Click on the **Advanced Settings** button to access additional reader settings and controls.

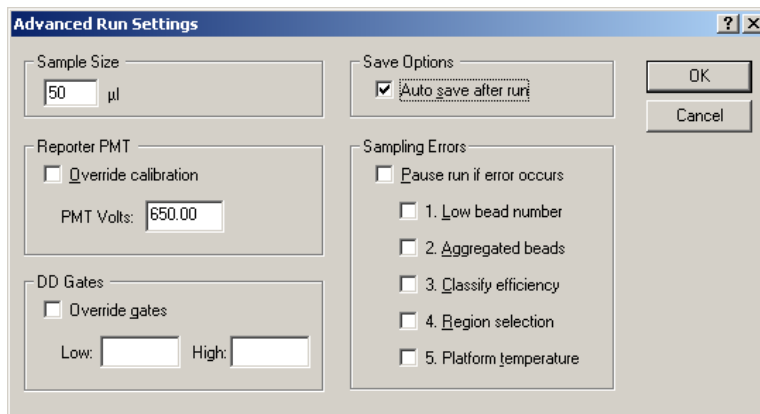


Figure 72. *Advanced Run Settings* dialog.

Sample Size

The **Sample Size** is the amount of sample that will be sucked from the microplate well by the reader. The default sample size of **50 µl** is recommended. You can specify a sample size as small as 10 µl or as large as 200 µl.

-
- To avoid air uptake, make sure that each microplate well to be read contains at least 125 µl of sample. The maximum sample volume per well is approximately 140 µl. (The array reader rinses an additional 160 µl into each well after a reading is complete; the maximum well capacity is 300 µl.)
-

Automatically Saving Results

Auto Save After Run will automatically save the Results file generated by the reading. With this checkbox selected, when you click on **Start** to begin a reading, you will be prompted to specify a name for the Results file.

Sampling Errors

During a reading, the Bio-Plex array system is continuously monitoring the flow of beads, the bead count, the bead regions, and the platform temperature. If the system detects a problem in any of these areas, it can stop the reading and trigger a warning.

This feature is enabled if any or all of the checkboxes under *Sampling Errors* are selected.

Select **Pause run if error occurs** to select all the possible error conditions, or select/deselect the individual error condition checkboxes. By default, these are not selected (i.e., the run will not pause for any error condition).

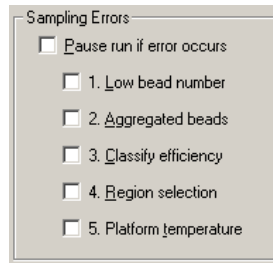


Figure 73. Error condition checkboxes

-
- All errors will be logged in the *Raw Data* and *Report* tables, whether or not the checkboxes are selected (see page 106).
-

The reading errors and their possible causes are listed in the following table:

Error	Possible Cause*
1. Low bead number	Too few beads in the assay; buffer volume in well is too low; plate was not shaken properly before analysis; microbubbles in the cuvette.
2. Aggregated beads	Bead clumping in the assay; sheath fluid is empty; waste reservoir is overfull.
3. Classify efficiency	Microbubbles in the cuvette; beads have become photobleached; percentage of beads outside the DD gate range is too high.
4. Region selection	Incorrect beads were selected in the Protocol or assay; too few beads are present in the assay.
5. Platform temperature	Platform temperature has fluctuated +/- 2°C during the reading.

* See Appendix 1 on troubleshooting on page 133 for a detailed list of reading errors, causes, and suggested solutions.

With the checkboxes selected, the reading will stop and an alert box will describe the problem and suggest steps for correcting it. After you have performed the suggested steps, you can rerun the reading.

-
- If you are rerunning wells in rerun recovery mode, you may want to disable the **Low bead number** checkbox since there is a lower bead number in a well that has already been read once.
-

PMT Voltage

Users who want to manually set the PMT voltage on the reporter detector can select the **Override calibration** checkbox in the *Advanced Run Settings* dialog and enter a new voltage in the **PMT Volts** field.

-
- Note that the PMT voltage is set during calibration, and should only be adjusted by users who know the precise PMT value that they want to specify. If you change the PMT, you will no longer have day-to-day and unit-to-unit reproducibility of results. The PMT voltage will be reset to the calibrated values after the plate is read and the results generated.
-

Doublet Discriminator Gate Range

During a reading, a doublet discriminator (DD) channel measures the amount of light scatter from particles that flow past the red laser. The light scatter is directly proportional to particle size, and an internal DD “gate” is designed to identify particles that are smaller or larger than a single microsphere, including microspheres that are clumped or aggregated.

In the **Advanced Run Settings** dialog, the default **DD Gates** range (**Low** = 4335; **High** = 7990) is designed to accommodate the standard 5.5- μ m beads distributed by Bio-Rad. If you are using different-sized beads, you can change the size of the gate by selecting the **Override gates** checkbox and entering new values in the **Low** and **High** fields.

You can also change the range by dragging the red lines on the *Run Protocol* window histogram (see page 97).

-
- The values in the fields are channel numbers corresponding to the amount of light scatter from a particle. Beads of 5.5 μ m fall squarely within the default range. If you want to change the range, you will need to know the approximate channel numbers corresponding to the size of the beads you are using.
 - If you are developing your own assay, check to ensure that the default DD gates are correct. In certain assays, the position of the DD peak may shift and thus the DD gates may need to be shifted accordingly. However, for Bio-Plex assays and other Luminex-type assays, the default DD gate settings will be correct.
-

Plate Loading Guidelines

Before starting the run, note the following plate handling guidelines and warnings:



Protect your assay microspheres from light. Once photobleached, the beads are no longer usable. Proper care of microspheres must be observed to maintain the benefits of your product warranty.



When using a filter plate, avoid touching the bottom of the wells when wrapping the plate. Sample may wick or leak from the bottom of the well, causing problems when reading the plate.



Make sure that you have added at least 125 μ l of sample to all the wells specified in your plate template before starting a reading. If the Bio-Plex array reader attempts to draw sample from an empty well, air will be sucked into the sample loop and injected into the flow chamber. As a result, bubbles will form in the cuvette and interfere with the analysis. If this happens, perform a **Remove Bubbles** procedure (page 32) and rerun the Protocol.



Shake the microplate on an plate shaker for 30 seconds prior to performing a reading.

Running the Protocol

When you have selected and reviewed all your Protocol options and prepared your microplate, click on the **Start** button. A dialog box will prompt you to eject the microplate platform tray (**Eject/Retract Plate**), load your microtiter plate, and begin the run.

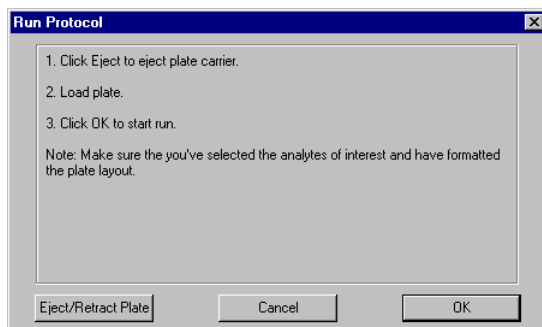


Figure 74. Instructions for inserting the microtiter plate.

Click on **OK** to begin the reading. The following conditions may delay the start of a reading:

- If the array reader optics have not warmed up, the reading will be delayed until warm up is complete.
- If the pressure is too low in the fluidics system, you will receive a warning message prompting you to check the sheath fluid level, hose connections, etc.
- If the platform heater is not at the specified temperature, a warning message will appear and you will be prompted to wait for the temperature to change.

When the reading begins, the sample needle will lower, draw sample from a plate well, and retract to its original position. The sample will be drawn from the needle into the sample loop and the reading will begin.

After initial system pressurization (5–20 seconds), a typical well reading may take up to 2 minutes; in most cases, individual wells are read in a matter of seconds.

Status Bar

As the sample is analyzed, the status of the reading will be displayed in the status bar at the bottom of the Bio-Plex Manager window.

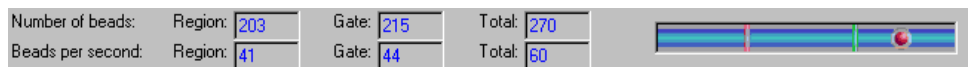


Figure 75. Status bar during a reading.

The status bar provides a running count of both the **Number of beads** and the **Beads per second**. Within these categories, the array reader keeps track of the following information:

- The *Region* fields monitor only those beads that fall within one of the regions you selected in the Protocol (page 51).
- The *Gate* fields monitor all beads that pass an internal discriminator "gate" designed to eliminate particles that are smaller or larger than a microsphere (including microspheres that are clumped or aggregated).
- The *Total* fields report all particles measured by the array reader, including those that do not pass the internal discriminator gate.

Manually Stopping a Reading

While the sample is being read, the **Start** button will change to a **Stop** button in the Protocol window. You can click on **Stop** to cancel the reading. An alert box will notify you that you can generate a report with the incomplete results or rerun the Protocol as described on page 107.

Generating Results from a Protocol

The reading stops when you click on **Stop**, when an error stops the reading, or when all the samples have been analyzed. At the end of the reading, the sample needle lowers and the syringe pump purges the sample loop and needle with sheath fluid, sending a fraction of sheath fluid (approximately 160 µl) back into each well.

At the end of a complete reading, the Results will be automatically calculated and displayed.

Note that the Results file will be automatically saved if you selected **Auto save after run** (page 88) in the Protocol. Otherwise, you must save your Results as a separate step (see the following chapter).

To generate a new Results file from a Protocol, select **New Results from Protocol**



from the *Run Protocol* toolbar or the *File* menu.

Bead Map and Histogram

The histogram and bead map provide a graphical display of the raw data from a reading. These graphs are continuously updated during the reading in the *Run Protocol* window of a Protocol file. They also can be displayed after a reading in the *Raw Data* window of a Results file.

The histogram and bead map share window space with the *Raw Data* table (see page 104). If the histogram and bead map are not visible, click on the **Show/Hide**

Histogram/Bead Map button  to display them.

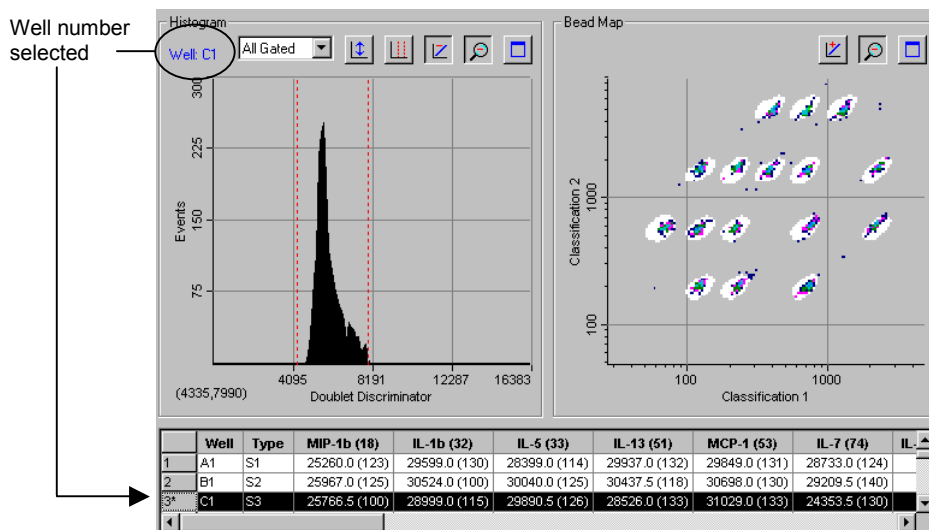


Figure 76. Histogram and bead map.

You can see some of the rows of the *Raw Data* table below the histogram and bead map display if you maximize the window.

During a run, the well number being read is displayed in blue in the upper left corner of the histogram. After the run, select the different wells to display by selecting rows in the table below the graphs.

When viewing the histogram and bead map, you can choose to display all analytes, each analyte separately, or all beads that pass through the discriminator gate (see page 91). Make your selection using the pulldown menu above the histogram.

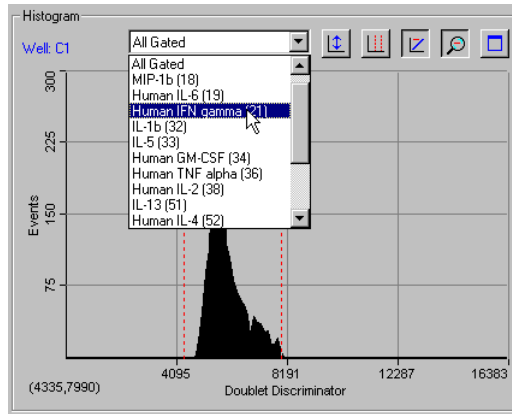


Figure 77. Selecting the analyte(s) to display.

To print the histogram and/or bead map, click on the **Print Histogram/Bead Map** button . This opens a small dialog in which you can select the histogram, bead map, or both to print.

Histogram

The histogram plots the number of events per channel number for the selected well, analyte(s), and channel type. An event is generated when a particle such as a bead or aggregated beads passes through the path of the lasers. Each event generates signals in different channels:

- The fluorochromes embedded in the bead(s) generate signals in the Classification 1 and Classification 2 channels.
- The fluorescent molecules bound to the analytes generate a signal in the Reporter 1 channel.
- The light scatter of the bead(s) generates a signal in the Doublet Discriminator channel. The amount of light scatter is directly proportional to bead size.

The signals are reported as channel numbers. In the histogram, the Y-axis is the events, and the X-axis is the channel numbers from 1 to 32766 for the selected channel type.

DD Gate Range

The **DD Gates** range specified under the *Run Protocol Advanced Settings* (see page 91) is shown on the histogram as two red lines perpendicular to the X-axis. In the Protocol window, you can change the DD gate range by positioning your cursor over the red lines and dragging them to the right or left. Note that this will also change the range as listed in the **Advanced Settings** dialog.

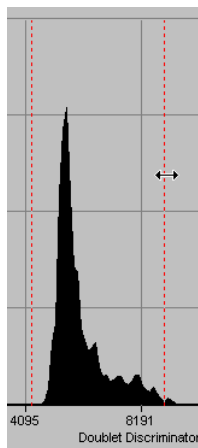


Figure 78. Changing the DD gate range.

-
- You can only change the DD gate range in a Protocol—not in a Results file.
 - You can only set the gates on the DD channel, not the RP1, CL1, or CL2 channel.
 - If you are running a Luminex-type assay from another manufacturer, you do not need to change the DD gates. Calibrate the Bio-Plex reader using the Bio-Plex DD target and use the default DD gate range (4335–7990) in the protocol.
-

You can show or hide the gate lines by right-clicking on the histogram and selecting **Show** or **Hide** from the **Gate** context menu.

To reset the gates to their default values, click on the **Restore Default Gates** button  above the histogram.

Selecting the Channel Type

You can change the histogram to display different channel types. Click on the histogram with the right mouse button and select **X-Axis** from the context menu.

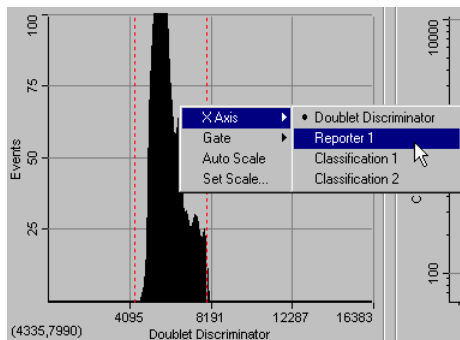


Figure 79. Histogram right-click context menu.

From the **X-Axis** submenu, select from the following channel types:

- The **Doublet Discriminator** channel measures the forward light scatter from particles as they pass through the red laser. Light scatter correlates to particle size; this channel allows you to discriminate single beads (“singlets”) from aggregated beads in the histogram. This is the default selection.
- The **Reporter 1** channel measures fluorescence from the reporter molecules bound to the analytes on each bead as they pass through the green laser. The amount of fluorescence per bead type is directly proportional to the amount of analyte present in the assay. This can be useful for viewing the distribution of the fluorescence signal of a particular analyte.
- The **Classification 1** channel measures fluorescence from the first classification dye embedded in each bead as it passes through the red laser.
- The **Classification 2** channel measures fluorescence from the second classification dye embedded in each bead as it passes through the red laser.

Note that the changing the channel type will change the channel numbers and the histogram display.

Scaling the Histogram Data

If the histogram is too big or too small to fit comfortably within the display range, click on the **Auto Scale** button  on the histogram toolbar. The height of the Y-axis will change to better fit the data. You can also select this command from the right-click context menu.

Select **Set Scale** from the right-click histogram context menu to manually set the range of the Y-axis. In the pop-up box, enter the maximum number of events to display on the Y-axis.

The default view of the data is log scale. From the toolbar, click on the **Log/Linear** button  to change the display from log to linear. Click again to return to log.

Magnifying and Resizing the Histogram

Click on the **Zoom** button  on the histogram toolbar to magnify the display. With **Zoom** selected, the cursor will change to a “hand” symbol  when you position it over the histogram. Drag the hand cursor to view different magnified regions of the histogram.

Click on the **Maximize** button  to maximize the histogram portion of the window within the larger Protocol window. Click on the **Restore** button  to return the histogram to its default state.

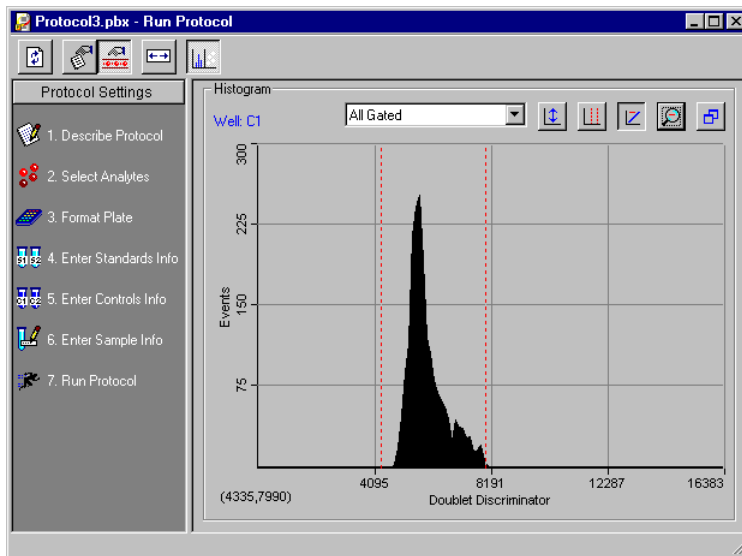


Figure 80. Histogram maximized in the Protocol window.

Bead Map

The bead color map is a density dot plot of the events in a reading. Different colored dots in the map represent different numbers of events at those data points.

In the default bead map, the X-axis is the **Classification 1** channel and the Y-axis is the **Classification 2** channel. These channels measure the embedded fluorochromes in each bead, which are used to identify the bead set and corresponding analyte. The resulting data point clusters in the map represent the different bead sets in the assay and their associated analytes

-
- Each bead set generates a cluster of data points, rather than a single point, because of minute variations in fluorochrome levels and intensity among the beads in a set.
-

The white areas on the map indicate the expected regions of the selected analytes. If you move your cursor over these areas, the name of the analyte will appear in a pop-up. The data point clusters should fall within these regions.

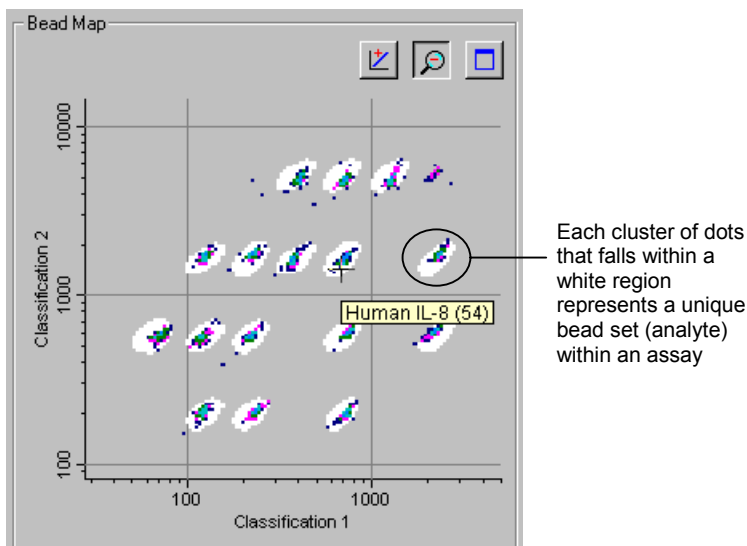


Figure 81. Bead map, with pop-up indicating the analyte region under the cursor.

If one or more clusters do not fall within the white areas, it may indicate a problem with the beads such as photobleaching, or the analyte selections in the Protocol may be incorrect. If all the data clusters have shifted out of the white regions, there may be microbubbles in the cuvette. See Appendix 1 on page 133 for guidance with troubleshooting.

In this way, the map can be used to confirm that the bead sets are correctly selected, measured and identified.

Changing the X- and Y-Axes of the Map

You can change the channel types of the axes. Right-click on the bead map and select one channel from the **X-Axis** submenu, and the other channel from the **Y-Axis** submenu. The available channels are:

- **Doublet Discriminator**—This channel measures light scatter from the beads as they pass through the red laser. Light scatter correlates to particle size.
- **Reporter 1**—This channel measures the reporter molecules bound to the analytes on each bead.

- **Classification 1**—This channel measures the first classification dye embedded in each bead.
- **Classification 2**—This channel measures the second classification dye embedded in each bead.

If you are developing your own assay, you may want to plot the Reporter 1 channel against the Doublet Discriminator channel to evaluate the distribution of the reporter signal. This yields information about the specificity of the antibodies in the assay.

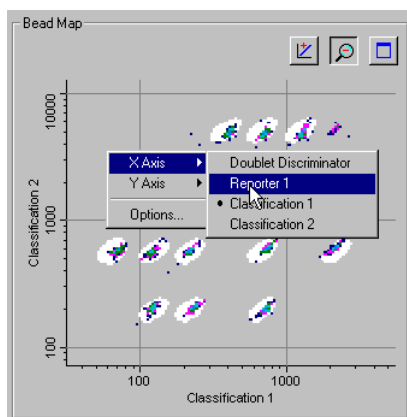


Figure 82. Changing the X-axis.

Scaling the Bead Map Data

The default view of the data is log scale. From the toolbar, click on the **Log/Linear** button  to change the display from log to linear. Click again to return to log.

Magnifying and Resizing the Bead Map

Click on the **Zoom** button  in the bead map toolbar to magnify the display. With **Zoom** selected, the cursor will change to a “hand” symbol  when you position it over the map. Drag the hand cursor to view different magnified regions of the bead map.

Click on the **Maximize** button  to maximize the bead map portion of the window within the larger Protocol window. Click on the **Restore** button  to return the bead map to its default state.

Filtering the Bead Map Display

Each colored dot in the bead map represents a certain number of events measured by the reader at that data point. Different colors are used to distinguish between different numbers of events. You can adjust the display so that data points with low numbers of events are not shown, or a certain number of events is required before the data point color changes. To access these settings, right-click on the bead map and select **Options**.

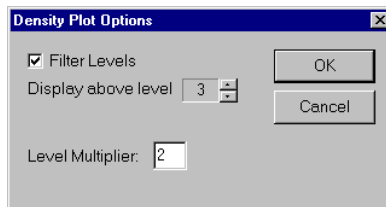


Figure 83. Bead map filter level settings.

-
- These settings only change the bead map display, and do not affect the underlying data in any way.
-

First, enter a number in the **Level Multiplier** field. This number is used exponentially to determine the number of events required before a data point changes color. For the default value of 2, the color level will change at 2 events, 4 events, 8 events, 16 events, etc.

If you want to filter lower numbers of events from the display, select the **Filter Levels** checkbox and use the arrow keys next to **Display above level** to select the exponent level below which no data points will be displayed. For example, if the **Level Multiplier** is set to 2, and **Display above level** is set to 3, data points are not displayed until 8 or more events are registered; the first color level occurs at 8 events, the second color level occurs at 16 events, the third color level occurs at 32 events, etc.

The color levels in the display are, in order, dark blue, purple, dark green, light blue, blue, light green, orange, black, and brown.

Raw Data Table

The *Raw Data* table displays the raw numerical data of a reading. This table is available in both the *Run Protocol* window and in the *Raw Data* window of the Results file. (Note that you can only print and export the *Raw Data* table from a Results file, not a Protocol window. See page 113.)

The first few lines of this table are visible below the bead map and histogram. Alternatively, hide the bead map/histogram display by clicking on the **Show/Hide**

Histogram/Bead Map button  to maximize the table.

	Well	Type	Human IL-2 (38)	Human IL-4 (52)	Human IL-6 (19)	Human IL-8 (54)	Human IL-10 (56)	Hu
1	A1	S1	30624.5 (114)	27006.5 (134)	29012.0 (116)	28236.0 (171)	30960.0 (100)	
2	B1	S2	31651.0 (121)	28427.0 (121)	28489.0 (117)	29672.0 (125)	31771.0 (115)	
3	C1	S3	30068.0 (113)	25496.0 (120)	18464.0 (107)	29496.0 (155)	29109.5 (112)	
4	D1	S4	12722.5 (112)	2619.5 (114)	2679.0 (103)	16132.0 (131)	8667.5 (119)	
5	E1	S5	721.5 (114)	165.0 (139)	215.0 (109)	1092.5 (163)	1356.0 (133)	
6	F1	S6	178.5 (128)	103.5 (154)	74.5 (122)	179.0 (147)	655.0 (157)	
7	G1	S7	148.0 (126)	104.5 (122)	55.5 (101)	132.0 (113)	621.0 (104)	
8	H1	B	113.0 (117)	104.0 (101)	42.0 (103)	85.5 (130)	618.5 (119)	
9	A2	S1	31739.0 (107)	28795.0 (127)	30648.0 (100)	30168.0 (130)	31875.5 (100)	
10	B2	S2	32018.0 (102)	29070.5 (112)	29775.0 (116)	30915.5 (132)	32022.0 (100)	
11	C2	S3	30433.0 (101)	24834.0 (132)	17395.0 (130)	29556.0 (119)	29054.0 (123)	
12	D2	S4	11596.0 (136)	2526.5 (150)	2273.0 (110)	15478.5 (110)	8633.0 (105)	
13	E2	S5	733.0 (115)	169.0 (134)	230.0 (112)	1065.0 (125)	1414.0 (110)	
14	F2	S6	202.0 (108)	104.5 (110)	74.0 (114)	197.0 (111)	716.0 (126)	
15	G2	S7	122.0 (120)	94.0 (127)	49.5 (112)	126.0 (154)	512.5 (116)	
16	H2	B	115.5 (140)	96.0 (141)	45.0 (121)	81.5 (140)	534.5 (100)	
17	A3	X1	28695.0 (115)	24950.5 (126)	25251.0 (130)	26184.5 (159)	28006.0 (101)	
18	B3	X2	28028.5 (102)	25142.5 (127)	24400.0 (135)	26554.0 (139)	27278.0 (107)	
19	C3	X3	30522.0 (117)	14236.5 (126)	9452.0 (102)	30237.0 (140)	25728.0 (111)	
20	D3	X4	6089.0 (143)	942.5 (126)	941.0 (127)	11316.0 (129)	4297.0 (125)	
21	E3	X5	376.5 (100)	116.0 (129)	91.0 (131)	764.0 (149)	451.0 (129)	

Figure 84. Raw data from a reading.

During a reading, the table is filled in line by line as each well is read. After the reading is complete, all the data is displayed.

In the table, the fluorescent intensities (FI) of the analytes you selected are listed for each well number.

-
- During a reading, the Bio-Plex array reader will always detect *all* the analytes in the sample, including any that you have not selected. However, analytes that were not selected in the Protocol will not be included in the table, even if they were detected. After a reading, you can go back and change your selections under *Select Analytes*, and the new analytes, if detected, will appear in the table.
-

For on-line Help information about the *Raw Data* table, click on the **Display Help for Table Legends** button .

Bead Statistics

To display statistical data for the individual beads as well as the entire well, click on the **Show/Hide Bead Statistics Columns** button  on the table toolbar. The number of beads collected for each analyte will be displayed in parentheses after the analyte fluorescent intensity.

Human IL-6 (19)	Human IFN gamma (21)
10744.00 (65)	11739.00 (75)

 
 Fluorescent intensity Bead count

Figure 85. Analyte fluorescent intensity and bead count.

With **Show/Hide Bead Statistics Columns** selected, the following columns are also displayed at the end of each row:

- **Total**—Total bead count in the well.
- **Region**—Number of beads that fell within defined regions.
- **Gate**—Number of beads that passed the DD gate (see page 91).
- **% Agg Beads**—Percentage of aggregated beads in the well ($100 - (\text{Gate}/\text{Total} \times 100)$).

Region	Gate	Total	% Agg Beads
2058	2128	2385	11

Figure 86. Well data.

-
- The number in the **Region** column will be flagged by an asterisk (*) if that well did not acquire the number of beads per region specified for the reading.
-

Sampling Errors

During a reading, the Bio-Plex system is continuously monitoring the flow of beads, the bead count, the bead regions, and the platform temperature. The *Sampling Errors* column reports errors in any of these areas for each analyte. These errors will be reported in the table even if you chose not to pause the run due to an error condition (see page 88).

The sampling error codes in the column are listed below.

Error Code:	Indicates:
1	Low bead number detected in the well
2	Aggregated beads detected in the well
3	Bead classification efficiency problem detected in the well
4	Region selection problem detected in the well
5	Platform temperature problem detected

Other Table Formatting

To display the incubator temperature for each well in the plate, click on the **Show/Hide Platform Temp** button . The *Temp (°C)* column will be displayed in the table.

To display the description of each sample in the table, click on the **Show/Hide Description Column** button  above the table. The *Description* column will be displayed.

To automatically size the columns fit the width of the text, click on the **Autofit** button  on the toolbar.

Copying the Raw Data

You can copy the contents of the table to the Windows clipboard. Select rows, columns, cells, or the entire table, then select **Copy** from the *Edit* menu. The values you copy can be pasted into other Windows® applications.

Rerun/Recovery Mode

After a reading has stopped—because of an error or user intervention, or because the reading is complete—the **Rerun/Recovery Mode** checkbox in the *Run Protocol* window will be selected. This allows you to rerun some or all of the samples.

With **Rerun/Recovery Mode** selected, checkboxes will appear next to each well number in the *Run Protocol* table. Select the checkboxes next to the wells that you want to read again.

The screenshot shows the 'Run Protocol' window. At the top, there is a 'Beads' field set to '100' and a 'per region' dropdown. To the right is an 'Advanced Settings...' button. Below these is a 'Start' button and a checked checkbox for 'Rerun/Recovery Mode'. Below the checkbox is a table with columns: Well, Type, Hu IL-2 (38), Hu IL-4 (52), Hu IL-6 (19), Hu IL-8 (54), Hu IL-10 (56), and Hu GM-CS. The table has 5 rows of data, each with a checkbox in the 'Well' column.

	Well	Type	Hu IL-2 (38)	Hu IL-4 (52)	Hu IL-6 (19)	Hu IL-8 (54)	Hu IL-10 (56)	Hu GM-CS
1	☐ A1	S1	174.5	241.0	146.0	804.0	164.0	
2	☒ B1	S2	266.0	283.0	174.5	1497.0	337.0	
3	☐ C1	S3	663.0	532.5	275.0	3628.0	995.0	
4	☒ D1	S4	2108.0	1587.0	607.0	10216.0	3118.0	
5	☐ E1	S5	7687.0	7166.0	1949.5	24728.0	10410.0	

Figure 87. Selecting wells to rerun.

- To rerun wells in a filter plate, you must first vacuum the excess fluid from the wells and resuspend the remaining beads in 125 μ l. (The beads will remain on top of each filter plug after vacuuming.)

Vacuum the wells of the filter plate using the vacuum filter apparatus and resuspend the beads in each well with 125 μ l of resuspension buffer. See the Bio-Plex Cytokine assay manual for detailed instructions.

Each well may be read twice without adversely affecting the results.

Because there are fewer beads in a well that has already been run once, be sure to select 100 beads per region or fewer in the *Run Protocol* settings (see page 87). Also, click on the **Advanced Settings** button and deselect the **Low bead number** error condition checkbox (page 87). Otherwise, a low bead error may stop the reading.

When you are ready, click on **Start** to begin the reading.

8. Analyzing the Results

After a reading is complete, a **Results file** is generated and automatically displayed. Note that the Results file is automatically saved if you selected **Auto save after run** in the Protocol window. Otherwise, you must save your Results file as a separate step.

To generate a Results file from an incomplete or previous Protocol reading, open the Protocol file and select **New Results from Protocol**  from the *Run Protocol* toolbar or the *File* menu.

Opening a Results File

To open an existing Results file, click on the **Open** button  on the *Quick Guide* or main toolbar or select **Open** from the *File* menu. A standard Windows *Open* dialog will be displayed, which you can use to locate and select the file. Results files have the extension *.rbx.

Saving a Results File

To save a Results file, with the file open click on the **Save** button  on the main toolbar or select **Save** or **Save As** from the *File* menu.

Reducing the Size of a Results File

Results files can be very large (>5 megabytes), and much of this file size is due to data necessary to generate the bead map and histogram displays. You can save a Results files *without* this bead map and histogram information, but with all the other numerical data from a reading intact. Note that the histogram and bead map data is not necessary for calculating results or generating reports; it is only used for displaying these graphs.

In the *Save As* dialog box, select the **Compressed mode** checkbox to save the file without the bead map and histogram data. All other data will be preserved.

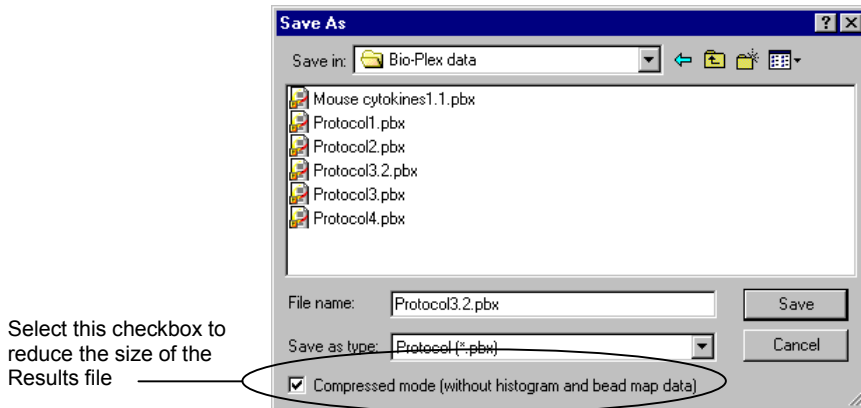


Figure 88. *Save As* dialog for Results files.

Results Window

The Results window is divided into subwindows, each containing different options for viewing, formatting, and calculating your data.

Click on these buttons to review/change the original Protocol settings.

Click on these buttons to view and calculate results.

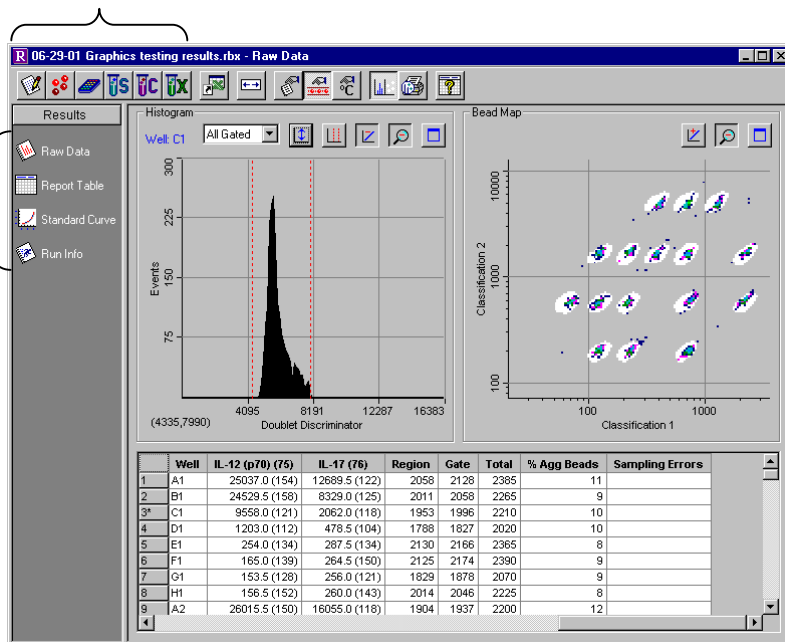


Figure 89. Results window.

Navigate among the different subwindows using the buttons along the left side of the Results window:

- Raw Data  (see page 113).
- Report Table  (see page 115).
- Standard Curve  (see page 125).
- Run Info  (see page 131).

Viewing/Changing the Protocol Settings

In the Results window, you can review and in some cases change the plate formatting, analyte selections, sample concentrations, etc., from the original Protocol file to display different data or recalculate data in the Results file tables. These controls are located in the toolbar across the top of the Results window.

-
- Any changes that you make in the Results file will not change the original Protocol file or the data from the reading. They will only change how that data is displayed in the Results file tables.
-

Click on the **Describe Protocol** button  in the Results toolbar to review or change the description of the Protocol, as described on page 50.

Click on the **Select Analytes** button  in the Results toolbar to review the analyte selection or select different analytes, as described on page 50.

-
- During a reading, the Bio-Plex array reader will always detect *all* the analytes in the sample, including any that you have not selected. However, analytes that were detected but not selected will not appear in the final reports and tables. After a reading, you can go back and correct your analyte selections, and the detected analytes will appear in the tables.
-

Click on the **Format Plate** button  in the Results toolbar to review or change the formatting of the plate. Plate formatting is described on page 56.

-
- Only formatted wells are read by the Bio-Plex array reader. You must define all the wells that you want to read before running a Protocol. However, you can change the formatting of a defined well (e.g., from unknown to standard) after a reading, and the analysis of that well will change.
-

Click on the **Enter Standards Info** button  in the Results toolbar to review or change the concentrations of the standards, as described on page 68.

Click on the **Enter Controls Info** button  in the Results toolbar to review or change the concentrations of the controls, as described on page 79.

Click on the **Enter Sample Info** button  in the Results toolbar to review or change the sample information, as described on page 81.

Raw Data

The *Raw Data* window in the Results file includes the same histogram and bead map and numerical *Raw Data* table as in the *Run Protocol* window. These features are described in the previous chapter on the following pages:

- Histogram and bead map—page 94.
- *Raw Data* table—page 104.

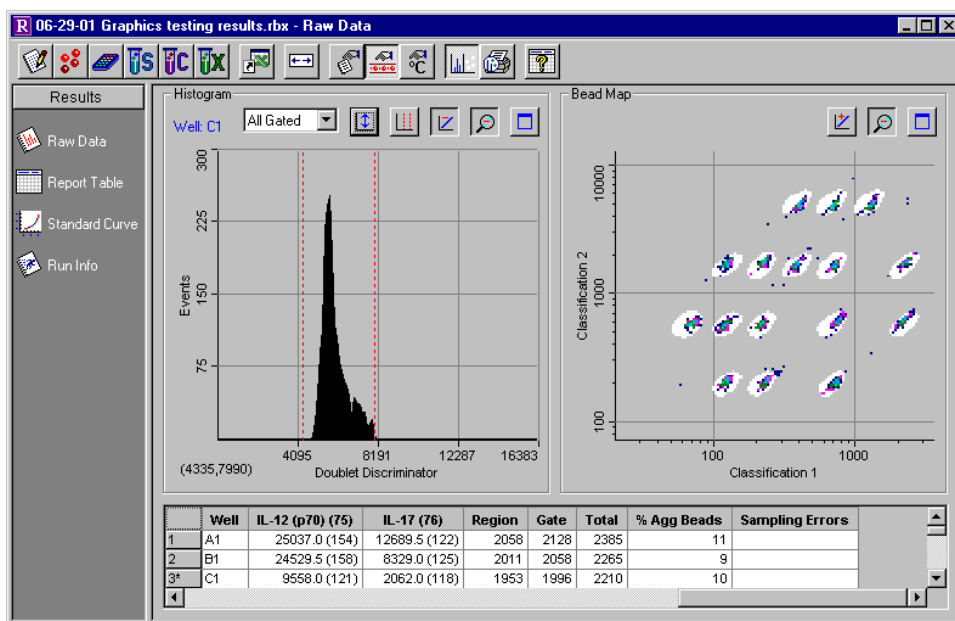
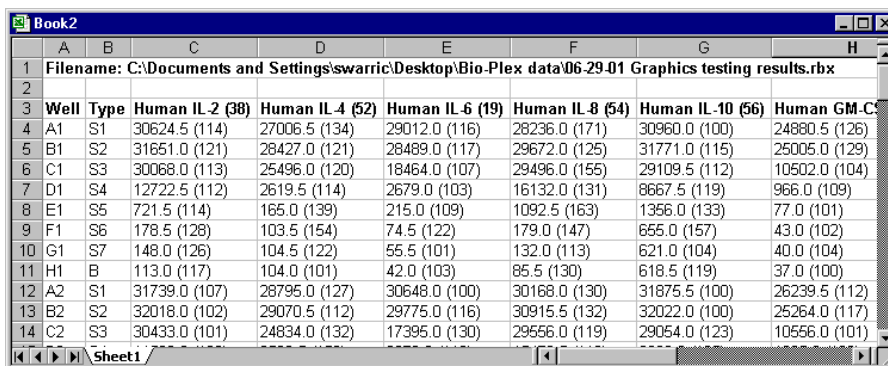


Figure 90. *Raw Data* window in Results file.

The *Raw Data* window in the Results file includes additional exporting and printing functions not available in the Protocol window.

Exporting Raw Data

To automatically export your raw data to Microsoft® Excel, click on the **Export to Excel** button  in the toolbar or select the command from the *File* menu. This command will automatically launch Excel and display a spreadsheet containing your data.



	A	B	C	D	E	F	G	H
1	Filename: C:\Documents and Settings\swarric\Desktop\Bio-Plex data\06-29-01 Graphics testing results.rbx							
2								
3	Well	Type	Human IL-2 (38)	Human IL-4 (52)	Human IL-6 (19)	Human IL-8 (54)	Human IL-10 (56)	Human GM-CSF (100)
4	A1	S1	30624.5 (114)	27006.5 (134)	29012.0 (116)	28236.0 (171)	30960.0 (100)	24880.5 (126)
5	B1	S2	31651.0 (121)	28427.0 (121)	28489.0 (117)	29672.0 (125)	31771.0 (115)	25005.0 (129)
6	C1	S3	30068.0 (113)	25496.0 (120)	18464.0 (107)	29496.0 (155)	29109.5 (112)	10502.0 (104)
7	D1	S4	12722.5 (112)	2619.5 (114)	2679.0 (103)	16132.0 (131)	8667.5 (119)	966.0 (109)
8	E1	S5	721.5 (114)	165.0 (139)	215.0 (109)	1092.5 (163)	1356.0 (133)	77.0 (101)
9	F1	S6	178.5 (128)	103.5 (154)	74.5 (122)	179.0 (147)	655.0 (157)	43.0 (102)
10	G1	S7	148.0 (126)	104.5 (122)	55.5 (101)	132.0 (113)	621.0 (104)	40.0 (104)
11	H1	B	113.0 (117)	104.0 (101)	42.0 (103)	85.5 (130)	618.5 (119)	37.0 (100)
12	A2	S1	31739.0 (107)	28795.0 (127)	30648.0 (100)	30168.0 (130)	31875.5 (100)	26239.5 (112)
13	B2	S2	32018.0 (102)	29070.5 (112)	29775.0 (116)	30915.5 (132)	32022.0 (100)	25264.0 (117)
14	C2	S3	30433.0 (101)	24834.0 (132)	17395.0 (130)	29556.0 (119)	29054.0 (123)	10556.0 (101)

Figure 91. Example of an Excel spreadsheet with exported data.

You can also export the raw data to a text file. With your raw data displayed, go to the *File* menu and select **Export Table to File**. This command will open a *Save As* dialog box in which you can specify a file name and location for the text file.

- You can copy any or all values in the table to the Windows clipboard. Select the entire table, or the desired rows, columns or entries, and select **Copy** from the *Edit* menu. The values you copy can be pasted into other Windows applications.

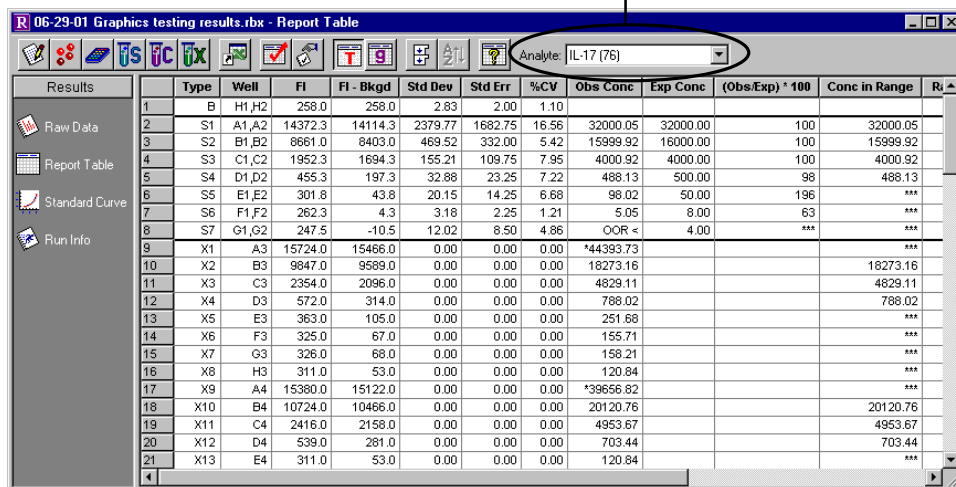
Printing Raw Data

You can print the *Raw Data* table using the **Print** command  on the main toolbar or *File* menu. **Print Preview** on the *File* menu displays the table as it will appear in the printout.

Report Table

The *Report* table provides detailed information about each of the analytes in your samples. Click on the **Report Table** button  to open the *Report* table window.

Select analyte from the pulldown list



	Type	Well	FI	FI - Bkgd	Std Dev	Std Err	%CV	Obs Conc	Exp Conc	(Obs/Exp) * 100	Conc in Range	Ri
1	B	H1,H2	258.0	258.0	2.83	2.00	1.10					
2	S1	A1,A2	14372.3	14114.3	2379.77	1682.75	16.56	32000.05	32000.00	100	32000.05	
3	S2	B1,B2	8661.0	8403.0	469.52	332.00	5.42	15999.92	16000.00	100	15999.92	
4	S3	C1,C2	1952.3	1894.3	155.21	109.75	7.95	4000.92	4000.00	100	4000.92	
5	S4	D1,D2	455.3	197.3	32.88	23.25	7.22	488.13	500.00	98	488.13	
6	S5	E1,E2	301.8	43.8	20.15	14.25	6.68	98.02	50.00	196	***	
7	S6	F1,F2	262.3	4.3	3.18	2.25	1.21	5.05	8.00	63	***	
8	S7	G1,G2	247.5	-10.5	12.02	8.50	4.86	OOR <	4.00	***	***	
9	X1	A3	15724.0	15466.0	0.00	0.00	0.00	*44393.73			***	
10	X2	B3	9847.0	9589.0	0.00	0.00	0.00	18273.16			18273.16	
11	X3	C3	2354.0	2096.0	0.00	0.00	0.00	4829.11			4829.11	
12	X4	D3	572.0	314.0	0.00	0.00	0.00	788.02			788.02	
13	X5	E3	363.0	105.0	0.00	0.00	0.00	251.68			***	
14	X6	F3	325.0	67.0	0.00	0.00	0.00	155.71			***	
15	X7	G3	326.0	68.0	0.00	0.00	0.00	158.21			***	
16	X8	H3	311.0	53.0	0.00	0.00	0.00	120.84			***	
17	X9	A4	15380.0	15122.0	0.00	0.00	0.00	*39656.82			***	
18	X10	B4	10724.0	10466.0	0.00	0.00	0.00	20120.76			20120.76	
19	X11	C4	2416.0	2158.0	0.00	0.00	0.00	4953.67			4953.67	
20	X12	D4	539.0	281.0	0.00	0.00	0.00	703.44			703.44	
21	X13	E4	311.0	53.0	0.00	0.00	0.00	120.84			***	

Figure 92. *Report* table.

First, select the analyte that you want to review from the **Analyte** pulldown list above the table. All the data for that analyte in each defined well of the plate will be displayed in the table.

For on-line Help information about the *Raw Data* table, click on the **Display Help for Table Legends** button .

Report Table Columns

The columns in the table show the data for the selected analyte for each well or replicate group of wells.

The available *Report* table columns are listed below:

- **Type**—Sample type (unknown, standard, blank, etc.) and number, based on plate formatting.
- **Well**—Well number; all wells are listed for a replicate group (see next section).
- **Outlier**—Contains a checkbox for flagging the well or group of wells as an outlier. See page 119.
- **Description**—The description you entered for each standard, control, or sample in the Protocol. Click on the **Show/Hide Description Columns** button  to view or hide this column.
- **FI**—The fluorescent intensity of the selected analyte.
- **FI – Bkgd**—The fluorescent intensity minus background (measured using the defined blank wells) of the selected analyte.
- **Std Dev**—The standard deviation of a replicate group of wells.
- **Std Err**—The standard deviation of a replicate group of wells divided by the square root of the number of replicates.
- **%CV**—The coefficient of variation of a replicate group of wells. If this value is high, you may want to review the individual FI values and select the appropriate member of the replicate group as an outlier (see next section).
- **Obs Conc**—The observed concentration of the selected analyte, calculated from its fluorescent intensity and the standard curve. Note that the observed concentrations of standards are back-calculated from their fluorescent intensities and the standard curve.
- **Exp Conc**—The expected concentration of the selected analyte standard or control, as specified in the *Enter Standards Info* window or *Enter Controls Info* window, respectively.
- **(Obs/Exp)*100**—The recovery rate for the observed versus expected concentrations of the standard or control. See page 120.
- **Conc in Range**—An observed concentration (see above) that falls within the range of standard concentrations given a specified recovery range (see page 77). Values outside the range are flagged as ***.
- **Group**—The defined group of the selected well (see page 65).
- **Ratio**—The ratio of the FI values of a member/reference or reference/member in a group (see page 65).
- **Dilution**—The specified dilution factor of the selected analyte.
- **Sampling Errors**—Error codes for a reading. See page 106.

Report Table Options

To select the columns to display in the *Report* table, as well as other display settings, click on the **Report Table Options** button .

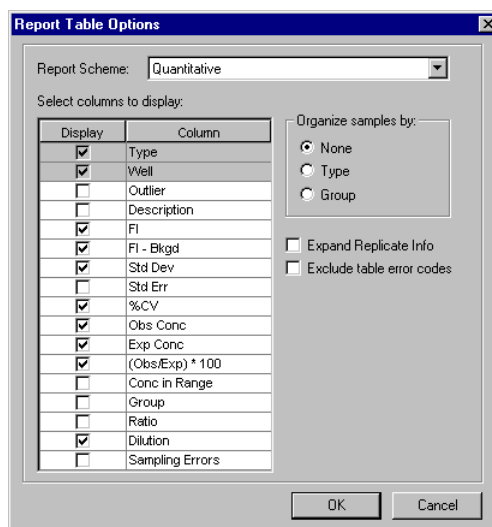


Figure 93. *Report Table Options* dialog.

In the *Report Table Options* dialog, select the columns to display using the checkboxes. You can also choose between predefined report settings by selecting **Quantitative** or **Qualitative** in the *Report Scheme* field; these selections will display different sets of columns.

-
- The predefined set of Quantitative columns are designed to be used with cytokine assays, while the Qualitative columns are designed to be used with phosphoprotein assays.
-

Under *Organize samples by*, you can group the rows in the *Report* table by **Type** (standards, controls, unknowns, etc.), **Group** (if defined), or **None**. Note that **Type** and **Group** selections will override the **Sort** function in the *Report* table (see page 121). For example, if you select **Type** organization, any sorting you do will be within each type.

To review the data for each well in a replicate group (see page 117), select the **Expand Replicate Info** option.

To hide error codes in the table such as OOR, ***, and ----, as well as * before concentration values, select the **Exclude table error codes** checkbox.

Replicate Groups

If you have defined multiple wells as a replicate group in the *Format Plate* window, the wells will be listed together in the **Well** column, and mean data values for those wells will be reported in the relevant columns for the selected analyte (unless the wells have been flagged as outliers; see page 119).

S1	A11,A12	23692.00
----	---------	----------

Figure 94. Example of a standard with two replicate wells (A11 and A12).

To review the data for each well in a replicate group, click on **Report Table Options**  and select the **Expand Replicate Info** checkbox.

	Type	Well	FI	FI - Bkgd
1	B	A1,A2	70.25	70.25
2		A1	69.00	69.00
3		A2	71.50	71.50
4	S1	A11,A12	23692.00	23621.75
5		A11	23706.00	23635.75
6		A12	23678.00	23607.75
7	S2	B11,B12	21500.00	21429.75
8		B11	21155.00	21064.75
9		B12	21845.00	21774.75
10	S3	C11,C12	3538.50	3468.25
11		C11	3594.00	3523.75
12		C12	3483.00	3412.75

Figure 95. Expanded data for replicate wells.

The mean values as well as the values for the individual wells will be reported for each replicate group.

Selecting Outliers

In some cases, you may want to exclude a well or replicate group of wells from statistical calculations (e.g., when the CV% of a set of replicates is high, or the observed concentration of an analyte is out of range). The **Outlier** column in the *Report* table window allows you to flag a well or replicate group of wells as an outlier.

Click on the **Show/Hide Outliers Column** button  in the toolbar to show or hide the column.

Each row in the column contains a checkbox. When you select the checkbox, the data for the selected analyte in the well is excluded and the table is automatically recalculated.

Note that flagging a well as an outlier for the selected analyte does not flag it as an outlier for all analytes. To do this, right click on the outlier checkbox for the well, and select **Set Outlier (All Analytes)** from the context menu.

	Type	Well	Outlier	Description	FI	FI
1	B	H1, H2	☐		148.8	
2	S1	A1, A2	☐		25526.3	
3	S2	B1, B2	☐		24290.8	
4	S3	C1, C2	☐		9940.5	
5	S4	D1, D2	☐		1246.5	
6	S5	E1, E2	☐		251.0	
7	S6	F1, F2	☐		164.3	
8	S7	G1, G2	☐		155.0	
9	X1	A3	☒			
10	X2	B3	☐			
11	X3	C3	☐			
12	X4	D3	☐			
13	X5	E3	☐			
14	X6	F3	☐			

Set Outlier
Clear Outlier
Set Outlier (All Analytes)
Clear Outlier (All Analytes)

Figure 96. Selecting a well as an outlier for all analytes.

Now the well data for all analytes will be excluded from calculations. To remove the outlier flag for all analytes, right-click on the well select **Clear Outlier (All Analytes)** from the context menu.

If you flag a replicate group of wells as an outlier, data for the entire group is excluded. If you click on the **Expand Replicates Info** button , you can flag each well in the group as an individual outlier. If you do this, the outlier checkbox for the entire group will appear checked and grayed out, indicating that part of the group has been flagged.

Type	Well	Outlier
B	H1,H2	☒
	H1	☐
	H2	☒

Group checkbox appears gray and selected

Figure 97. Flagging a single well in a replicate group as an outlier.

Obs/Exp*100 Column

Because the standard curve (page 125) is critical for calculating the concentrations of unknowns, the **Obs/Exp*100** column in the *Report* table allows you to assess how well the curve fits the standards. This column is also used with controls (page 79) to determine the overall accuracy of an assay.

To display this column, click on **Report Table Options**  and select the **(Obs/Exp)*100** checkbox.

The **(Obs/Exp)*100** column shows the recovery rate for the observed versus expected concentration of each standard. This number should fall within the recovery percentage range selected in the *Enter Standards Info* window (see page 77). For example, if you selected a recovery range of 80–120%, the observed (back-calculated) concentration of a standard should be within 80–120% of the expected concentration. Values outside this range are flagged in the **Conc in Range** column.

Conc in Range Column

The concentration in range column (**Conc in Range**) provides a method for determining which unknown concentrations are reliable based on the recovery range of the standards (see previous section). This column shows only observed concentration values that fall within the range of valid standards; values that do not fall in the range are displayed as ***. (The concentration in range values of the back-calculated standards are also shown, for comparison purposes.)

Obs Conc	Exp Conc	(Obs/Exp) * 100	Conc in Range
	32000.00		
7937.63	16000.00	50	***
3999.99	4000.00	100	3999.99
500.00	500.00	100	500.00
49.97	50.00	100	49.97
8.84	8.00	111	8.84
2.73	4.00	68	***
1556.37			1556.37
1662.21			1662.21
7232.12			***
335.06			335.06
38.24			38.24
7.20			***
*2.00			***

Back-calculated standard concentrations within a specified recovery percentage range of 80–120%

Concentrations of unknowns within the range of valid back-calculated standard concentrations

Concentrations of unknowns outside the range of valid back-calculated standard concentrations

Figure 98. *Conc in Range* column.

For example, if only standard concentrations from 1.9 to 8,100 pg/ml fall within the specified recovery range, then only unknown concentrations that fall within this range will be displayed in the **Conc in Range** column.

Table Formatting

To sort the data in the table alphanumerically, click on the heading of the column that you want to sort by, then click on the **Sort** button . Clicking twice will invert the order.

To highlight the divisions between the unknowns, standards, controls, etc. in the report, click on the **Organize by Type** button . With this button selected, any sorting you do will be within each type.

To highlight the divisions between defined groups (see page 65) in the report, click on the **Organize by Group** button . With this button selected, any sorting you do will be within each group.

To automatically size the columns fit the width of the text, click on the **Autofit** button  on the toolbar.

Exporting the Report Table

To export the data in the *Report* table to a different file, click on the **Export Report Table** button  in the toolbar.

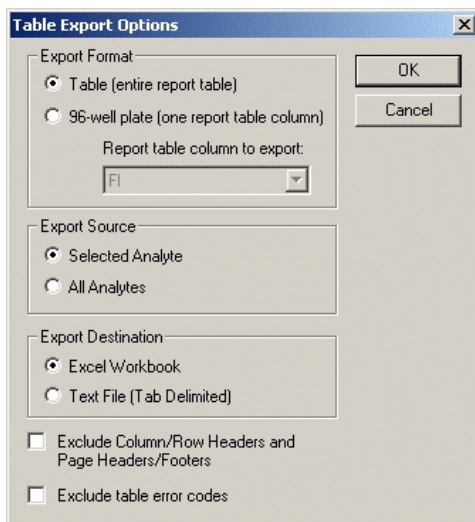


Figure 99. *Table Export Options* dialog.

You can export the *Report* table data in either a table format or a 96-well plate format (i.e., the data is arranged like the wells in a 96-well plate). If you export in 96-well plate format, you can export only one column of data from the report table at a time, and the data for each well in that column will be displayed in a plate layout.

In the *Export Format* section, select **Table** to export the entire table, or **96-well plate** to export in plate format. If you select **96-well plate**, specify the column to export under the *Report table column to export* pulldown list.

C:\Program Files\Bio-Rad\Bio-Plex Manager\Examples\Human Cytokine 16 plex.rbx
 FI - Bkgd
 Hu TNF alpha (36)

	1	2	3	4	5	6	7	8	9	10	11	12
A	31.00	31880.00	364.00	27041.00	32043.00	32214.50	32045.50	32735.00	32666.00	32735.00	32735.00	32735.00
B	33.00	29213.50	380.00	26205.00	31977.50	31616.00	31204.00	32735.00	30125.00	29382.00	27725.00	30507.50
C	29.00	29646.00	46.00	20294.00	20248.00	15367.00	23669.00	30275.00	4824.50	3129.00	3249.00	3843.00
D	---	28781.00	45.00	1370.50	1441.50	2077.00	1830.00	3462.50	268.50	206.50	205.50	257.00
E	30630.00	6489.00	47.00	113.00	87.00	119.00	140.00	267.50	38.50	31.00	31.00	37.00
F	31636.50	6372.00	20.00	11.00	13.00	24.00	14.00	38.00	18.00	18.00	15.00	10.00
G	31561.50	6606.00	20.00	4.00	1.00	9.00	5.50	19.00	-4.00	-4.00	0.50	-1.00
H	31243.00	372.00	20.00	-7.00	-7.00	-8.00	-5.50	0.00	32735.00	32735.00	32735.00	32735.00

Hu IL-4 (52)

	1	2	3	4	5	6	7	8	9	10	11	12
A	31.00	31880.00	364.00	27041.00	32043.00	32214.50	32045.50	32735.00	32666.00	32735.00	32735.00	32735.00
B	33.00	29213.50	380.00	26205.00	31977.50	31616.00	31204.00	32735.00	30125.00	29382.00	27725.00	30507.50
C	29.00	29646.00	46.00	20294.00	20248.00	15367.00	23669.00	30275.00	4824.50	3129.00	3249.00	3843.00
D	---	28781.00	45.00	1370.50	1441.50	2077.00	1830.00	3462.50	268.50	206.50	205.50	257.00
E	30630.00	6489.00	47.00	113.00	87.00	119.00	140.00	267.50	38.50	31.00	31.00	37.00
F	31636.50	6372.00	20.00	11.00	13.00	24.00	14.00	38.00	18.00	18.00	15.00	10.00
G	31561.50	6606.00	20.00	4.00	1.00	9.00	5.50	19.00	-4.00	-4.00	0.50	-1.00
H	31243.00	372.00	20.00	-7.00	-7.00	-8.00	-5.50	0.00	32735.00	32735.00	32735.00	32735.00

Figure 100. FI - Bkgd data for multiple analytes exported to a text file, plate format.

To export only the data for the currently selected analyte in the *Report* table, select **Selected Analyte**. To export the data for all the analytes, select **All Analytes**. If you export all analytes to Excel using the plate export format, each analyte will be exported to a separate worksheet within a single workbook. If you export to a text file, each analyte will appear consecutively as a separate plate layout table in the file (see Figure 100).

Under *Export Destination*, you can export to a Microsoft® Excel file (**Excel Workbook**) or a tab-delimited text file (**Text File**).

Select the **Exclude Column Headers/Footers and Page Headers/Footers** to exclude this information from the exported file.

The error codes symbols in the *Report* table (OOR<>, ***, ---, * before concentration values) can interfere with macros in some types of spreadsheets, including Excel spreadsheets. Select the **Exclude table error codes** checkbox to exclude these symbols from the exported file. If this option is selected, table cells with error codes will be exported as empty cells.

If you are exporting to Excel, Microsoft® Excel will automatically launch when you click on **OK** and a spreadsheet will be created.

If you are exporting to a text file, a *Save As* dialog will open when you click on **OK**. Select the desired directory and name and click on **Save** to complete the operation.

Microsoft Excel - Book1

File Edit View Insert Format Tools Data Window Help

Figure 101. Example of an Excel spreadsheet with exported data.

Copying the Report Table

You can copy the contents of the table to the Windows clipboard. Select rows, columns, cells, or the entire table, then select **Copy** from the *Edit* menu. The copied values can be pasted into other Windows applications.

Printing the Report Table

You can print the *Report* table for the selected analyte using the **Print** command  on the main toolbar or *File* menu. You can print the *Report* table for all analytes using the **Print All Analytes** command  on the main toolbar or *File* menu.

Print Preview displays the table as it will appear in the printout.

Standard Curve

The *Standard Curve* window displays the standard curve for each standard analyte in the Results file, and allows you to change the regression model as well as various curve display controls for each analyte. (Curve settings are initially selected in the *Enter Standards Info* window in a Protocol; see page 68.)

Click on the **Standard Curve** button  to open the *Standard Curve* window.

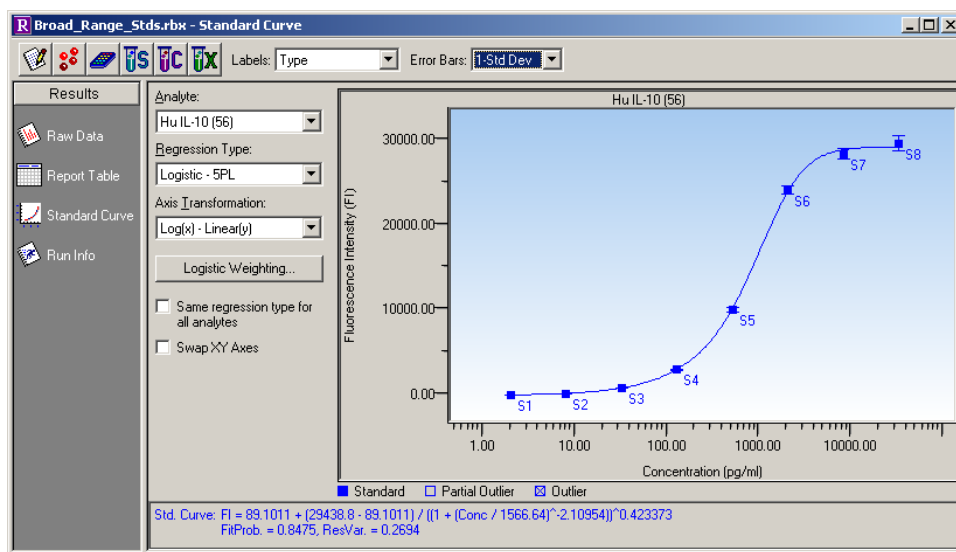


Figure 102. Standard curve.

In the graph, standards defined on the current plate will be displayed as blue squares, while external standards will be displayed as red circles. If both types of standards are used in the *Results* file, both will be displayed on the curve. Outliers and partial outliers will be displayed on the curve as indicated in the legend below the curve.

The equation for the selected regression method as well as some curve fit statistics (R^2 , fit probability, residual variance) will be displayed below the standard curve, if possible. Equations for standards defined on the current plate are shown in blue,

while equations for external standards are displayed in dark red. Any curve fitting errors will also be displayed below the curve in red.

Use the **Analyte** pulldown list to display the curve for a selected analyte.

Above the curve, use the **Labels** pulldown list to select from a variety of different labels for the points in the curve. If both external and current standards are displayed in the graph, labels will only be displayed for the external standards.

Use the **Error Bars** pulldown list to select from several different error bars (e.g., 1-Std Dev, 2-Std Err) to plot for each point in the curve. If there is only one replicate for a point, the bar will have a value of zero. If both external and current standards are displayed in the graph, error bars will only be displayed for the external standards.

Use the **Regression Type** pulldown list to choose from seven different regression models to generate the standard curve (see next section).

-
- Changing the regression type will automatically recalculate all the data in the *Report* table. If you are unsure whether your data has been recalculated, select **Recalculate** from the *Edit* menu.
-

Use the **Axis Transformation** pulldown list to select the standard curve axis scale. Four different combinations of linear and logarithmic are available.

-
- Changing the axis transformation will only change the display of the curve—not the calculated data.
-

You can use a different regression model for each analyte. Deselect the **Same regression type for all analytes** checkbox, select a different analyte from the pulldown list, and select the regression type to use for that analyte.

Select the **Swap XY Axes** checkbox to change the X and Y axes of the curve.

To display all the data for each point in the curve, position your cursor over the point. A pop-up box will list the data.

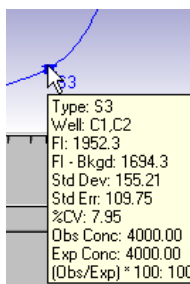


Figure 103. Pop-up data for a point in the curve.

Regression Methods

There are seven regression methods for calculating the standard curve:

- Logistic-5 PL
- Logistic-4 PL
- Linear
- Cubic spline
- Point-to-point
- Linear (semi-log)
- Point-to-point (semi-log)

The minimum number of non-zero standards required for using each type of regression method is shown in the following table:

Method	Minimum number of standards
Logistic-5PL	6
Logistic-4PL	5
Cubic spine	4
Linear (linear and semi-log)	2
Point to point (linear and semi-log)	2

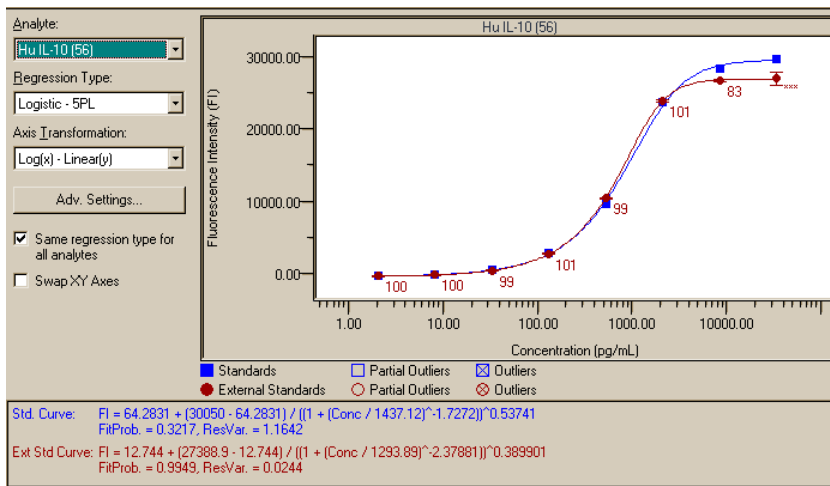


Figure 104. Example of Logistic-5PL curves for standards on the current plate and external standards.

If you do not have enough points for the selected method, an error message will be displayed in red below the standard curve.

Logistic-5PL

This version of the five parameter logistic equation has been developed by Brendan Scientific, creator of StatLia immunoassay software. The Logistic-5PL regression type yields the best results for most assays; therefore, this is the default in Bio-Plex Manager. However, there may be cases where the Logistic-4PL equation gives slightly better results.

The Logistic-5PL equation is:

$$y = d + \frac{a - d}{\left[1 + \left(\frac{x}{c} \right)^b \right]^g}$$

where:

x is the concentration,

y is the response,

a is the estimated response at zero concentration,

b is the slope of the tangent at midpoint,

c is the midrange concentration or midpoint,

d is the estimated response at infinite concentration, and

g is an asymmetry factor.

Logistic-4PL

In certain cases, where the curve is a perfectly shaped “S,” the Logistic-4PL regression may give better results. However, the Logistic-5PL regression has been shown to be the most robust routine for most Bio-Plex assays.

The Logistic-4PL equation is:

$$y = d + \frac{a - d}{1 + \left(\frac{x}{c}\right)^b}$$

where:

x is the concentration,

y is the response,

a is the estimated response at zero concentration,

b is the slope of the tangent at midpoint,

c is the midrange concentration or midpoint, and

d is the estimated response at infinite concentration.

Linear

The Linear equation is:

$$y = a + bx$$

Where a is the intercept and b is the slope of the line.

The simplest method for determining concentrations from a standard curve is to construct a plot using the linear portion of the response curve. The R-squared value may be used to determine the overall goodness of the linear fit. A linear regression with an R-squared value of greater than 0.99 is considered a very good fit.

The primary advantage of this method is that it is extremely simple. The primary disadvantage is that the linear range of concentrations is relatively small compared to that which may be obtained using nonlinear regression.

The Linear curve is available in linear and semi-log versions.

Cubic Spline

Cubic spline curves are smooth curves that go through every data point. The model is a cubic polynomial on each interval between data points. In some cases, a spline curve can work well as a standard curve for interpolation. However, because the curve is calculated individually for every pair of points, it does not correspond to any single equation.

The cubic spline curve is applicable to a wide range of assays; however, there are no built-in biochemical restrictions on the curve shape, and values cannot be extrapolated.^{1,2}

Point-to-Point

A single equation is not available for the point-to-point method. The slope of each segment of the curve between data points is calculated independently.

The Point-to-Point curve is available in linear and semi-log versions.

Linear versus Logistic Regression Methods

In terms of performance, the primary difference between linear and logistic regression models is that the linear range in which concentrations can be accurately predicted is much smaller. In most cases, the overall response of the assay is best

¹ The Immunoassay Handbook. David Wild, Editor. Nature Publishing Group, New York, NY. (2nd Edition, 2001).

² The GraphPad Guide to Nonlinear Regression. Harvey Moltusky, Ph.D. GraphPad Software, San Diego, CA.

modeled using a logistic fit and the Logistic-5PL model will yield the best results. For more information, see Bio-Rad bulletin 2861, available on our Web site (www.bio-rad.com) or from Technical Support (see page 12).

Nevertheless, in certain cases you may want to use linear regression to analyze the data. For most cytokines, the linear portion of the curve is well within the biological range of concentrations expected for patient samples. Thus, for serum the linear regression may be useful. The main advantage of a linear regression is that fewer points (as few as 2) can be used.

Copying the Standard Curve

To copy a bitmap image of the curve to the Windows clipboard, click in the *Standard Curve* window and select **Copy** from the *Edit* menu. You can then paste this image into your spreadsheets or other documents using the **Paste** command in your other applications.

You can also copy the regression method equation from below the standard curve by dragging your cursor over the equation to highlight it, and then selecting **Copy** from the *Edit* menu or right-click context menu.

Printing the Standard Curve

You can print the standard curve for the selected analyte using the **Print** command  on the main toolbar or *File* menu. You can print the curve for all analytes using the **Print All Analytes** command  on the main toolbar or *File* menu.

When you select a print command, a dialog box will open allowing you to specify the page layout of the printed curve.

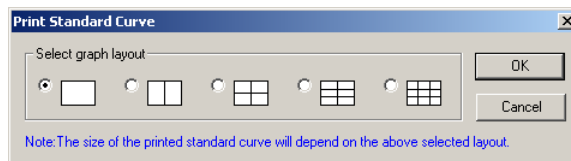


Figure 105. *Print Standard Curve* dialog.

The layout diagrams show the number of curves that can be printed on a page. If you are printing the curve for a single analyte, that curve will be sized based on the layout you select (e.g., the  layout will size the curve at one-quarter the full size of the page). The regression type will be printed below each curve.

Select a layout and click on **OK** to display the standard Windows® *Print* dialog.

Run Info

The *Run Info* window displays information about the reading, including date and time, sample size, PMT voltage, analytes selected for the reading, DD gates, etc.

Click on the **Run Info** button  to open the *Run Info* window.

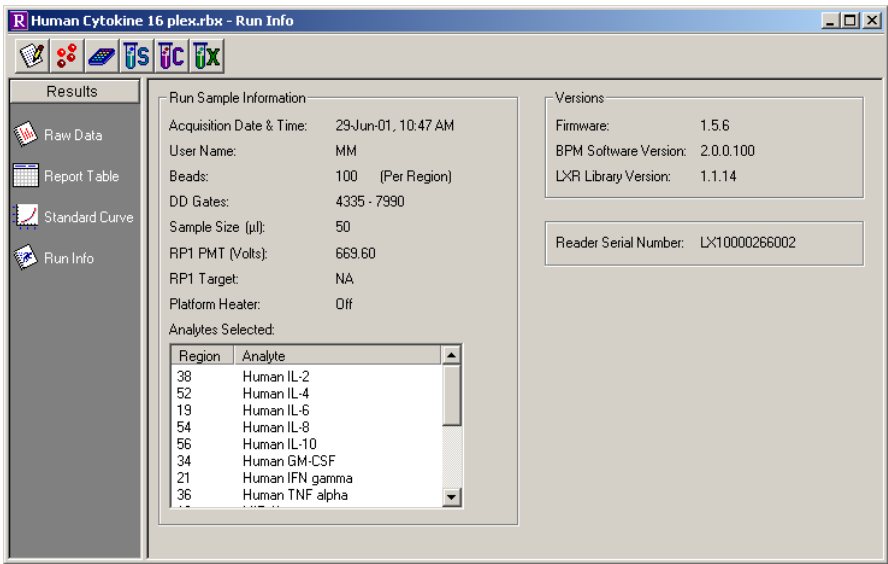


Figure 106. *Run Info* dialog.

- Note that you cannot change any of the information in the *Run Info* window. This information is for documentation purposes only.

Appendix 1.

Troubleshooting Guide

Message/Problem	Cause	Solution
"Bio-Plex Manager has detected a problem with low bead number"	Most Likely:	
	Too few beads in the assay (5,000 per region recommended).	Check bead number calculations in assay.
	Plate not shaken for 30 seconds before analysis.	Remove plate from array reader and shake for 30 seconds.
	Buffer volume in wells is too low (must be at least 125µl).	Re-suspend wells in 125µl. Perform REMOVE BUBBLES.
	Microbubble in cuvette.	Perform REMOVE BUBBLES. Perform UNCLOG to verify fluidics integrity.
	Low/no sheath fluid.	Refill sheath fluid, check sheath connections. Perform START UP.
	Possible clog.	Perform UNCLOG and rerun. If unsuccessful, repeat. If still unsuccessful, contact technical support.
	Less Likely:	
	Incorrect needle height.	Adjust needle height.
	Incompatible plate type used.	Replace with flat bottom or filter plate and adjust needle height.
	Vacuum system not calibrated.	Calibrate vacuum system (see hardware manual).
	Red laser failure.	Contact technical support.
	Filter plate not flat.	Check filter plate flatness (see hardware manual).
	Incompatible suspension buffer used.	Check buffer compatibility (see hardware manual).

Message/Problem	Cause	Solution
"Bio-Plex Manager has detected a problem with selection of regions"	Incorrect bead regions were selected in the Protocol.	Compare bead regions in the assay with those selected in the Protocol.
	Incorrect regions were selected when preparing the assay.	Verify regions chosen during assay preparation.
	Too few beads in the assay in one or more regions.	Verify that the correct number of beads was used by referring to the specific assay preparation procedure.
" Bio-Plex Manager has detected a possible problem due to microbubbles in the cuvette or photobleached beads "	Most likely:	
	Calibration was performed before the array reader was warmed up.	Perform 30-minute WARM UP and recalibrate.
	Calibrated with incorrect CL1 or CL2 target values.	Check that the target values of the CAL beads match values entered in the software. If different, recalibrate.
	Array reader was calibrated with a dirty MCV Plate II.	Clean MCV Plate II and recalibrate.
	Misalignment of optics.	Perform OPTICS VALIDATION. Contact technical support if values are not within range.
	Less likely:	
	Microbubbles present in cuvette.	Perform REMOVE BUBBLES.
	Calibration beads are photobleached (do not expose to light for more than 1 hour).	Recalibrate with new CAL1 beads.

Message/Problem	Cause	Solution
"Bio-Plex Manager has detected a problem with bead aggregation"	Incorrect DD gate values used.	View histogram in Run Protocol window and make sure DD peak falls within the gate range. If necessary, adjust gates (see page 97).
	Incorrect DD target value used for calibration.	View histogram in Run Protocol window and make sure that the DD peak falls within the gate range. If running a Luminex-type assay from another manufacturer, use only Bio-Plex DD settings (targets and gates).
	Clumped beads present.	Vortex plate at 900 rpm for 1 minute.
	Sheath reservoir is empty.	Refill sheath reservoir, then perform START UP.
	Waste reservoir is overfilled.	Empty waste and reconnect.
"Check Link" in status bar of software	Incompatible suspension buffer used.	Check hardware manual for buffer compatibility.
	Array reader or microplate platform is not turned on.	Turn on array reader and microplate platform.
	Software is not communicating with array reader.	Close and restart Bio-Plex Manager.
"Pressurizing" in status bar of software	Cables between computer and array reader or microplate platform are loose/not connected.	Check cables for proper connections.
	Leak in sheath bottle or cap.	Tighten sheath cap.
Needle stuck in down position	Protective assay plate covering was not removed.	See hardware manual for procedure for raising needle stuck in down position. Then remove cover from assay plate.
	Needle guide is not screwed all the way in.	Tighten needle guide by turning tube clockwise until tight.
	Sample needle is bent.	Replace bent needle with a new needle (see hardware manual for procedure).

Message/Problem	Cause	Solution
No Assay Signal Detected	Most likely:	
	Error in assay preparation.	Make sure that streptavidin and cytokine standards have been added. Consult assay preparation procedure.
	Incorrect target values used in calibration.	Check that target values in the <i>Calibrate</i> dialog match the values on CAL1 and CAL2 bottles.
	Waste container overfilled.	Empty waste. Reconnect waste. Perform UNCLOG.
	Sheath reservoir low or empty.	Refill sheath reservoir. Perform START UP. Proceed with run.
	Less likely:	
	Waste line not connected properly.	Check waste line connection. Make sure that it clicks into place.
	Laser not functioning.	Perform calibration. If no signal detected, call technical support.
"Bio-Plex Manager has detected a change in sheath pressure"	Most likely:	
	Sheath reservoir cap not on securely.	Tighten sheath cap. Message should disappear within two minutes.
	Sheath bottle lines are not connected properly.	Make sure that all hoses are connected to the appropriate ports, and that they have clicked into place.
	Less likely:	
	Sheath fluid level above the AIR port on the sheath container.	Adjust sheath fluid level so that sheath fluid is below the AIR port of sheath bottle.
	Sheath bottle has a leak.	Try new sheath bottle. Call technical support for further assistance.
	High-throughput fluidics (HTF) system not turned on.	Turn on HTF and check to ensure that message disappears.
	Air Compressor not working.	Listen for air pump to turn on when WARM UP is selected. Contact technical support for further assistance.

Message/Problem	Cause	Solution
Bio-Plex Manager has detected a change in the temperature of the array reader. Please calibrate before running an assay to ensure accurate results.	Room temperature has changed.	Calibrate array reader.
The calibration was unsuccessful. Please repeat calibration. If calibration fails a second time, consult Troubleshooting Guide.	Calibration procedure failed.	Make sure CAL1 beads and CAL2 beads are placed in the appropriate wells (CAL1 in red well and CAL2 in green well). Repeat calibration. If problem persists, contact technical support.
The calibration was unsuccessful. Bio-Plex Manager has detected a problem with low bead number. Please repeat calibration.	Calibration procedure failed due to low bead number.	Run UNCLOG procedure, then repeat calibration. If problem persists, contact technical support.
Optics Validation Procedure shows value(s) outside of acceptable range(s).	Problem with optical component of array reader.	Repeat validation procedure. If values are still out of range, contact technical support.
Fluidics Validation Procedure shows value(s) outside of acceptable range(s).	Problem with fluidics component of array reader.	Repeat validation procedure. If values are still out of range, contact technical support.
Reporter Validation Procedure shows value(s) outside of acceptable range(s).	Problem with optical component of array reader.	Repeat validation procedure. If values are still out of range, contact technical support.
Classify Validation Procedure shows value(s) outside of acceptable range(s)	Problem with calibration or optical component of array reader.	Repeat validation procedure. If values are still out of range, contact technical support.

Message/Problem	Cause	Solution
Optics, Fluidics, Reporter, or Classify Validation Procedure fails due to low bead number	Beads not resuspended.	Resuspend beads and repeat validation procedure. If procedure still fails, contact technical support.
	Clog in instrument.	Perform UNCLOG operation and repeat validation procedure. If procedure still fails, contact technical support.
Platform heater not at target temperature	Insufficient warm-up time allowed.	Allow at least 15 minutes for platform heater to reach target temperature. If heater does not reach target in 30 minutes, contact technical support.
	Platform heater not functioning properly.	Contact technical support.
Platform heater not at ambient temperature	Insufficient cool-down time allowed.	Open platform door to facilitate cooling. An ice pack can also be placed on the platform plate holder to facilitate cooling.

Appendix 2.

Glossary

Accuracy	The closeness of a measured value to the actual value or the difference between expected and measured values. Using the Reporter Validation Kit, accuracy is defined as the percent difference between the measured regression curve and the expected or actual data points.
Aggregated microspheres	Microspheres that have associated. Two microspheres that have associated are called a doublet.
Analyte	The substance being measured, identified, quantitated or otherwise analyzed in an experiment. In the Bio-Plex system, analytes are molecules that are being quantitated.
Antibody	Immunological protein capable of binding a unique antigen via a specific binding site or epitope. Used in immunological assays to quantitate the amount of a specific analyte present.
Assay	The bead and reagent mixture that, when combined with sample, is loaded into microplate wells and is read by the Bio-Plex array reader.
Background (noise)	Fluorescent intensity measured by the reporter channel that is due to non-specific binding of a fluorochrome and the electronic noise of the array reader. If you include blank wells in your microplates, you can calculate the average background intensity and subtract it from the intensity readings of your standards and unknowns.
Bead	A microsphere.
Calibration	A process that uses microspheres (calibrators) with a stable fluorescent intensity to adjust the gain settings in the Bio-Plex array reader's detectors for optimal and consistent microsphere classifications and reporter readings, over time and across different array readers.
Calibrators	Microspheres with stable fluorescent intensities used to standardize the signal output of the Bio-Plex array reader.

Capture antibody	An antibody used in sandwich immunoassays that binds or "captures" a specific analyte for quantitation.
Carboxylated microspheres	Microspheres with chemically modified surfaces such that they are covered with carboxyl functional groups (COOH) that allow covalent biomolecular attachment.
Classification channels	Fluorescent detection channels of the Bio-Plex array reader that detect the fluorescent intensities of two dyes embedded in the Bio-Plex microspheres. The ratio of the two dyes identifies each bead as a unique analyte within a sample.
Classify validation	A validation procedure for measuring the ability of the Bio-Plex array reader to efficiently classify assay beads into specified regions.
Clumping	The aggregation of two or more microspheres.
Conjugated microspheres	A microsphere set that has been chemically modified so that its surfaces are covered with molecules that bind with the target analytes in an assay system.
Coupled microsphere	See conjugated microsphere.
Cytokine	An immunological protein that can signal a response in the immune system.
Detection antibody	An antibody used in capture sandwich immunoassays that binds a captured analyte to a fluorochrome, allowing for quantitation of the protein. Detection antibodies may be biotinylated, which allows for addition of a streptavidin-bound fluorochrome, or the detection antibody may be directly labeled with a fluorochrome.
Doublet	Two microspheres that have associated. The resulting signal of the associated microspheres is twice that of a single (singlet) microsphere.
Doublet discriminator	A channel that measures the amount of light scatter from particles that flow past the red laser. Light scatter is directly proportional to particle size, and the channel is designed to identify particles that are smaller or larger than a single microsphere, including microspheres that are clumped or aggregated.

Dynamic range	The calculated number of decades covered by the log amplifier and the reporter channel scale of the Bio-Plex array reader. Range is calculated using the Reporter Validation Kit.
Emission spectrum	The wavelength range of light emitted by an excited fluorochrome when its electrons fall from a higher to a lower energy state. Expressed in nanometers (nm).
Ex/Em	A common way of reporting a molecule's excitation and emission wavelengths (for example, Ex/Em = 488/520 nm).
Excitation spectrum	The wavelength range that excites a molecule's electrons to a higher energy state. Expressed in nanometers (nm).
External standard	A standard that has been imported from a different file.
Fit probability	A statistical measure of goodness of fit that incorporates the residual SSE. A fit probability of 1 indicates a perfect fit, while a fit probability of 0 indicates no fit.
Fluidics validation	The process of verifying the well-to-well carryover of beads in the Bio-Plex array reader.
Fluorescence	The emission of light that occurs when the electrons of a fluorochrome drop to a lower energy state.
Fluorochrome	A fluorescent molecule.
Gate	An electronic method of discriminating or eliminating aggregated microspheres in the analysis of an assay. A gate separates singlet beads from aggregated beads.
Group	A group of wells defined on a plate. You can calculate the ratio of the fluorescent intensities of the group's member wells to the fluorescent intensity of a Reference well in the group.
Immunofluorescence	A technique that uses a covalently linked fluorochrome-antibody complex to detect or quantify a particular antigen.
Kinase	An enzyme that functions to phosphorylate specific proteins. Kinases are responsible for the activation of a variety of proteins through the process of phosphorylation.

Laser	A highly purified source of light used to excite fluorochrome electrons. An acronym for Light Amplification by Stimulated Emission of Radiation.
Linearity	A measurement of the coefficient of determination or R^2 value of a set of standard beads. An R^2 value of 0.995 or greater is an acceptable linearity value using the Bio-Plex validation kit.
Logistic regression	A regression model for binary (dichotomous) outcomes. The data are assumed to follow binomial distributions with probabilities that depend on the independent variables.
Median fluorescent intensity	A relative measurement of fluorescent intensity based on the median, a robust statistical measure.
Microparticle	A solid substance with a diameter in the micrometer range. Often used as a synonym for a microsphere.
Microsphere set	A set of Bio-Rad multianalyte microspheres containing a unique mixture of two distinct fluorochromes to distinguish them from other multianalyte microspheres.
Microspheres	Latex spheres with a diameter in the micrometer range. Also called beads.
Multianalyte or multiplex	An analysis of several assays or tests performed simultaneously in the same reaction container.
Operational qualification	The process of determining that an instrument is fit for its intended use. The Bio-Plex Validation Kit is a tool for operational qualification.
Optics validation	The process for verifying the alignment of the optics assembly of the Bio-Plex array reader.
Outlier	A well or replicate group of wells whose measured value is outside the normal range. You can exclude outliers from statistical calculations.
Panel	A group of related bead sets. For example, the Human Cytokine panel includes bead sets for Human IL-2, Human IL-4, Human IFN gamma, and other human cytokines. You can select multiple bead sets from the same panel or sets from different panels in the same multiplex assay.

PE	See phycoerythrin.
Phosphoprotein	An enzyme that functions to phosphorylate specific proteins. Phosphoproteins are responsible for the activation of a variety of proteins through the process of phosphorylation.
Photobleaching	A chemical reaction caused by exposure to light, in which a fluorochrome is converted into a differently fluorescent or non-fluorescent compound.
Photomultiplier tube	A light detector typically used in fluorescence detection systems, designed to convert a fluorescent signal into an electronic signal that can be quantitated.
Phycoerythrin (PE)	The fluorochrome used as the reporter molecule in Bio-Plex assays. Excitation = 546 nm, emission = 575 nm.
PMT	See photomultiplier tube
Precision	A measure of the reproducibility of replicate readings, usually represented by the coefficient of variation (CV%).
Protocol file (*.pbx)	A file containing the settings of a reading, including the microplate wells to read, the analytes to detect, sample size, etc. To perform a reading, you open a Protocol file, select the settings, and then run the Protocol. You can then save the Protocol settings and reuse or modify them. After a reading, a Results file is created containing the settings from the Protocol as well as the data results. The raw data from the most recent reading is also stored in the Protocol file.
Quality control	Procedures and guidelines that determine conformity to requirements.
Recovery percentage	A mechanism for assessing the fit of the standard curve to the actual standards. For each analyte standard, an observed concentration is back-calculated from the standard curve and the fluorescent intensity. This is divided by the expected concentration and multiplied by 100 to give the recovery percentage.

Recovery Range	The range of acceptable recovery percentages (see above). For example, a recovery range of 80–120% means that the observed concentration of a standard should be within 80–120% of the expected concentration. Concentrations of standards and unknowns that fall outside this range are flagged as unreliable.
Reference	A well within a group of wells defined on a plate that is used as a reference. You can calculate the ratio of the fluorescent intensity of the Reference well to the fluorescent intensities of the other wells in the group.
Region	The region of a fluorescent color map used to identify a particular bead set. Each bead set is embedded with specific quantities of two fluorescent dyes; the combination of these fluorochromes, as detected by the Bio-Plex array reader, places the bead set within a unique region on the color map, thereby identifying the set and its associated analyte.
Reporter	A fluorescent molecule that is incorporated into an assay in such a way that the fluorescence intensity is directly proportional to the amount of analyte in an assay.
Reporter channel	The channel of the Bio-Plex array reader that detects the fluorescent signal of the reporter molecule (phycoerythrin).
Reporter system	Any combination of molecules in solution that serves to quantitate the analytes in a particular assay.
Reporter validation	A standard method for verifying the performance of the reporter channel of the Bio-Plex array reader. The five primary parameters that verify accurate performance of the reporter channel include linearity, dynamic range, sensitivity, accuracy and slope of the response.
Residual variance	A statistical measure of goodness of fit. This is the weighted sum of the squared errors between the observed response and the response predicted by the fitted curve (residual SSE), divided by the number of degrees of freedom. A small residual variance indicates a good curve fit.

Results file (*.rbx)	A file containing the results of a reading, including the raw data, the settings information from the Protocol, and tools for analyzing and exporting the data (tables, a standard curve, export functions, etc.). Each reading generates a new Results file.
Sample	A solution of analyte standards or unknowns that, when added to an assay, will bind to the microspheres in that assay and be identified and quantitated by the array reader.
Sampling error	A measure of the degree to which a sample differs from the entire population.
Sandwich immunoassay	Immunoassay for the quantitation of a specific analyte through the use of monoclonal antibodies (Ab). A sandwich is formed when an analyte of interest binds to a capture Ab, and then a second antibody (detection Ab) binds to a different epitope of the protein.
Sensitivity	The lowest detectable signal above instrument noise. Noise may be attributed to the lasers, the detectors and the amplification electronics of the array reader.
Signal	The detectable measurement unit of a reporter molecule.
Signal-to-noise ratio	The ratio of a specific assay signal to the underlying noise in the assay. Signal to noise is typically used to measure the sensitivity of an assay.
Singlet(s)	A single microsphere or a population of single microspheres. Compare to doublets which are two microspheres associated together.
Slope	Defined as the rise over the run when plotting a series of points to form a line. Using the Reporter Validation Kit, the slope of the response is related to the dynamic range of the array reader and yields information about the response of the photomultiplier tube (PMT).
Spectral address	The unique fluorescent emission spectra of a microsphere.
Standard deviation	The spread in individual data points (e.g., in a replicate group) to reflect the uncertainty of a single measurement.
Standard error	The standard deviation of a set of replicates divided by the square root of the number of replicates.

Unconjugated microspheres	A microsphere set that has been chemically modified so that its surfaces are covered with some functional group, but that has not been chemically prepared for use as a component in an assay system. These base microspheres do not carry biomolecular reactant.
Uniformity	The reproducibility of a signal over a series of replicate measurements.
Validation	A formal process for documenting that an instrument is fit for its intended use and is kept in an appropriate state of maintenance and calibration.
Validation kit	A set of reagents and procedures designed to verify the performance of the Bio-Plex array reader apart from an assay. Validation Kit 3.0 assesses the alignment of the optics assembly, measures the performance of the reporter channel, evaluates the fluidics system, and verifies the classify efficiency of the Bio-Plex array reader.
Variance	The mean square deviation of a variable around the average value.

Appendix 3.

Table Error Codes

Certain data that cannot be measured or calculated will be flagged in the *Raw Data* and *Report* tables. Columns and their possible error codes are listed below:

Column	Error Code	Cause
FI	***	No data present
	---	Marked as an outlier
FI – Bkgd	***	No data present (negative values, if any, will be displayed)
	---	Marked as an outlier
SD	***	No data present
CV%	***	No data present
Obs Conc (4PL/5PL)	***	No data present
	OOR>	Out of Range - Above asymptote of equation
	OOR<	Out of Range - Below asymptote of equation
	OOR	Out of Range - Math Error
	*Value	Above or below standards FI
Obs Conc (Linear))	***	No data present
	OOR	Negative extrapolated data
	*Value	Positive extrapolated data or Data above/below the standards FI
Exp Conc	***	No data entry - this value is input by user
Obs/Exp*100	***	OOR displayed in Obs Conc column
Conc In Range	***	Obs Conc is not within the specified recovery range of standards conc (the recovery range is specified by the user; see page 78)
Sampling Error	1	Low bead number detected in the well
	2	Aggregated beads detected in the well
	3	Classify efficiency problem detected in the well
	4	Region selection problem detected in the well
	5	Platform temperature problem detected in the well

Appendix 4.

Basic Concepts

The following is a brief overview of the basic principles of the Bio-Plex array system. For more information, see *Practical Flow Cytometry*, 3rd edition, by Howard M. Shapiro, M.D. (New York: Wiley-Liss Inc., 1995).

Microspheres

Bio-Plex microspheres are highly uniform, 5.5-micron polystyrene beads that have been crosslinked during polymerization for physical and thermal stability. Microspheres are grouped into sets; each set is then embedded with specific quantities of two fluorescent dyes. The ratio of these two dyes gives each set of microspheres a unique spectral address.

Each set of microspheres is then conjugated with a different reactant specific for a particular target analyte. Reactants can include enzyme substrates, receptors, antigens, and antibodies. A conjugated bead-reactant mixture can be mixed with a sample to create, for example, a capture sandwich immunoassay.

-
- To ensure the stability of the spectral address, it is essential to protect the microspheres from light. Do not subject microspheres to prolonged high temperatures, and protect them from freeze/thaw manipulations.
-

Reporter fluorochromes

When you mix your samples with the microspheres, the target analytes bind to the reactants on the surface of the beads. The amount of analyte that binds to the beads is then quantitated through the use of a fluorochrome, R-phycoerythrin. The intensity of this fluorochrome as detected by the Bio-Plex array reader is directly proportional to the amount of analyte present. The excitation wavelength of the Bio-Plex array reader reporter channel is 532 nm, while the emission wavelength is $575 \pm 12\text{nm}$. Specific information on R-phycoerythrin is shown in the following table:

	R-Phycoerythrin
Formula weight (Daltons)	240,000
Absorbance max (nm)	480, 546, 565
Extinction max (M-1cm-1)	1,960,000
Emission max (nm)	578
Quantum yield	0.82

Fluidics

There are two fluidic paths in the Bio-Plex array reader. The first path includes a syringe-driven mechanism that controls the sample uptake. This mechanism permits small sample uptake volumes from small reaction volumes. The syringe-driven system transports a user-specified volume of sample from a microplate well to the cuvette. The sample is injected into the cuvette at a steady rate, and the microspheres in the sample are aligned in a single file through the path of the two Bio-Plex lasers.

Following analysis, the first sample path is automatically purged with sheath buffer fluid via the second fluidics path. This process effectively removes residual sample in the tubing, valves, and sample needle.

Excitation

There are two lasers in the Bio-Plex array reader: one for classifying each microsphere and its associated analyte, and the other for quantitating the amount of analyte bound to each microsphere. The first laser, called the classification or "red" laser, excites the dyes embedded in each microsphere; the fluorescent signal is discriminated with selective emission filters and converted into intensity units by fluorescence detectors and a digital signal processor, and the microsphere is classified. The second laser, called the reporter or "green" laser, excites the fluorochromes bound to the target analytes on the surface of the microspheres; again, the fluorescent signal is discriminated with selective emission filters and converted into intensity units by fluorescence detectors and a digital signal processor, and the amount of analyte is quantitated.

Appendix 5.

Microsphere Handling

Microsphere Dispersion

Bio-Plex microspheres will settle and aggregate if left undisturbed. You should always ensure that they are evenly distributed before dispensing. Gentle vortexing is the most effective method of mixing for most coated microsphere preparations; however, sonication can also be used to separate aggregated microspheres prior to coupling. After sonication, microspheres generally remain dispersed for about 1 hour. Sonication is most effective if applied to a centrifuged pellet of microspheres. In order to maintain bead concentration, we suggest that you centrifuge your stock beads before sonication.

The following two indirect sonication methods are effective at separating aggregates in closed containers of microsphere preparations.

Inserting a sonicator probe into preparations is not recommended

Probe Sonicator

To disperse a microsphere pellet using a probe sonicator:

1. Place probe tip in a small bath of water.
2. Insert the end of the microsphere tube near the tip of the sonicator probe, without touching it. Do not immerse the tube closure.
3. Adjust sonication for optimal disruption and pulse sonicate until the microsphere pellet is dispersed.

Bath Sonicator

To disperse a microsphere pellet using a bath sonicator:

1. Turn on the bath sonicator and examine the surface for an area of maximum disruption.

2. Insert the end of the tube into the sonicator bath near the area of maximum disruption.
3. Sonicate until the microsphere pellet is dispersed.

Enumeration of Microsphere Suspensions

Bio-Plex microspheres are provided at standard concentrations. The microsphere yield after a coupling process may be less than the starting concentration due to loss during wash steps. The microsphere loss can vary according to operator technique, coupled reactant properties, and scale of coupling.

It is not advisable to construct an assay without defining the concentration of the subject-coupled microsphere preparation. The total surface area (total number of microspheres) represented in an assay is a critical variable, so the optimization and control of this variable begins with manipulation of known microsphere numbers. Microspheres can be counted with a hemocytometer. Please follow counting and calculation methods outlined in the instructions for your hemocytometer.

Microsphere separation methods

Bio-Plex assay development often requires separation of microspheres from an aqueous matrix.

Centrifugation

Microsphere coupling protocols require separation of microspheres from reaction mixtures or wash buffers by means of centrifugation. Generally, a tabletop microcentrifuge that can spin 1.5 ml microcentrifuge tubes can pellet Bio-Plex microspheres in 1–2 minutes. The g-forces vary in these instruments from approximately 6,000 to 13,000 $\times$ g.

Vacuum

A vacuum separation method can be used for routine high throughput washing required for washed assay procedures. In this case, the microsphere reactions are performed in microtiter filter-bottom 96-well plates. A vacuum is applied to the plate allowing the liquid to filter through while retaining the microspheres on the filter. Resuspension is accomplished by adding adequate fluid to each well and by repetitive vigorous pipetting.

Microsphere Agitation During Assay

Agitation of Bio-Plex assays with a plate shaker during incubation steps prevents microspheres from settling during extended incubation periods. The benefit of agitation may not be discernible for some assay conditions and should be evaluated according to the requirements of a given application. It is important to resuspend the beads in an assay for 30 seconds to performing a Bio-Plex reading.

Microsphere Stability and Storage

Bio-Plex microspheres are light sensitive and should be protected from light during all stages of storage and usage. In addition, freezing conditions and organic solvents should be avoided.

Appendix 6.

Product Numbers

Software Packages (includes CD-ROM, User Guide, and Hardware Protection Key)

Bio-Plex Manager—Workstation version	171-000001
Bio-Plex Manager 3.0—upgrade from 2.0	171-
Bio-Plex Manager software—Desktop version	171-001014
Bio-Plex Manager software—Network Desktop version, 5 seat	171-001015
Bio-Plex Manager software—Network Desktop version, 10 seat	171-001016
Bio-Plex Manager software—Network Desktop version, 20 seat	171-001017

General System Accessories

Sheath Fluid, 20L	171-000055
Communications Cable, 5 ft., DB9	171-002001
Hardware Instruction Manual	4006025
Bio-Plex Manager User Guide	4000182

Array Reader Accessories

Sheath Fluid Bottle, 1 L, polypropylene, including 2 ports and tubing	171-002010
Sheath Waste Bottle, 2 L, polypropylene, including 1 port and tubing	171-002012
Sample Needle, 11.7 cm/4.6 in.	171-002020
Needle Adjustment Tool	171-002026
Protective Shield	171-002030

Microplate Platform Accessories

Needle Guide	171-002023
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Validation and Calibration Accessories

MCV Plate II	171-203031
Validation Kit 3.0	171-203000
Calibration Kit	171-203060

Appendix 7.

Reporting Problems

Included in your Bio-Plex Manager installation is a program called Solobug. We would like you to use this program to request features, request design changes, or report noncritical problems with Bio-Plex.

-
- For critical problems, contact Bio-Rad technical support at the numbers listed on page 12.
-

To use Solobug, select **Solobug** from the *Start* menu, enter your information and a description of the request in the Solobug window, and click on **Save**. This will create a report that you can attach to an e-mail and send to Bio-Rad.

Please send your Solobug files to Bio-Rad at:

LSG.TechServ.US@Bio-Rad.com (in the U.S.)

LSG.TechServ.Intl@Bio-Rad.com (International)

Appendix 8.

Software Warranty

Bio-Rad Laboratories warrants that Bio-Plex Manager software shall substantially conform, in all operational features, to Bio-Rad's current specifications as published in Bio-Rad's user and installation guides and that, when properly installed, it will be free of material defects which affect system performance.

The Purchaser must notify Bio-Rad in writing, within 30 days of delivery of the software (not including delivery of any subsequent modifications to the software), of any defect. If the software is found to be defective by Bio-Rad, Bio-Rad's sole obligation under this warranty is to remedy the defect in a manner consistent with Bio-Rad's regular business practices. For a defect which adversely affects the performance of the software, Bio-Rad shall use its best efforts to cure such defect as soon as reasonably practicable after receipt of Purchaser's notice. For minor defects, Bio-Rad shall use its best efforts to correct such minor defects in the next release of its software. If, however, Bio-Rad is unable to cure a major defect within 90 days of receipt of Purchaser's notice, Purchaser shall have the option to cancel this agreement, whereupon Bio-Rad shall refund only the software fees paid.

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Index

Aggregated beads.....	88	Ejecting plate.....	42
Air bubbles, removing	32	General cautions	21
Alcohol		Miscellaneous functions	43
Flush.....	44	Optics	150
Wash	44	Principles of operation.....	149
Analytes	1	Removing bubbles	32
Custom, creating	52	Sample needle.....	17
Editing.....	54	Shutting down.....	44
Panels	51	Start up	23
Selecting in protocol	50	Status	42
Array Reader.....	<i>See</i> Bio-Plex Array	Unclogging	33
Reader		Warm up	24
Assay reading.....	<i>See</i> Reading	Washing.....	31
Auto save after run.....	88	Bio-Plex Array Reader Validation ...	<i>See</i>
Autofill		Validation	
Settings.....	59	Bio-Plex Manager	
Turning off.....	60	Installation	10
Back flush	44	Interface.....	16
Bead map	100	Overview	5
Analytes selected.....	95	Starting	13
Changing axes	101	Versions of software.....	5
Display options	103	Bio-Plex System.....	1
Displaying	94	Components.....	9
Level multiplier.....	103	Concepts	149
Log/linear view	102	Controls	21
Magnifying.....	102	Product numbers	155
Resizing	102	BioPlex.mdb file.....	11
Beads.....	<i>See Also</i> Microspheres	Blank wells, defining.....	64
Aggregated.....	88	Calibration	25
Low number	88	Adding new control numbers	26
Photobleached.....	88	Log file	30
Beads per region counted.....	87	Log viewer.....	30
Bio-Plex Array Reader.....	1	Monitoring.....	29
Calibration.....	25	Preparing plate.....	28
Cancelling an operation	44	Target values	26
Connecting	14	Cancelling an instrument operation...	44
Controls.....	21	Cautions.....	21
Disconnecting	15	COM port settings	14

Combining panels of analytes	56	Glossary	139
Compressed mode	47, 109	Grouping	<i>See</i> Microplate format, grouping
Computer		Hardware Protection Key (HPK)	11
Screen resolution	13	HASP key	11
Concentration in range	120	Heater	40
Connecting to the reader	14	Histogram	96
Control wells, defining	63	Analytes selected	95
Controls		Channel selection	98
Dilution series	81	DD gate lines	97
Entering information	79	Displaying	94
General system	21	Magnifying	99
Recovery rate	120	Resizing	99
Wells	63	Scaling data	99
Copying		HPK	11
Raw data	106	Installation	
Report table	124	Software	10
Standard curve	131	Instrument information	42
Custom analytes and panels	52	Level multiplier	103
DD gate range	91	Loading plate	91
Displayed in histogram	97	MCV plate	23, 24
Density plot	94	MCV Plate II	22
Describe protocol	50	Menus	16
Desktop version	5	Microplate	
Disconnecting from the reader	15	Loading	91
Doublet discriminator channel	91	Microplate format	56
Displaying	98	Deleting/changing	65
Draining the fluidics system	43	Formatting wells	58
Eject/retract plate	42	Group reference	67
Errors in sampling	88	Grouping	65
Excitation	150	Printing	68
Exporting		Microplate wells	
Raw data	114	Autofill	59
Report table	122	Blanks	64
External standards	70	Controls	63
Fluidics	150	Deleting formatting	65
Draining	43	Grouping	65
Priming	43	Numbering	59
Removing bubbles from	32	Replicate groups	59
Sanitizing	43	Resequencing	62
Unclogging	33	Standards	62
Washing	31	Unknowns	58
Format plate	<i>See</i> Microplate format	Microsoft.net	11
Gate range	<i>See</i> DD gate range		

Microspheres.....	149	Window	49
Calibration.....	26	Quick Guide	17
Handling.....	151	Ratio of member value to reference ..	65
Per region counted	87	Raw data.....	114
Network Desktop version	5	Copying	106
Opening		Exporting.....	114
Protocols	47	Printing.....	114
Results.....	109	Sampling errors	106
Optics		Raw data table.....	104
Shut down	24	Reading	<i>See Also</i> Run
Warm up.....	24	Information about.....	132
Outliers	119	Parameters	47
Panels of analytes		Raw data.....	104
Combining.....	56	Rerun/recovery	107
Custom, creating	52	Results	109
Custom, deleting	55	Sampling errors	88
Custom, renaming	55	Status	93
Editing.....	54	Stopping manually.....	94
Selecting.....	51	Recovery percentage	78
Phycoerythrin.....	150	Reference.....	65
Plate loading	91	Region	
Platform heater.....	40	Beads counted per	105
Temperature error flag	88	Number.....	51
PMT voltage	90	Regression methods.....	127
Priming the fluidics system.....	43	Selecting in protocol.....	76
Printing		Remove bubbles	32
Raw data.....	114	Replicate groups.....	59
Report table	124	Changing	62
Standard curve	131	Defining.....	59
Protocol.....	47	In report table	118
Analytes selection	50	Report table	115
Creating.....	47	Columns	115
Describing.....	50	Conc in Range column	120
Entering controls info.....	79	Copying	124
Entering unknown samples info....	82	Error conditions.....	147
Format plate .. <i>See</i> Microplate format		Exporting.....	122
Generating results	94	Formatting	121
New protocol from results.....	47	Obs/Exp*100 column	120
Opening.....	47	Options	117
Rerunning.....	107	Outliers	119
Running.....	92	Printing.....	124
Saving	47	Replicate groups	118
Standards info	68	Reporter fluorochromes.....	149

Rerun/recovery mode	107	Copying.....	131
Resequencing microplate wells	62	Equations	128
Results	109	Log versus linear.....	130
Analytes selection.....	112	Logistic weighting	77
Calculated values.....	115	Printing	131
Controls info.....	113	Regression method, selecting in	
Exporting raw data	114	protocol.....	76
Exporting report table.....	122	Regression methods	127
Generating from protocol	94	Standard wells, defining	62
Opening	109	Standards	68
Plate format	112	Back-calculation	120
Printing raw data	114	Concentration units	75
Protocol description.....	112	Current plate	71
Raw data	104, 114	Curve.....	125
Report table	115	Entering concentrations	73
Reviewing/changing protocol		External.....	70
settings.....	112	Recovery percentage range	78
Run information	132	Recovery rate	120
Samples info	113	Wells.....	62
Saving.....	109	Start up.....	23
Standard curve	125	Starting Bio-Plex Manager	13
Standards info	112	Status bar	16, 93
Run	See Also Reading	Stopping a reading	94
Information.....	132	System	See Bio-Plex System
Protocol	85	System Requirements	10
Settings	86	Technical support.....	12
Sample grouping <i>See</i> Microplate format,		Toolbars.....	16
grouping		Total beads counted	87
Sample needle, adjusting.....	17	Triggers.....	<i>See</i> Errors in sampling
Sample size.....	88	Troubleshooting	133
Sampling errors		Unclog	33
Raw data	106	Unknown samples	
Sanitizing the fluidics system.....	43	Defining wells.....	58
Saving		Entering information.....	82
Automatically after reading.....	88	Validation	34
Compressed mode	47, 109	Kit	34
Protocols.....	47	Loading Plate	36
Results	109	Log.....	38
Screen resolution	13	Results.....	37
Select analytes	50	Set Up	35
Shut down.....	44	Warm up	24
Solobug	157	Warranty	159
Standard curve.....	125	Wash	

Alcohol.....	44	Wells	<i>See</i> Microplate wells
Basic.....	43	Workflow	7
Between plates	31	Workstation version	5